CAUSAL GRAPH DISCOVERY AND GRAPH-GUIDED MIXTURE-OF-EXPERTS FOR CLIMATE AND REGIONAL WEATHER FORECASTING

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Abbreviations

ACC	–	Anomaly Correlation Coefficient
ADWIN	–	ADaptive WINdowing (drift detector)
BH	–	Benjamini–Hochberg (false-discovery-rate procedure)
BSISO	–	Boreal Summer Intraseasonal Oscillation
CI	–	Conditional Independence
DAG	–	Directed Acyclic Graph
DIR	–	Discovering Invariant Rationales
ENSO	–	El Niño–Southern Oscillation
EOF	–	Empirical Orthogonal Function
ERA5	–	Fifth-generation ECMWF atmospheric reanalysis
FDR	–	False Discovery Rate
FFT	–	Fast Fourier Transform
GNN	–	Graph Neural Network
h850	–	Geopotential height at 850 hPa
IMDAA	–	Indian Monsoon Data Assimilation and Analysis (regional reanalysis)
ISD	–	Integrated Surface Database (NOAA)
ISO	–	Intraseasonal Oscillation
JJAS	–	June–July–August–September
MAE	–	Mean Absolute Error
MCI	–	Momentary Conditional Independence
MJO	–	Madden–Julian Oscillation
MLP	–	Multi-Layer Perceptron
MoE	–	Mixture of Experts
MSE	–	Mean Squared Error
MSLP	–	Mean Sea-Level Pressure
NWP	–	Numerical Weather Prediction
OLR	–	Outgoing Longwave Radiation
ParCorr	–	Partial-correlation conditional-independence test
PC	–	Peter–Clark (algorithm)
PC1	–	PC condition-selection stage restricted to one combination per order
PCMCI	–	PC algorithm followed by Momentary Conditional Independence tests
PW	–	Precipitable Water (total column water vapour)

RH	–	Relative Humidity
RMSE	–	Root-Mean-Square Error
RQ	–	Research Question
RSI	–	Regime Shift Index (STARS)
S2S	–	Subseasonal-to-Seasonal
SST	–	Sea Surface Temperature
STARS	–	Sequential T-test Analysis of Regime Shifts
u200	–	Zonal wind at 200 hPa
u850	–	Zonal wind at 850 hPa

List of Symbols

R	–	Number of regions in a spatial partition ($R = 50$; $R = 100$ for the persistence check)
$\mathcal{V}$	–	Set of climate variables $\{\mathrm{SST}, \mathrm{h850}, \mathrm{u200}, \mathrm{PW}, \mathrm{u850}, \mathrm{OLR}\}$
N	–	Number of time-series nodes, $N = R\,\lvert\mathcal{V}\rvert$ (300 at $R = 50$)
$i = (r, v)$	–	Node index: variable v averaged over region r
X_t^i	–	Standardised anomaly of node i on day t
$\mathbf{X}_t$	–	State vector $(X_t^1, \ldots, X_t^N)^\top \in \mathbb{R}^N$
Y_t^r	–	OLR target of region r on day t (equal to $X_t^{(r,\mathrm{OLR})}$)
L	–	Forecast lead time in days ($L = 7$ primary, $L = 1$ secondary)
τ	–	Lag in days between a source and a target value
$\tau_{\min}, \tau_{\max}$	–	Minimum and maximum lags tested by PCMCI (1 and 60)
$\mathcal{T}_{\mathrm{dev}}, \mathcal{T}_{\mathrm{test}}$	–	Development period (1979–2012) and held-out test period (2013–2022)
$\mathcal{S}$	–	Set of BSISO-season target days
$\mathcal{P}(X_t^j)$	–	Set of lagged parents of X_t^j
$\hat{\mathcal{P}}^{\mathrm{PC}}(X_t^j)$	–	Parent superset returned by the PC1 screening stage
$\hat{\rho}(\cdot, \cdot \mid \cdot)$	–	Sample partial correlation
$\hat{\rho}^{\mathrm{MCI}}_{i \to j}(\tau)$	–	MCI partial correlation of the link $X_{t-\tau}^i \to X_t^j$ (link strength)
$p,\ q$	–	Raw p-value and Benjamini–Hochberg adjusted q-value
$\alpha_{\mathrm{PC}}, \alpha_{\mathrm{FDR}}$	–	Significance levels of the screening stage and of the FDR control (0.05 each)
$\mathcal{G} = (\mathcal{N}, \mathcal{E})$	–	Lagged causal graph with node set $\mathcal{N}$ and edge set $\mathcal{E}$
$e = (i, j, \tau)$	–	Directed lagged edge from node i to target node j at lag τ
$\tilde{X}_t^d$	–	Phase-randomised surrogate (decoy) series
D	–	Number of decoys (250)
δ	–	Decoy rate: fraction of decoys selected as a parent of any target
$d(r, r')$	–	Great-circle distance between the closest cells of regions r and r'
s_e	–	Implied propagation speed $d(r, r')/(\tau\,\Delta t)$ of edge e
$s_v^{\max}$	–	Maximum admissible propagation speed of source variable v
k	–	Number of outside-region edges retained per target by top-k selection
Π_r	–	Persistence score of target region r across resolutions

$J(A, B)$	–	Jaccard similarity $\lvert A \cap B \rvert / \lvert A \cup B \rvert$
$\pi_r(r')$	–	Graph prior: strength assigned to source region r' for target r
$\mathcal{C}_r$	–	Candidate source-region pool of target r
$a_{r'r}$	–	Splitter score of candidate edge $r' \to r$ (sigmoid output)
ρ_{top}	–	Fraction of candidate edges selected as causal by the splitter
λ	–	Weight of the variance-across-interventions penalty
$\beta,\ \gamma$	–	Graph-initialisation scale and graph-regularisation weight
S_{pers}	–	MSE skill score relative to persistence
R^2_{clim}	–	MSE skill score relative to climatology
σ_a	–	Standard deviation of a target's mean splitter scores (discrimination measure)
Δ_w	–	Causal-signature distance between consecutive windows

Abstract

Data-driven forecasting models select their predictors by statistical association, yet in the tropical atmosphere most associations are inflated by autocorrelation, shared seasonal cycles and common drivers. This thesis asks whether a *validated* graph of lagged climate dependencies, found by causal discovery and frozen before any test year is seen, can tell a forecasting model which source regions to use. The setting is daily outgoing longwave radiation (OLR), a proxy for deep tropical convection, forecast for 50 tropical regions centred on the Indo-Pacific during the boreal summer intraseasonal oscillation (BSISO) season.

Phase 1, the primary contribution, represents the climate system as 300 time-series nodes (50 regions × six variables: SST, 850 hPa geopotential height, 200 and 850 hPa zonal wind, precipitable water and OLR) over 1979–2022, with 1979–2012 reserved for discovery and training and 2013–2022 held out. A BSISO index reproduced from the project's own data (leading modes explaining 11.2% and 7.7% of variance, close to published values) defines an activity-based season that restricts target days only. PCMCI with partial-correlation tests and Benjamini–Hochberg false-discovery-rate control is run over 18,000 lagged candidates per OLR target (lags 1–60 days) using a vectorised implementation verified link-for-link against Tigramite. The resulting graph of 536 links is passed through four independent checks: phase-randomised decoys (0 of 250 decoys selected), a physical propagation-speed mask (3 of 536 links rejected), data-driven top-k selection, and persistence at doubled spatial resolution (mean persistence 0.62; 33 of 48 scored targets at or above 0.5). The validated graph is sparse, short-lag and largely regional: a median outside-link lag of one day and about two source regions per target.

The frozen graph is then used as a source-selection prior for an invariance-trained causal splitter and per-region expert within a Mixture-of-Experts forecasting framework. In the full 50-region candidate pool the graph-restricted splitter becomes discriminative (informative splitters rise from 0 to 24 of 50 regions at a 7-day lead and from 0 to 36 at a 1-day lead). Seven-day forecast skill is unchanged relative to the no-prior model (mean skill against persistence +0.416 versus +0.415), whereas at a 1-day lead skill improves in 32 of 50 regions (mean skill −1.33 versus −1.53; mean R^2 against climatology 0.745 versus 0.720). Because the graph links act on time scales of one to two days, the evidence locates the graph's value at short leads, and attributing the gain to the graph's choice of regions, rather than to pool size, requires the matched random- and correlation-prior controls defined in the experimental design. A sliding-window analysis shows that changes in the causal structure do not align with ENSO phase changes at one- and three-season window lengths.

Phase 2 extends the framework to hourly, kilometre-scale near-surface forecasting over India, combining a shared backbone, causally gated regional experts and drift-triggered expert adaptation, and specifies the data, baselines and shift-aware evaluation it requires. The thesis contributes a reproducible build–validate–freeze–use protocol for causal graphs in forecasting, a validated graph with full diagnostics, and an evaluation design that separates a graph's credibility from its predictive usefulness.

Keywords: causal discovery, PCMCI, BSISO, outgoing longwave radiation, surrogate testing, mixture of experts, graph prior, subseasonal forecasting, concept drift.

Chapter 1

Introduction

1.1 Climate Variability and the Need for Useful Regional Forecasts

Weather forecasts are traditionally separated into two regimes. At short range, up to roughly ten days, numerical weather prediction (NWP) integrates the equations of atmospheric motion forward from an analysed initial state, and its skill has grown steadily for four decades [1]. At seasonal range, predictability comes mainly from slowly varying boundary conditions such as sea surface temperature (SST), most prominently through the El Niño–Southern Oscillation (ENSO) [2]. Between the two lies the *subseasonal* window of about two weeks to two months. It is too long for the initial atmospheric state to be remembered in detail, because small errors grow chaotically [3], and too short for the boundary forcing to dominate. This window has been called a predictability desert, and it has become a priority of the operational and research community through coordinated efforts such as the Subseasonal-to-Seasonal (S2S) prediction project [4, 5].

The window matters in practice. Agricultural planning, reservoir operation, heat-health warnings and disaster preparedness need advance notice of weeks rather than days, and the most valuable information is regional: whether convection and rainfall will be enhanced or suppressed over a particular part of South Asia or the Maritime Continent in the coming week. In the tropics, the main physical source of skill in this window is organised, slowly propagating convection: the Madden–Julian Oscillation (MJO) in boreal winter [6–8] and the Boreal Summer Intraseasonal Oscillation (BSISO) in boreal summer [9–11]. A forecasting system that recognises where these organised disturbances are, and which regions they are about to influence, has a physical basis for skill beyond persistence.

Machine learning has transformed medium-range forecasting. Graph neural networks and transformers trained on reanalysis now match or exceed operational NWP on many headline scores [12–15]. These models are, however, trained to minimise an average error over all their inputs. They do not state which inputs carry information about a given target, and they are not designed to detect when that dependence changes. For scientific understanding, for trust and

for adaptation to a changing climate, it is useful to have an explicit, testable statement of *which* source regions and variables influence a target, and with what delay. This thesis investigates whether such a statement can be built from data, validated independently, and then used to improve a forecasting model.

1.2 The Boreal Summer Intraseasonal Oscillation and Tropical Convection

The BSISO is the dominant mode of intraseasonal variability over the Asian summer monsoon region between roughly May and October [11, 16]. Its convective envelope forms over the equatorial Indian Ocean and then propagates both eastward along the equator and northward over the Indian subcontinent and the Bay of Bengal, producing a characteristic north-west to south-east tilted rain band that also moves northward over the western North Pacific [10, 17, 18]. Its dominant period is 30–60 days, with a secondary 10–20-day component [11, 19]. Over India it modulates the alternation of active and break spells of the monsoon [19–21].

The BSISO is related to, but distinct from, the boreal-winter MJO. The MJO propagates predominantly eastward along the equator and is well described by the Wheeler–Hendon index [22]. The BSISO's northward propagation, its coupling to the monsoon trough and its seasonal timing make a summer-specific description necessary. Lee *et al.* [11] therefore proposed BSISO indices based on a multivariate empirical orthogonal function (EOF) analysis of outgoing longwave radiation (OLR) and 850 hPa zonal wind over 10°S–40°N, 40°–160°E during May–October. This thesis reproduces that index from its own data (Chapter 3) and uses it to define the season in which forecasts are evaluated. Throughout the thesis, the boreal-summer phenomenon is not conflated with winter intraseasonal variability; where results from MJO studies are used, the transfer is stated explicitly.

1.3 OLR as a Forecasting Target and a Proxy for Convection

OLR is the thermal infrared radiation leaving the top of the atmosphere, observed by polar-orbiting satellites since the 1970s [23]. Its value depends on the temperature of the emitting surface. Over the tropical oceans, deep convective clouds have cold, high tops and emit little, so low OLR indicates active deep convection. Clear skies or shallow clouds allow emission from the warm lower troposphere and surface, so high OLR indicates suppressed convection. Because of this relation, OLR is a standard proxy for large-scale tropical convective rainfall [24] and was the primary field in the original detection of intraseasonal convection [18].

OLR is a proxy and not a measurement of precipitation. Thick non-precipitating cirrus also lowers OLR, and the relation between OLR and rainfall varies with region and cloud regime. In this thesis an OLR forecast is therefore described as a forecast of convective activity, and

statements about rainfall are interpretations rather than verified results. OLR is a demanding target: its day-to-day autocorrelation decays quickly, so after a week a naive "no change" forecast retains almost no information and any skill must come from the evolution of the larger system.

1.4 Challenges of Multivariate, Lagged Climate Dependencies

The future OLR of a region depends on the past state of many variables in many regions. Moisture is advected by low-level winds; convective heating excites equatorial Kelvin and Rossby waves that carry signals thousands of kilometres within days [25–27]; SST anomalies modify the surface fluxes that sustain convection over weeks. A forecasting model that is given every variable of every region at every lag faces a very large and highly redundant input space. In the representation used here, 50 regions with six variables give 300 time series. If each of them is considered as a possible source for each OLR target at each lag between 1 and 60 days, there are $300 \times 60 = 18{,}000$ lagged candidates per target, $50 \times 18{,}000 = 900{,}000$ candidates for the 50 OLR targets, and $300 \times 299 \times 60 \approx 5.4$ million ordered source–target–lag combinations if every distinct node pair were considered (Section 3.2.3).

Three properties of climate data make selection among these candidates difficult. First, every series is strongly *autocorrelated*, so the effective sample size is much smaller than the number of days and spurious correlations between unrelated series are common [28, 29]. Second, many series share *common drivers*: an active BSISO phase simultaneously changes winds, moisture and OLR over large areas, so two regions can be correlated at a lag without either influencing the other. Third, relations are *non-stationary*: the monsoon–ENSO relationship, for example, has varied over decades [30].

1.5 Limitations of Correlation-Based Source Selection

The simplest way to choose predictors is to rank candidate sources by their lagged correlation with the target. Runge *et al.* [31] showed that lagged cross-correlation in climate data is ambiguous in both strength and delay: autocorrelation of the source and target inflates and broadens the correlation function, indirect pathways appear as direct links, and a common driver produces correlation at a lag determined by the driver's own delays rather than by any interaction between the two series. A correlation-ranked shortlist therefore tends to contain redundant sources that carry the same information and sources whose apparent relevance arises only from a common driver. Such sources may still forecast well within the training period, but their relevance can disappear when the driver's behaviour changes.

1.6 Motivation for Causal Discovery and Graph-Guided Forecasting

Causal discovery for time series replaces "is X correlated with Y at lag τ?" with "does $X_{t-\tau}$ still carry information about Y_t after conditioning on the other relevant past values?" [32, 33]. The PCMCI algorithm [34] implements this idea at scale. It first screens each target's candidate parents with a fast condition-selection stage and then tests each surviving candidate with a momentary conditional-independence (MCI) test that conditions on the parents of both the target and the source. PCMCI has been used to study Arctic–midlatitude interactions [35], monsoon teleconnections [36, 37] and climate-model evaluation [38], and it is the reference method in reviews of causal inference in the Earth sciences [39, 40].

A conditional-independence graph is not a proof of causation. It depends on assumptions such as causal sufficiency, stationarity and the adequacy of the test [33]. Its value as a *forecasting prior* is also not guaranteed: a graph can be statistically credible and still not help a model whose bottleneck lies elsewhere. This thesis therefore separates two questions that are often merged: whether the graph is credible, which is tested by independent validation checks, and whether it is useful, which is tested by a controlled forecasting comparison.

1.7 Problem Statement

The research problem addressed by this thesis is:

> *Can a data-driven, independently validated graph of lagged climate dependencies, frozen before the test period, improve the selection of source regions for OLR forecasting during the BSISO season, relative to a model without such a prior and to random and lagged-correlation priors of matched size?*

Answering this question requires a reproducible pipeline that builds the graph only from development years, validates it with checks that can fail, freezes it, and then evaluates it on held-out years under matched forecasting conditions.

1.8 Aim and Objectives

The aim of this thesis is to design, implement and evaluate a *build–validate–freeze–use* protocol for causal graphs in forecasting, applied to BSISO-season OLR, and to set out how its findings shape an India-focused regional forecasting extension. The objectives are:

O1. Data representation. Construct a 300-node daily representation (50 regions, six variables, 1979–2022) with a strict development/test separation and an activity-based BSISO season derived from the project's own data.

O2. Graph discovery. Apply PCMCI to all 50 OLR targets over a full candidate pool of 300 nodes and lags of 1–60 days, with false-discovery-rate control, using an implementation fast enough for this scale and verified against a reference implementation.

O3. Graph validation. Test the graph with four independent checks in a fixed order: surrogate decoys, physical lag masking, data-driven top-k selection, and persistence across spatial resolution.

O4. Graph use. Convert the validated graph into a region-level prior for a graph-guided Mixture-of-Experts (MoE) source splitter, implement the restrict, initialise and regularise modes of use, and define no-prior, random-prior and correlation-prior controls.

O5. Forecast evaluation. Evaluate the effect of the prior on held-out 2013–2022 BSISO-season days at 7-day and 1-day leads, measuring both forecast skill and the discrimination of the splitter's source scores.

O6. Temporal stability. Measure whether the causal structure changes over time in step with known climate regimes, as a candidate signal for adaptive experts.

O7. Extension design. Translate the Phase 1 findings into the design of an India-focused, hourly forecasting system with causal gating and drift-triggered expert adaptation.

1.9 Research Questions

The objectives are organised around five research questions. Table 1.1 lists, for each, the method that addresses it and the evidence that would count as a positive, neutral or negative answer. Defining these outcomes before reporting results protects against reinterpreting a weak result as a success.

RQ1 – Causal discovery

Can PCMCI identify meaningful conditional lagged dependencies among the six climate variables across the 50 regions?

RQ2 – Graph credibility

How robust are the discovered relationships under surrogate-decoy testing, physically justified lag checks, top-k selection and spatial-resolution comparison?

RQ3 – Predictive usefulness

Does a validated causal graph improve seven-day OLR forecasting when used as a prior for the MoE splitter?

RQ4 – Value of causal structure

Does the causal prior outperform a no-prior model, a comparable random prior and a lagged-correlation prior under matched evaluation conditions?

RQ5 – Generalisation and adaptation

How can the Phase 1 findings inform an India-focused forecasting system that responds to temporal distribution shift and changing regional climate relationships?

Table 1.1: Research questions, methods and the evidence that determines their answer.

RQ	Method	Positive outcome	Neutral or negative outcome
RQ1	PCMCI with ParCorr and BH-FDR (Ch. 4)	A sparse set of significant links whose fields, lags and distances are physically interpretable	Very dense graph (FDR failing), or links dominated by implausible fields or lags
RQ2	Checks B1–B4 (Ch. 5)	Decoy rate below the pre-set warning line; few speed violations; links persist at doubled resolution	Decoys selected at rates comparable to real nodes; many impossible speeds; low persistence
RQ3	Graph prior in the splitter, 7-day lead (Ch. 6)	Higher held-out skill with the prior in most regions	No change (neutral) or lower skill (negative)
RQ4	Random and correlation priors of equal size	Causal prior better than both controls	Random or correlation prior equally good: the gain is due to pool size or association, not causal structure
RQ5	Design synthesis and drift analysis (Ch. 8)	Clear transfer of validated components; a drift signal aligned with known regimes	Components that do not transfer; a drift signal dominated by sampling noise

1.10 Testable Hypotheses

The research questions are made operational by the following hypotheses. Each is stated so that it can be rejected by the experiments described in Chapter 6.

Hypothesis 1 (Sparsity and credibility)**.** After FDR control, PCMCI selects a graph in which phase-randomised decoys are selected at a rate below 30% and in which fewer than 5% of links require a propagation speed above the per-variable limit.

Hypothesis 2 (Resolution persistence)**.** For the majority of targets with outside drivers, at least half of the validated outside links reappear, with the same field and a lag within ± 2 days, when the regions are split to double the spatial resolution.

Hypothesis 3 (Predictive usefulness)**.** Restricting the splitter's candidate pool to the validated graph increases held-out forecast skill relative to the no-prior model at the lead for which the forecast is designed (7 days).

Hypothesis 4 (Causal specificity)**.** Any skill gain from the causal prior exceeds the gain from a random prior and from a lagged-correlation prior of the same size per target.

Hypothesis 5 (Regime sensitivity)**.** The distance between causal parent sets in consecutive seasonal windows is larger when the ENSO phase changes between the windows than when it does not.

1.11 Scope and Technical Boundaries

- **Target and season.** Phase 1 forecasts the regional mean OLR anomaly of each of 50 regions. Only target days inside the BSISO active season are scored; source history before the season is retained.
- **Spatial domain.** The data lie on a 2.5° grid covering 30°S–30°N at all longitudes. The 50 regions tile this tropical belt; the Indo-Pacific sector, where the BSISO is active, is the physical focus.
- **Statistical model of dependence.** Graph discovery uses linear partial correlation. Nonlinear dependence and contemporaneous (lag-0) links are outside the scope of Phase 1.
- **Causal claims.** All links are statistical statements of conditional dependence at the scale of region means. They are not claims of mechanism.
- **Forecast model.** The forecaster is deliberately small, so that the effect of source selection is not hidden by model capacity. Building a competitive operational OLR forecast is not an aim of Phase 1.
- **Phase 2.** The India-focused extension is specified as a design with its data, components and evaluation protocol. Its experiments are future work, and global-scale forecasting is discussed only as a long-term direction.

1.12 Contributions

The contributions of this thesis, and the evidence that supports each, are:

1. **A build–validate–freeze–use protocol** for using causal graphs in forecasting, in which the graph is built and validated only on development years and frozen before any test year is used (Chapters 4–5).
2. **A fast, verified PCMCI implementation.** A vectorised PCMCI with partial-correlation tests and Benjamini–Hochberg FDR that agrees link-for-link with Tigramite on test cases and processes 18,000 candidates per target for 50 targets in about 23 minutes (Chapter 6).
3. **A validated lagged graph for BSISO-season OLR.** 536 links, of which none involve a decoy, 3 violate the propagation-speed limits, and 62% of outside links on average persist at doubled resolution (Chapter 7).
4. **A reproduced BSISO index and activity-based season** computed from the project's own fields, with leading-mode variances of 11.2% and 7.7%, close to the published 10% and 7% (Chapter 3).
5. **A first controlled assessment of the graph as a splitter prior.** The restrict prior makes full-pool splitter scores discriminative and improves 1-day skill over the no-prior

model in 32 of 50 regions, with no measurable effect at 7 days (Chapter 7).

6. **A negative result on causal-structure drift.** Seasonal changes in parent sets do not track ENSO phase changes at one- and three-season windows (Chapter 7).
7. **A design for the India-focused extension** with explicit transfer of the validated components (Chapter 8).

Contributions 1–4 and 6 are fully supported by the experiments reported here. Contribution 5 is an established comparison against the no-prior model; whether the gain is specific to the causal graph is the subject of the matched-control experiments defined in Section 6.8.

1.13 Overview of the Complete System

Figure 1.1 shows the two phases of the research programme. Phase 1 builds the climate graph (Stage A), validates it (Stage B), uses it as a source-selection prior (Stage C), and measures its stability over time (Stage D). Phase 2 carries the components that pass these tests into an India-focused hourly forecasting system with causal gating and adaptive experts.

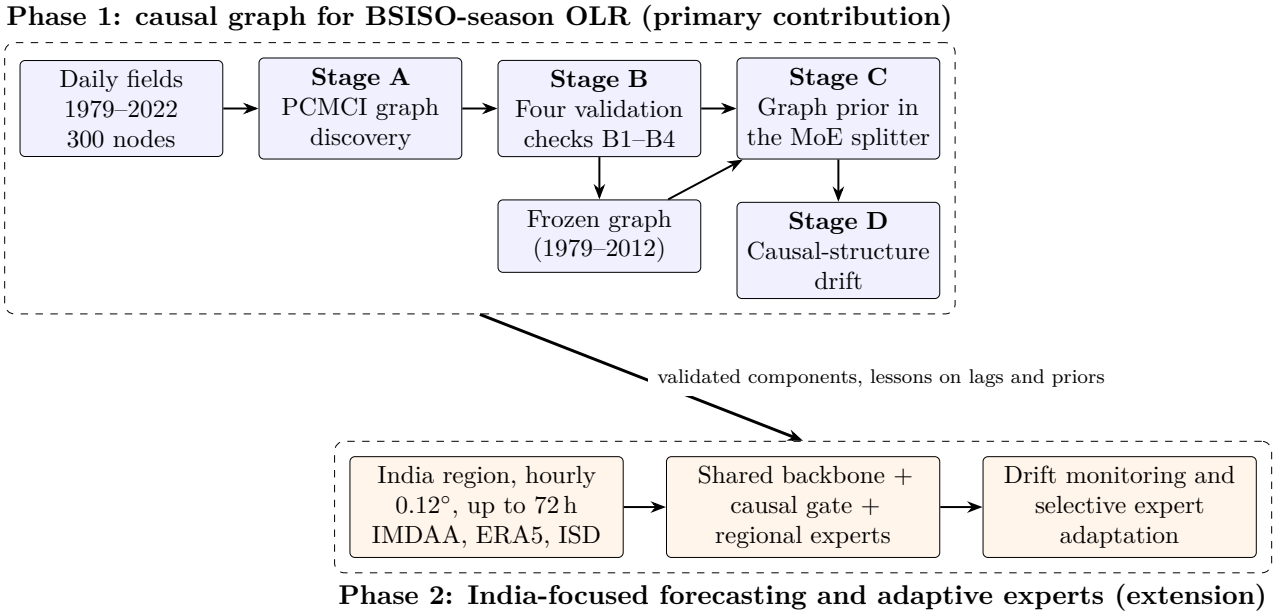


Figure 1.1: Two-phase research roadmap. Phase 1 builds, validates, freezes and uses a causal graph for BSISO-season OLR forecasting and measures its stability. Phase 2 transfers the components that pass these tests to an India-focused hourly forecasting system. Global-scale forecasting is treated as future work.

1.14 Organization of the Thesis

- **Chapter 2** reviews tropical convection and the BSISO, causal discovery for time series, multiple testing and surrogate methods, Mixture-of-Experts models, machine-learning weather forecasting and drift detection, and identifies the research gaps.

- **Chapter 3** formalises the forecasting task and describes the data, regional partition, BSISO season, preprocessing and leakage controls.
- **Chapter 4** presents the PCMCI methodology, its statistical tests, FDR correction and computational design.
- **Chapter 5** develops the four validation checks and the graph-guided MoE, including the three modes of using the graph and the control priors.
- **Chapter 6** describes the implementation, the experimental grid, the metrics and the statistical analysis.
- **Chapter 7** reports and discusses the results, organised by research question, together with threats to validity.
- **Chapter 8** presents the India-focused Phase 2 extension.
- **Chapter 9** summarises the findings, assesses each objective and outlines future work.
- **Appendices A–C** give the notation, derivations, configuration tables, experiment registry, reproducibility instructions and decision log.

Chapter 2

Literature Review and Technical Background

2.1 Introduction and Review Scope

This thesis lies at the intersection of four bodies of work that are rarely combined: the dynamics of tropical intraseasonal convection, causal discovery for multivariate time series, conditional-computation models such as the Mixture of Experts, and data-driven weather forecasting under distribution shift. Each field has its own standards of evidence. Atmospheric dynamics asks whether a relationship is physically possible; causal discovery asks whether it survives conditioning on alternative explanations; machine learning asks whether it improves out-of-sample prediction; and the drift literature asks whether it remains valid over time. The review below is organised around these questions rather than paper by paper. It closes with a comparative table (Table 2.1) and a statement of the research gaps that the thesis addresses.

2.2 Tropical Convection and OLR

Deep convection in the tropics is organised on scales from individual cumulonimbus to planetary-scale envelopes. Satellite OLR has been the principal large-scale observable of this organisation since the interpolated NOAA record became available [23]. The physical basis of OLR as a convection proxy is the dependence of emitted infrared flux on cloud-top temperature: the colder the top, the deeper the cloud. Arkin and Meisner [24] established the empirical link between cold-cloud fraction and large-scale convective rainfall, and this link underlies the use of OLR in intraseasonal indices [11, 22].

The convection–moisture relationship gives a second physical constraint relevant to this thesis. Bretherton *et al.* [41] showed that tropical oceanic precipitation rises sharply, approximately exponentially, with column water vapour once it exceeds a threshold. Precipitable water (PW) is therefore both a precursor and a consequence of convection, and its relation with OLR is expected to be strong at short lags within a region. The 850 hPa zonal wind (u850) captures low-level moisture convergence and the westerly flow beneath active monsoon convection; the

200 hPa zonal wind (u200) captures upper-level outflow and the large-scale baroclinic structure of the convective envelope; the 850 hPa geopotential height (h850) captures the low-level pressure and wave response; and SST supplies the slowly varying surface boundary condition. The six Phase 1 variables were chosen because each has a documented role in the life cycle of intraseasonal convection.

2.3 BSISO Dynamics and Propagation

Early observational studies identified northward-moving cloud bands over India with periods of 30–50 days [9] and 40–50-day OLR variability in northern summer [18]. Wang and Xie [10] provided a theoretical model in which an equatorial Kelvin–Rossby wave packet, modified by the monsoon mean flow, produces the observed north-west to south-east tilted band and its northward movement. Jiang *et al.* [17] attributed the northward propagation to vertical wind shear and moisture–convection feedback. Goswami and Ajaya Mohan [20] and Annamalai and Slingo [19] linked these oscillations to the active and break phases of the Indian monsoon.

For monitoring and prediction, Lee *et al.* [11] defined two indices from a multivariate EOF of OLR and u850 anomalies over 10°S–40°N, 40°–160°E in May–October. BSISO1, given by the first two principal components, describes the canonical 30–60-day northward and eastward propagating mode; BSISO2, from the third and fourth components, describes the 10–30-day mode. The first two modes explain about 10% and 7% of the filtered variance in their analysis, and the index is constructed without a non-causal band-pass filter by removing the mean of the preceding 120 days, which makes it usable in real time. Kikuchi *et al.* [16] proposed a bimodal representation distinguishing the MJO-like winter mode from the BSISO summer mode. Sperber and Annamalai [42] documented that coupled models struggle to reproduce BSISO propagation, which motivates statistical and data-driven approaches.

Two facts from this literature shape the methodology. First, the BSISO is seasonal, so a graph estimated over all days of the year mixes summer and winter regimes. Second, its envelope propagates at a few metres per second while equatorial waves embedded in it travel faster, which sets the physical speed scales used in Chapter 5.

2.4 Climate Teleconnections and Delayed Dependencies

Teleconnections are dependencies between distant regions. Classical theory explains how a localised tropical heat source excites a Kelvin-wave response to its east and a Rossby-wave response to its west [25, 26], and how tropical SST anomalies force extratropical wave trains [2]. Wheeler and Kiladis [43] and Kiladis *et al.* [27] documented convectively coupled equatorial waves in OLR spectra, with Kelvin waves travelling eastward at roughly 15–20 $\mathrm{m\,s^{-1}}$when coupled to convection and faster when dry [27]. These results matter because they establish that legitimate dependencies between regions thousands of kilometres apart can act within one or two days, and that a geographic-distance cutoff would wrongly remove them.

Teleconnections also change over time. Krishna Kumar *et al.* [30] reported a weakening of the ENSO–Indian monsoon relationship in the late twentieth century. Any data-driven graph is therefore an estimate for a particular period, and its stability must be tested.

2.5 Correlation, Autocorrelation and Conditional Independence

Atmospheric statistics has long recognised that autocorrelation reduces the effective number of independent samples and inflates the apparent significance of correlations [28, 29, 44]. Runge *et al.* [31] made a sharper point for lagged relationships: the lagged cross-correlation between two autocorrelated series is not a reliable estimate of either the strength or the delay of their interaction, because it mixes the coupling with the memory of each series. They proposed measures based on graphical models that condition on the past of both series.

Conditional independence is the formal tool behind this idea. Two variables X and Y are conditionally independent given Z, written $X \perp\!\!\!\perp Y \mid Z$, if knowing X adds no information about Y once Z is known. Under the causal Markov and faithfulness assumptions, the conditional-independence relations among variables identify the edges of the underlying causal graph up to an equivalence class [32, 45]. For time series with only lagged links, time order fixes the edge direction, which removes much of that ambiguity [33]. Granger causality [46] is the time-series special case in which X "Granger-causes" Y if the past of X improves the prediction of Y beyond the past of Y and other variables; it coincides with conditional dependence on the full past under linear-Gaussian assumptions.

2.6 Causal Discovery in Multivariate Time Series

Methods for causal discovery in time series fall into four families [39, 40, 47].

Granger and regression approaches. Vector autoregression with significance tests, possibly with lasso regularisation [48], estimates all lagged coefficients jointly. With 300 nodes and 60 lags there are 18,000 regressors per target and the regression is ill-conditioned; regularisation helps but makes the significance of individual links hard to interpret. Neural Granger causality [49] extends the idea to nonlinear models with sparsity penalties.

Constraint-based methods. The PC algorithm [32] removes edges by testing conditional independence with conditioning sets of increasing size. In autocorrelated time series a naive application has low power, because the strong autocorrelation of the source makes conditioning sets large and the test statistic noisy [34]. PCMCI addresses this; PCMCI$^+$ [50] adds contemporaneous links, and LPCMCI [51] relaxes causal sufficiency to allow latent confounders.

Score-based and continuous-optimisation methods. NOTEARS [52] recasts DAG learning as continuous optimisation with an acyclicity constraint, and DYNOTEARS [53] extends it to time series. These methods scale with gradient-based solvers but do not provide link-level significance and are sensitive to variable scaling.

Learned and amortised methods. Neural relational inference [54] and amortised causal discovery [55] train a network to infer interaction graphs from many examples of a system. They can be fast at inference time but need training data with known graphs, which do not exist for the real atmosphere.

Nonlinear state-space methods. Convergent cross mapping [56] detects coupling in deterministic dynamical systems, and transfer entropy [57] is an information-theoretic analogue of Granger causality. Both are attractive for nonlinear systems but are expensive and data-hungry at the scale considered here.

For this thesis the decisive requirements are link-level significance with error control, explicit lags, scalability to thousands of candidates per target, and an established record in climate applications. PCMCI is the method that satisfies all four.

2.7 PCMCI and Related Methods in Climate Science

Runge *et al.* [34] introduced PCMCI and showed in numerical experiments that it achieves higher detection power than PC and lasso-Granger approaches at a controlled false-positive rate, particularly for strongly autocorrelated, high-dimensional systems. Its key idea is to separate a condition-selection step (PC1), which finds a small superset of each variable's parents, from the final momentary conditional-independence (MCI) test, which conditions on the parents of both the target and the lagged source. Conditioning on the source's parents removes the influence of the source's own autocorrelation, which is the main cause of false positives and of poorly calibrated tests in climate data. The method is described in detail in Chapter 4.

Applications in climate science have used causal-effect networks built with PCMCI to study drivers of the winter stratospheric polar vortex [35], the tropical and mid-latitude drivers of Indian summer monsoon rainfall [36], and the time-scale dependence of tropical–extratropical interactions in boreal summer [37]. Nowack *et al.* [38] used causal networks among sea-level pressure modes as fingerprints to evaluate climate models. Saggioro *et al.* [58] extended the approach to regime-dependent graphs. Tibau *et al.* [59] built a stochastic climate model with known teleconnections to benchmark such methods. Earlier, Ebert-Uphoff and Deng [60] applied constraint-based graphical models to atmospheric indices.

Two features distinguish these studies from the present work. First, most target a small number of indices or a handful of dimension-reduced modes, typically tens of variables, whereas this thesis tests a full pool of 300 regional nodes at 60 lags for every target. Second, they use the graph as an end in itself, for understanding or for model evaluation, and do not test whether

it improves a forecasting model under matched controls.

2.8 Multiple Testing, Surrogate Tests and Graph Robustness

Multiple testing. When thousands of hypotheses are tested, a per-test threshold of 0.05 guarantees many false positives. Benjamini and Hochberg [61] introduced the false discovery rate (FDR), the expected fraction of false positives among rejections, and a step-up procedure controlling it under independence. Benjamini and Yekutieli [62] showed that the procedure remains valid under positive regression dependence and provided a conservative version for arbitrary dependence. In climatology, Ventura *et al.* [63] and Wilks [64] argued that FDR control should replace naive per-gridpoint testing, which routinely overstates field significance.

Surrogate data. Theiler *et al.* [65] introduced surrogate data to test a null hypothesis by generating series that share chosen properties of the data but are otherwise random. Phase-randomised surrogates preserve the power spectrum, and hence the autocorrelation function, of a series while destroying any phase relation with other series. Schreiber and Schmitz [66] and Lancaster *et al.* [67] reviewed refinements such as amplitude-adjusted and iterated surrogates and emphasised that the validity of any surrogate test depends on whether the preserved properties match the null hypothesis of interest. Block bootstrap methods [68, 69] provide an alternative for autocorrelated data.

Robustness of discovered graphs. Causal-discovery studies typically report the sensitivity of their graphs to significance levels, lag ranges or test choice [39]. Using surrogate series as *decoy nodes* inside the candidate pool, so that the discovery algorithm itself must reject them alongside the real candidates, is a less common check. It provides a direct empirical estimate of how often a series with realistic autocorrelation but no real coupling is selected under the full pipeline, including FDR correction. Spatial-scale robustness, testing whether links persist when the same field is aggregated differently, is rarely reported, although aggregation is known to change the dependence structure of climate fields [29].

2.9 Physical Constraints and Scientific Priors in Causal Discovery

Physical knowledge can enter causal discovery as constraints on admissible links, as priors on link probability, or as a post-hoc plausibility check. Di Capua *et al.* [36] used a "theory-guided" approach in which candidate variables were chosen from prior knowledge of monsoon dynamics. Karpatne *et al.* [70] and Reichstein *et al.* [71] argued more broadly for combining physical knowledge and data-driven learning in Earth system science. A distinctive question for lagged

graphs is whether the implied propagation speed of a link, the distance between source and target divided by the lag, is achievable by any physical process carried by the source variable. This turns wave-speed knowledge from equatorial dynamics [25, 27, 43] into a link-level test. The literature reviewed did not use it as a systematic check of a discovered graph; Chapter 5 develops it as a diagnostic.

2.10 Mixture-of-Experts Architectures and Gating Networks

The Mixture of Experts was introduced by Jacobs *et al.* [72] as a set of expert networks combined by a gating network that assigns input-dependent weights, so that each expert specialises in a region of input space. Jordan and Jacobs [73] extended it to hierarchies trained with expectation–maximisation. In deep learning, sparsely gated MoE layers [74], GShard [75] and Switch Transformers [76] route each token to one or a few experts, which decouples parameter count from computation; these works also documented load imbalance and the need for auxiliary balancing losses. For graphs, Graph MoE [77] routes nodes to experts with different aggregation behaviour.

In these architectures the gate is learned purely from data and is judged only by downstream loss. In the present work, the role of the gate is played by a *splitter* that decides which source regions feed each target's expert. Such a splitter can be given external information about which sources are plausible, and the question of whether a causal graph is a good form of that information is the subject of Phase 1.

2.11 Invariant Learning and Graph Rationales

Invariant causal prediction [78] rests on the observation that the conditional distribution of a target given its causal parents is invariant across environments, whereas relations with non-causal variables can change. Invariant risk minimisation [79] and risk extrapolation [80] translate this into training objectives that penalise variation of the risk across environments. Wu *et al.* [81] adapted the idea to graph neural networks as Discovering Invariant Rationales (DIR): a rationale generator splits each input graph into a causal and a non-causal subgraph, the non-causal parts are exchanged between examples to create interventional environments, and the model is trained to minimise the mean and the variance of the risk across those environments. The splitter used in this thesis adapts DIR from graph classification to spatiotemporal regression (Section 5.8.3).

2.12 Graph-Guided Learning and Structured Predictor Selection

Graph neural networks [82, 83] propagate information along a given graph and are widely used for spatiotemporal forecasting, for example in traffic [84]. When no graph is given, models such as MTGNN [85] learn an adjacency matrix jointly with the forecaster. Learned adjacencies optimise prediction but are not tested for statistical credibility, and fixed geographic adjacencies are credible but may omit remote influences. Structured sparsity methods such as the lasso [48] select predictors by penalised fitting. A causal graph offers a third option: a sparse, explicitly tested, lag-resolved candidate structure that is fixed before training. Whether it helps is an empirical question, and the answer may depend on how the graph enters the model: as a hard restriction, as an initialisation, or as a soft penalty.

2.13 Machine Learning for Weather Forecasting

Data-driven global weather models have advanced rapidly. WeatherBench [86] and WeatherBench 2 [87] provide common evaluation protocols. Keisler [12] showed that a graph neural network can produce competitive multi-day global forecasts; FourCastNet [88], Pangu-Weather [14] and GraphCast [13] matched or exceeded deterministic operational NWP on many variables and lead times; GenCast [15] extended this to probabilistic ensemble forecasting; and ECMWF has deployed a data-driven system, AIFS [89]. Ben Bouallègue *et al.* [90] assessed such models in an operational-like setting, and Schultz *et al.* [91] discussed the conditions under which deep learning could replace NWP. Oskarsson *et al.* [92] adapted graph-based neural forecasting to limited-area domains, which is directly relevant to the regional focus of Phase 2.

These models are trained on reanalyses such as ERA5 [93] and are evaluated mainly at medium range. Their inputs are entire global states, and the dependence between regions is learned implicitly. They neither expose a tested statement of which remote regions drive a given target nor adapt to changes in such dependence. The present work does not compete with them on scale; it studies a complementary question about explicit, validated source selection.

2.14 Distribution Shift, Regime Detection and Online Adaptation

Machine-learning models assume that training and deployment data share a distribution; when the joint distribution of inputs and targets changes, performance degrades [94]. In the streaming setting this is called concept drift [95, 96]. Detectors such as CUSUM [97] and ADWIN [98] monitor an error or statistic stream and signal when its distribution changes. In climatology, Rodionov's sequential t-test analysis of regime shifts (STARS) [99] detects shifts in the mean of a time series as they occur, with prewhitening to handle autocorrelation [100].

Adapting to drift raises the problem of catastrophic forgetting [101]. Progressive networks [102] and Expert Gate [103] avoid forgetting by adding new modules or experts rather than overwriting old ones, and route inputs to the most relevant expert. For climate forecasting, regimes such as ENSO phases recur, which suggests archiving experts and reactivating them when a familiar regime returns. A detector based on forecast error reacts only after skill has degraded; a detector based on the causal structure itself could in principle react earlier, but only if the estimated structure is stable enough between windows that real changes stand out from sampling noise. That condition is tested in Stage D.

2.15 Related Datasets and Benchmarks

The interpolated OLR record [23] and atmospheric reanalyses are the standard sources for BSISO research. ERA5 [93] provides hourly global fields at about 31 km from 1940 onward. For India, the IMDAA regional reanalysis [104] provides hourly fields at about 12 km for the satellite era. The NOAA Integrated Surface Database [105] provides hourly station observations worldwide, including Indian stations. WeatherBench 2 [87] is the most widely used benchmark for global data-driven models. These datasets are the basis of the Phase 2 design (Chapter 8).

2.16 Comparative Summary of Related Work

Table 2.1 compares representative works along the dimensions that matter for this thesis: whether the method discovers lagged structure, whether that structure is validated independently, whether it is used in a forecasting model, and whether the method addresses temporal change.

Table 2.1: Comparison of related work. Columns: **Lag** = discovers explicit lagged dependencies; **Val.** = validates the discovered structure with an independent check; **Fcst.** = uses the structure in a forecasting model; **Drift** = addresses temporal change. ✓ = yes; ∘ = partly; – = no.

Work	Approach	Lag	Val.	Fcst.	Drift	Relevance / gap
Granger [46]	Predictive causality via regression	✓	–	∘	–	Foundational; no control for common drivers beyond included variables
Runge et al. [31]	Graph-based coupling measures	✓	–	–	–	Shows lagged correlation is ambiguous in climate data

Work	**Approach**	**Lag**	**Val.**	**Fcst.**	**Drift**	**Relevance / gap**
Runge et al. [34]	PCMCI	✓	○	–	–	Core method; validated on synthetic systems
Runge [50]	PCMCI$^+$	✓	○	–	–	Adds lag-0 links; not needed for forecasting
Gerhardus & Runge [51]	LPCMCI	✓	○	–	–	Handles latent confounders; higher cost
Kretschmer et al. [35]	Causal effect networks	✓	○	○	–	Climate drivers from a few indices
Di Capua et al. [36]	Theory-guided CEN, monsoon	✓	○	–	–	Monsoon drivers; small variable set
Saggioro et al. [58]	Regime-dependent PCMCI	✓	○	–	✓	Regime-specific graphs; known regimes
Nowack et al. [38]	Causal fingerprints	✓	○	–	–	Uses graphs to evaluate models
Tank et al. [49]	Neural Granger	✓	–	○	–	Nonlinear; no link-level error control
Pamfil et al. [53]	DYNOTEARS	✓	–	–	–	Scalable; no significance testing
Kipf et al. [54]	NRI	○	–	✓	–	Learned interactions; needs graph-labelled data
Wu et al. [81]	DIR-GNN rationales	–	○	–	–	Invariance-trained selection; adapted here
Shazeer et al. [74]	Sparse MoE	–	–	✓	–	Data-learned gating only

Work	Approach	Lag	Val.	Fcst.	Drift	Relevance / gap
Wang et al. [77]	Graph MoE	–	–	✓	–	Experts on graphs; no causal prior
Wu et al. [85]	MTGNN learned graph	○	–	✓	–	Learned adjacency not tested for credibility
Lam et al. [13]	GraphCast	–	–	✓	–	Global ML NWP; implicit dependence
Price et al. [15]	GenCast	–	–	✓	–	Probabilistic ML NWP
Oskarsson et al. [92]	Limited-area graph NWP	–	–	✓	–	Regional ML forecasting
Rodionov [99]	STARS	–	–	–	✓	Regime-shift test used in Stage D
Bifet & Gavaldà [98]	ADWIN	–	–	○	✓	Error-stream drift detection
Aljundi et al. [103]	Expert Gate	–	–	✓	✓	Adding experts without forgetting
This thesis	PCMCI + four checks + graph prior in MoE + structural drift	✓	✓	✓	✓	Tests credibility and usefulness separately under matched controls

2.17 Research Gaps and Positioning

The comparison identifies four gaps that this thesis addresses.

G1. Scale of the candidate pool. Climate applications of PCMCI typically use tens of pre-selected indices or modes. Running PCMCI over a full pool of 300 regional nodes at 60 lags for every target, without pre-selecting variables by hypothesis, has practical cost and statistical-power implications that have not been documented for intraseasonal OLR.

G2. Independent validation of the discovered graph. Studies report sensitivity to parameters, but a fixed sequence of checks that can each fail, combining decoys inside the discovery pool, a propagation-speed test, data-driven sparsification and persistence across

spatial resolution, is not standard practice.

G3. Credibility versus usefulness. Causal graphs in climate science are used for understanding or model evaluation; MoE gates and learned adjacencies in machine learning are judged only by loss. No study among those reviewed tests whether a validated causal graph improves a forecaster's source selection against matched random and correlation priors.

G4. Causal structure as a drift signal. Drift detectors watch error streams, and regime-dependent causal discovery assumes regimes are known. Whether sliding-window changes in an estimated causal structure align with known climate regimes, and could therefore trigger expert adaptation, is an open empirical question.

The thesis is positioned as an applied methodological study. Its novelty lies not in a new discovery algorithm but in the protocol connecting discovery, validation and forecasting use, in the scale at which the protocol is applied, and in reporting the outcome honestly whichever way it falls.

2.18 Chapter Summary

This chapter reviewed the physical basis of OLR and the BSISO, the reasons why correlation is an unreliable guide to lagged dependence in climate data, the main families of causal-discovery methods and the reasons for choosing PCMCI, the statistical tools for multiple testing and surrogate validation, the Mixture-of-Experts and invariant-learning literature on which the forecasting model builds, recent machine-learning weather models, and drift detection. It identified four gaps. The next chapter formalises the forecasting problem and the data.

Chapter 3

Problem Formulation, Dataset, and Preprocessing

3.1 Formal Definition of the Forecasting Task

Let $t \in \{1, \ldots, T\}$ index days and let Y_t^r denote the standardised OLR anomaly averaged over region $r \in \{0, \ldots, R-1\}$ on day t, with $R = 50$. For a lead time L (in days), the forecasting task is to estimate Y_{t+L}^r from information available at the issue time t:

$$\hat{Y}_{t+L}^r = f_r\Big(\{X_{t-\ell}^i : i \in \mathcal{N}_r,\ \ell \in \Lambda\}\Big), \tag{3.1}$$

where X_t^i is the standardised anomaly of node i (Section 3.2), $\mathcal{N}_r$ is the set of nodes available to the forecaster of region r, and Λ is the set of input lags. The forecast is *direct*: a separate model maps the state at time t to the value at $t + L$, with no autoregressive roll-out. The primary lead is $L = 7$ days, which is the horizon of the research question; $L = 1$ day is used as a secondary horizon. The question studied in this thesis is how $\mathcal{N}_r$, the set of source regions available to each forecaster, should be chosen, and whether a validated causal graph is a good basis for that choice.

A forecast is *scored* only if its target day $t + L$ lies in the BSISO active season $\mathcal{S}$ (Section 3.6) and in the test period $\mathcal{T}_{\text{test}}$ = 2013–2022. The inputs may come from any earlier day.

3.2 Mathematical Representation of the 300-Node System

3.2.1 Variables

The six variables are listed in Table 3.1 with their physical interpretation and role in tropical convection. All are provided as daily anomalies on the native grid.

Table 3.1: The six Phase 1 variables, their typical physical units before anomaly processing, and their role in tropical intraseasonal convection. The data used here are standardised anomalies and are therefore dimensionless.

Symbol	Variable	Units	Physical role	Typical time scale
SST	Sea surface temperature	K	Surface boundary condition; modulates surface heat and moisture fluxes that sustain convection; responds to cloud shading and wind-driven evaporation	Weeks to months (slow)
h850	Geopotential height at 850 hPa	m (geopotential metres)	Low-level pressure field; marks the Rossby- and Kelvin-wave response to convective heating and monsoon troughs	Days to weeks
u200	Zonal wind at 200 hPa	$\mathrm{m\,s^{-1}}$	Upper-tropospheric outflow above convection; large-scale overturning circulation	Days to weeks
PW	Precipitable water	$\mathrm{kg\,m^{-2}}$ (mm)	Column moisture; strong precursor of deep convection above a critical value [41]	Days
u850	Zonal wind at 850 hPa	$\mathrm{m\,s^{-1}}$	Low-level monsoon westerlies and moisture convergence; one of the two BSISO index fields [11]	Days to weeks
OLR	Outgoing longwave radiation	$\mathrm{W\,m^{-2}}$	Convection proxy: low OLR indicates cold, high cloud tops [24]; the forecast target	Days (fast)

3.2.2 Nodes and the state matrix

A node $i = (r, v)$ is a variable $v \in \mathcal{V}$ averaged over a region r. With $R = 50$ regions and $|\mathcal{V}| = 6$ variables there are

$$N = R\,|\mathcal{V}| = 50 \times 6 = 300 \tag{3.2}$$

nodes. Let Ω_r be the set of grid cells assigned to region r and $\Omega_r^{\text{ocean}} \subseteq \Omega_r$ its ocean cells. The raw regional series is

$$Z_t^{(r,v)} = \frac{1}{|\Omega_r^v|} \sum_{c \in \Omega_r^v} A_{c,t}^v, \qquad \Omega_r^v = \begin{cases} \Omega_r^{\text{ocean}} & v = \text{SST}, \\ \Omega_r & \text{otherwise}, \end{cases} \tag{3.3}$$

where $A_{c,t}^v$ is the gridded anomaly of variable v in cell c on day t. SST is averaged only over ocean cells because it is undefined over land. Each node is then re-standardised with statistics computed only on the development period:

$$X_t^i = \frac{Z_t^i - \mu_i}{\sigma_i}, \qquad \mu_i = \frac{1}{|\mathcal{T}_{\text{dev}}|} \sum_{t \in \mathcal{T}_{\text{dev}}} Z_t^i, \quad \sigma_i^2 = \frac{1}{|\mathcal{T}_{\text{dev}}|} \sum_{t \in \mathcal{T}_{\text{dev}}} (Z_t^i - \mu_i)^2. \tag{3.4}$$

The system state on day t is $\mathbf{X}_t = (X_t^1, \ldots, X_t^N)^\top \in \mathbb{R}^N$, and the full data form the matrix $\mathbf{X} \in \mathbb{R}^{T \times N}$ with $T = 16{,}071$ days. Figure 3.1 shows the structure.

Region means in Eq. (3.3) are unweighted averages of the member cells. On a 2.5° grid within 30° of the equator, the cell area varies by at most a factor of $\cos 30° \approx 0.87$, so area

weighting would change the means only slightly; it is recorded as an implementation choice in Appendix C.4.

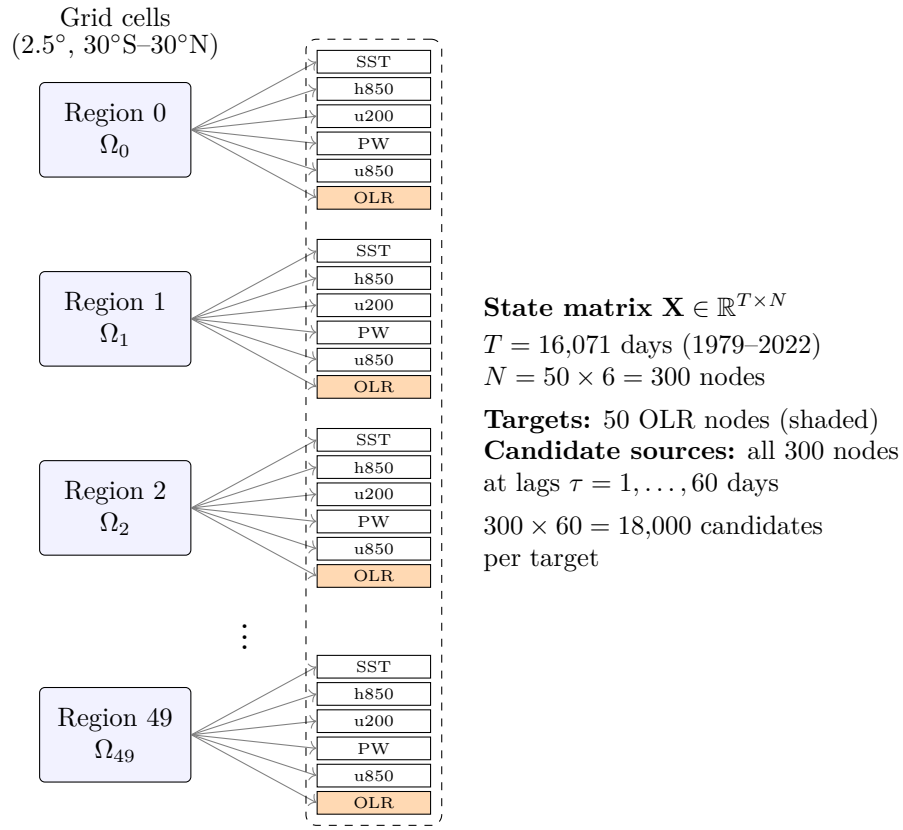


Figure 3.1: Schematic of the 300-node representation. Each of the 50 regions contributes six regional-mean variables. The 50 OLR nodes (shaded) are the forecast targets; all 300 nodes are candidate sources at lags of 1–60 days. The diagram is schematic and does not show the real region geometry.

3.2.3 Candidate source–target–lag combinations

For one OLR target node j, every node $i \in \{1, \ldots, 300\}$, including j itself, is a candidate source at every lag $\tau \in \{1, \ldots, 60\}$, which gives

$$|\mathcal{C}_j| = N\,\tau_{\max} = 300 \times 60 = 18{,}000 \tag{3.5}$$

lagged candidates per target and $50 \times 18{,}000 = 900{,}000$ for the 50 OLR targets. If instead every ordered pair of *distinct* nodes were considered with all 300 nodes as targets, the count would be

$$N(N-1)\,\tau_{\max} = 300 \times 299 \times 60 = 5{,}382{,}000. \tag{3.6}$$

These numbers count candidate hypotheses before screening, not statistical tests performed: PCMCI's screening stage (Chapter 4) removes most candidates after few tests, and the final MCI test is applied only to the survivors. They nonetheless explain why FDR control and an efficient implementation are essential.

3.3 Regional Partition

The native grid has $25 \times 144 = 3{,}600$ cells. Individual cells are noisy and too numerous for a 60-lag analysis, and a single 2.5° cell has no physical significance. The cells are therefore grouped into 50 regions by k-means clustering [106] applied to the cells' OLR time series over the development period, so that cells whose convection varies together form a region. In practice 46 of the 50 regions are single contiguous areas, and each region contains on average 72 cells. A region adjacency graph is derived from whether any two cells of different regions touch, with longitude wrapping around the date line. The regions are indexed 0–49; this numbering is used in all result figures.

For the resolution check (Section 5.5), each region is split into two sub-regions by k-means with $k = 2$ applied to its own cells, giving 100 nested regions. Because some sub-regions contain no ocean cells, the 100-region representation has 596 rather than 600 nodes.

Clustering on OLR co-variability and then searching for OLR dependencies between the clusters introduces a mild circularity: regions are, by construction, internally coherent in OLR, and the boundaries between them fall where OLR co-variability drops. The effect is to make between-region OLR links harder rather than easier to detect, but it does mean that the regions are not independent of the target variable. The point is revisited in the threats to validity (Section 7.14). Geographic adjacency, used to define local candidate pools for the forecaster (Section 5.8.2), is a separate rule based only on geometry.

3.4 Temporal Coverage and Data Provenance

Table 3.2 summarises the data. The fields come from a BSISO monitoring archive in which each variable is stored as a daily anomaly on a common 2.5° grid covering 30°S–30°N and all longitudes, from 1 January 1979 to 31 December 2022. The record has 16,071 days with no missing days. Each field was anomaly-processed upstream: a daily climatology was removed and the anomalies were normalised separately at each grid cell. A land–sea mask identifies ocean cells for SST.

3.5 Target Construction and Forecast Lead Time

The target of region r at lead L is $Y^r_{t+L} = X^{(r,\mathrm{OLR})}_{t+L}$. Two leads are studied:

- $L = 7$ days, the primary horizon of the research question, at which OLR retains almost no persistence;
- $L = 1$ day, a secondary horizon at which persistence is very strong and the graph's short-lag links act on the same time scale as the forecast.

The input to the forecaster of region r is built from source regions $r' \in \mathcal{C}_r$. For each source region, the six variables are taken at the seven input lags $\Lambda = \{0, 1, 2, 3, 5, 7, 10\}$ days before

Table 3.2: Phase 1 data summary. Items marked [confirm] must be checked against the original archive documentation before submission.

Property	Value
Archive	Pre-processed BSISO monitoring fields (`rt_preprocessed_archive`); originating reanalysis and OLR products [confirm]
Variables	SST, h850, u200, PW, u850, OLR
Grid	2.5° × 2.5°; 25 latitudes (30°N to 30°S) × 144 longitudes (0°–357.5°E)
Period	1979-01-01 to 2022-12-31; 16,071 daily values; no missing days
Pre-processing upstream	Daily climatology removed and anomalies normalised per grid cell; climatology base period 1979–2014
Auxiliary data	Ocean mask (25 × 144, Boolean)
Licence	[confirm with archive provider]
Development / test split	1979–2012 / 2013–2022

the issue time, and an ocean-fraction feature is appended, giving a 49-dimensional vector

$$\mathbf{h}_t^{r'} = \left[X_{t-\ell}^{(r',v)} \;:\; v \in \mathcal{V},\ \ell \in \Lambda \right] \,\|\, \phi_{r'} \;\in \mathbb{R}^{49}, \tag{3.7}$$

where $\phi_{r'}$ is the ocean fraction of region r' and $\|$ denotes concatenation. The irregular lag set gives dense coverage of the most recent days and sparser coverage further back. The splitter additionally sees a window of the last 10 days of each region's state.

3.6 BSISO Season and Activity-Based Target Selection

3.6.1 Why an activity-based season

A fixed calendar window such as May–October treats a year with an early and vigorous BSISO in the same way as a year in which the oscillation is weak until July. Because the graph is meant to describe dependencies that operate when the BSISO is active, the season is defined by the oscillation's own amplitude, with a margin so that onset and decay are included.

3.6.2 Reproducing the BSISO index

The index follows the BSISO1 recipe of Lee *et al.* [11], adapted to the available grid:

1. OLR and u850 anomalies are taken over 10°S–30°N, 40°–160°E. Lee *et al.* use 10°S–40°N; the data stop at 30°N.
2. The mean of the preceding 120 days is subtracted at each grid point to isolate intraseasonal variability without a non-causal band-pass filter.
3. Each grid point is weighted by $\sqrt{\cos\varphi}$, where φ is latitude, and each field is divided by its domain-mean standard deviation so that OLR and u850 contribute equally.
4. A combined EOF [107, 108] is computed on May–October days of the development years only (1979–2012), so that test years never influence the patterns.

5. The first two principal components are normalised to unit variance over the fit days, and the BSISO amplitude is

$$A_t = \sqrt{\mathrm{PC1}_t^2 + \mathrm{PC2}_t^2}. \tag{3.8}$$

3.6.3 Season definition

For each year, the 31-day centred running mean $\bar{A}_t$ of the amplitude is computed inside a search window from 1 April to 30 November. The season starts at the first day of the first run of at least 20 consecutive days with $\bar{A}_t > 1$, ends at the last day of the last such run, and is padded by 21 days on each side:

$$\mathcal{S}_y = \left[\, t_y^{\mathrm{on}} - 21,\ t_y^{\mathrm{off}} + 21 \,\right]. \tag{3.9}$$

A year with no qualifying run falls back to 1 May–31 October. The resulting season averages 208 days per year, ranges from 82 to 286 days, and begins between April and July. The development years contain 7,153 season days. Figure 7.1 in Chapter 7 shows the amplitude climatology and the season of each year.

The 31-day running mean is centred and therefore uses up to 15 days after a given day. This is acceptable because the season mask decides only *which* target days are scored; it is never an input to a forecast. The consequence is that evaluation is conditioned on the BSISO having been active, which is the intended scope.

3.6.4 Target days versus source history

The season restricts target days only. A source value at lag τ is read from day $t-\tau$ regardless of whether that day is in the season. For a target on 1 June, a lag-45 source value comes from mid-April. Removing all non-season days before building the lags would delete exactly the precursor information that the analysis is meant to find, and would create artificial discontinuities at the season boundaries. Formally, the PCMCI sample for target node j is

$$\mathcal{D}_j = \Big\{ \left(X_t^j,\ \{X_{t-\tau}^i\}_{i,\tau}\right) \ :\ t \in \mathcal{S} \cap \mathcal{T}_{\mathrm{dev}},\ t - 2\tau_{\max} \geq 1 \Big\}, \tag{3.10}$$

where the requirement $t - 2\tau_{\max} \geq 1$ ensures that the lagged conditioning values of the MCI test (Section 4.5) lie inside the record.

3.7 Climatology, Anomalies, Standardisation and Units

3.7.1 Why the seasonal cycle must be removed

Almost every climate variable has a strong annual cycle. Two unrelated series that both peak in boreal summer are strongly correlated at every lag shorter than a few months, purely because of their shared calendar dependence. If the annual cycle were left in the data, PCMCI would find dense, spurious lagged links driven by the calendar, and the forecaster would learn the seasonal

cycle rather than intraseasonal dynamics. The standard remedy is to subtract a smoothed daily climatology,

$$A^v_{c,t} = V^v_{c,t} - \bar{V}^v_c\big(d(t)\big), \qquad \bar{V}^v_c(d) = \sum_{h=0}^{H} \Big[a_h \cos(2\pi h d/365.25) + b_h \sin(2\pi h d/365.25)\Big], \quad (3.11)$$

where $V^v_{c,t}$ is the raw field, $d(t)$ is the day of year and $\bar{V}$ is a smoothed climatology, here written as the common choice of the mean and first H annual harmonics. The anomalies are then standardised.

3.7.2 Implementation in this work

The fields arrive as anomalies with respect to an upstream daily climatology computed over 1979–2014 and normalised per grid cell. The exact smoothing rule and leap-day treatment of the upstream climatology are those of the archive [confirm]. This work adds two steps: (i) the regional means of Eq. (3.3) are re-standardised with development-period statistics only (Eq. (3.4)); and (ii) the BSISO index removes the preceding 120-day mean before the EOF analysis.

The upstream base period overlaps the test period by two years (2013 and 2014). A climatology contains, for each calendar day, an average over 36 years; two of those years being test years means that roughly $2/36 \approx 6\%$ of the information in the climatology comes from the test period. The effect on forecast scores is expected to be small, because the climatology is a smooth function of calendar day and all models compared in this thesis share it, but it is a genuine departure from a fully clean split and is listed as a limitation (Section 7.14). Refitting the climatology on 1979–2012 requires reprocessing the gridded archive and is included in future work.

3.8 Missing Data and Quality Control

The archive contains no missing days. Quality control at the regional level consists of (i) checking that every region has at least one ocean cell for SST, which holds at 50 regions but not for 4 of the 100 sub-regions, whose SST nodes are dropped; (ii) checking that each node has a non-zero development-period variance before standardisation; and (iii) confirming that the season mask produces a non-empty season every year. Because there are no gaps, no imputation is performed. A future multi-source dataset (Phase 2) will need an explicit gap and outlier policy.

3.9 Lag Construction and Temporal Alignment

All series share a common daily time axis. For a target day t, the candidate value of node i at lag τ is $X^i_{t-\tau}$. Lag 0 is excluded from graph discovery: a link must point from the past to

the present, so its direction follows from time order, and contemporaneous dependence, which cannot be oriented without further assumptions [50], is outside scope. For forecasting at lead L, the newest usable value of any source is at the issue time t, so a source that is a parent of Y_{t+L} at lag τ can be used directly only if $\tau \geq L$. This forecast-time availability condition is central to interpreting the Stage C results and is discussed in Section 5.11.

3.10 Development and Holdout Split

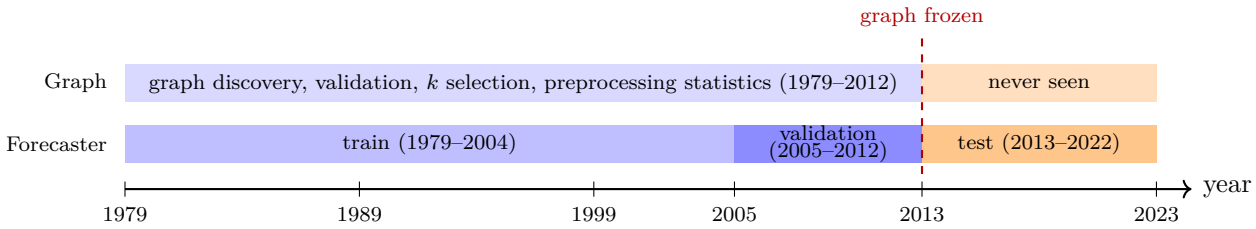


Figure 3.2: Phase 1 time split. Everything that shapes the graph or the forecaster, including the preprocessing statistics, the BSISO EOF patterns, PCMCI, the validation checks and the choice of k, uses 1979–2012 only. The graph is frozen before 2013–2022 is used, and test years are used only for scoring.

Figure 3.2 shows the split. The development period $\mathcal{T}_{\text{dev}}$ (1979–2012) is used for graph discovery and for training. Within it, the forecaster uses 1979–2004 for fitting and 2005–2012 for validation, and the top-k check fits on 1979–2004 and scores on 2005–2012 season days. The test period $\mathcal{T}_{\text{test}}$ (2013–2022) is held out: it provides 1,989 season target days for evaluation. A chronological split rather than random cross-validation is used because random folds of autocorrelated series place nearly identical days in training and test sets and overstate skill [109].

3.11 Leakage Prevention

Leakage occurs when information that would not be available at prediction time influences model construction [110]. Table 3.3 lists each possible channel and the control applied.

3.12 Unresolved Dataset Decisions

Three dataset decisions remain open and are recorded in the decision log (Appendix C.4): (i) confirmation of the originating products and licence of the archive; (ii) refitting of the anomaly climatology on 1979–2012 to remove the overlap with the test years; and (iii) the use of a second, independent reanalysis or OLR product to test whether the graph depends on the data source. None of them affects the internal consistency of the comparisons in this thesis, because every model is evaluated on identical inputs and targets.

Table 3.3: Leakage channels and controls in Phase 1.

Channel	Control
Node standardisation	Means and variances from 1979–2012 only (Eq. (3.4))
Regional partition	k-means fitted on the development span
BSISO EOF patterns	Fitted on May–October days of 1979–2012 only
Graph discovery and validation	PCMCI, decoys, lag mask, persistence on 1979–2012 season days only
Choice of k	Fit 1979–2004, score 2005–2012
Correlation-prior ranking	Lagged correlations computed on training rows only
Forecaster training and early choices	1979–2004 train, 2005–2012 validation
Climatology baseline	Mean of the training targets
Upstream anomaly climatology	Base period 1979–2014; two-year overlap acknowledged as a limitation
Season mask	Selects scored days only; never an input to a forecast

3.13 Chapter Summary

This chapter defined the direct seven-day OLR forecasting task, the 300-node representation built from 50 OLR-coherent regions and six variables, the 49-dimensional regional feature vectors, the activity-based BSISO season reproduced from the project's own data, and the development/test split with its leakage controls. The next chapter describes how PCMCI turns the 18,000 candidates per target into a sparse lagged graph.

Chapter 4

Proposed Methodology: PCMCI and Graph Discovery

4.1 Overview of the Causal-Discovery Pipeline

Stage A turns the 300-node state matrix into a lagged graph whose edges point into the 50 OLR targets. Figure 4.1 shows the sequence. For each target, a lagged design matrix is built from the season target days of the development period; the PC1 stage screens the 18,000 candidates down to a small superset of parents; the MCI stage retests each survivor while conditioning on the parents of both the target and the source; and the Benjamini–Hochberg procedure controls the false discovery rate over all candidates of that target. The output is a list of links, each with a source node, a lag, a signed strength and an adjusted q-value.

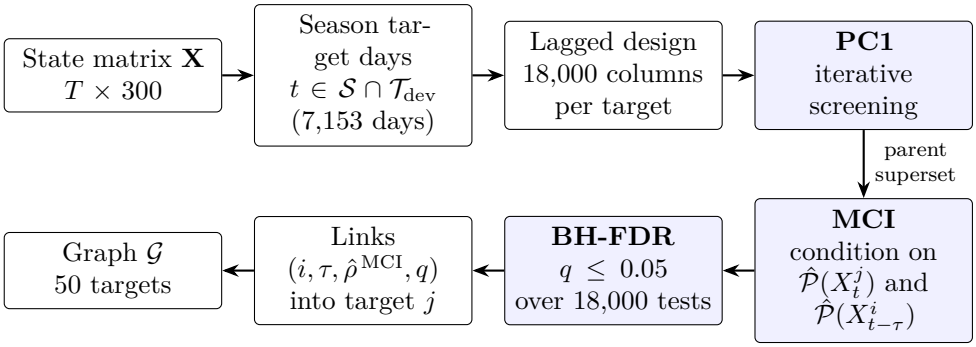


Figure 4.1: Stage A pipeline. For each OLR target node j, PC1 screens 18,000 lagged candidates down to a parent superset; MCI retests every survivor conditioned on the parents of both target and source; Benjamini–Hochberg correction controls the FDR over all candidates of the target. Only development-period season days are used.

4.2 Graph Notation and Node Definitions

We assume that the standardised node series follow a lagged structural model

$$X_t^j = g_j\Big(\mathcal{P}(X_t^j),\, \eta_t^j\Big)\,, \qquad j = 1, \ldots, N, \tag{4.1}$$

where $\mathcal{P}(X_t^j) \subseteq \{X_{t-\tau}^i : i = 1, \dots, N;\ \tau = 1, \dots, \tau_{\max}\}$ is the set of lagged parents of X_t^j, g_j is a function, and the noise terms η_t^j are mutually independent and independent of the past. The associated time-series graph $\mathcal{G}$ has a node for every pair (i, t) and a directed edge $X_{t-\tau}^i \to X_t^j$ whenever $X_{t-\tau}^i \in \mathcal{P}(X_t^j)$. Under stationarity the edge set is the same for every t, so $\mathcal{G}$ is summarised by the set of lagged links

$$\mathcal{E} = \left\{(i, j, \tau) : X_{t-\tau}^i \in \mathcal{P}(X_t^j)\right\}. \tag{4.2}$$

In this thesis only the parents of the 50 OLR nodes are estimated; the parents of the other 250 nodes are needed only as conditioning sets in the MCI stage.

4.3 Conditional-Independence Foundations

4.3.1 Why correlation is insufficient

Consider three cases in which $X_{t-\tau}^i$ and X_t^j are correlated although X^i has no direct influence on X^j:

- **Autocorrelation and indirect paths.** If $X_{t-\tau-1}^i \to X_t^j$ is a true link and X^i is autocorrelated, then $X_{t-\tau}^i$ is also correlated with X_t^j. The lagged correlation function of an autocorrelated source is smeared over many lags [31].
- **Common driver.** If $X_{t-5}^k \to X_{t-3}^i$ and $X_{t-5}^k \to X_t^j$, then X_{t-3}^i and X_t^j are correlated at lag 3 without any link between them. In the tropics, an active BSISO phase is such a driver for many regions simultaneously.
- **Mediated path.** If $X_{t-2}^i \to X_{t-1}^m \to X_t^j$, then X_{t-2}^i is correlated with X_t^j but has no direct link to it.

In each case the spurious dependence disappears once the right variables are conditioned on: the source's own past in the first case, the common driver in the second, the mediator in the third. Conditional independence is the formal statement of this idea.

4.3.2 Conditional independence and its testing

$X \perp\!\!\!\perp Y \mid \mathbf{Z}$ holds if $p(x \mid y, \mathbf{z}) = p(x \mid \mathbf{z})$. Under the causal Markov condition, a variable is independent of its non-descendants given its parents; under faithfulness, every conditional independence in the data is implied by the graph structure [32, 45]. Together, and assuming causal sufficiency (no unobserved common drivers among the included nodes), they imply that for lagged links

$$X_{t-\tau}^i \in \mathcal{P}(X_t^j) \iff X_{t-\tau}^i \not\perp\!\!\!\perp X_t^j \,\Big|\, \mathbf{X}_t^- \setminus \{X_{t-\tau}^i\}, \tag{4.3}$$

where $\mathbf{X}_t^-$ is the entire past of the system up to lag $\tau_{\max}$ [33]. Equation (4.3) cannot be tested directly with 18,000 conditioning variables and 7,153 samples. PCMCI tests an equivalent condition with much smaller conditioning sets.

4.3.3 The partial-correlation test (ParCorr)

The conditional-independence test used throughout is partial correlation, which is exact for linear-Gaussian dependence. Given samples of X, Y and a conditioning set $\mathbf{Z}$ of dimension $|\mathbf{Z}|$, the residuals of X and Y after least-squares regression on $\mathbf{Z}$ are

$$r_X = (\mathbf{I} - \mathbf{Q}\mathbf{Q}^\top)\,\mathbf{x}, \qquad r_Y = (\mathbf{I} - \mathbf{Q}\mathbf{Q}^\top)\,\mathbf{y}, \tag{4.4}$$

where $\mathbf{Q}$ is an orthonormal basis of the column space of the conditioning matrix $\mathbf{Z}$ (obtained by QR decomposition). The partial correlation and its test statistic are

$$\hat{\rho}(X, Y \mid \mathbf{Z}) = \frac{r_X^\top r_Y}{\|r_X\|\,\|r_Y\|}, \qquad t = \hat{\rho}\sqrt{\frac{n-2-|\mathbf{Z}|}{1-\hat{\rho}^2}}, \tag{4.5}$$

and under the null hypothesis $\rho = 0$ the statistic t follows a Student-t distribution with $n-2-|\mathbf{Z}|$ degrees of freedom, giving a two-sided p-value $p = 2\,\Pr(T_{n-2-|\mathbf{Z}|} > |t|)$. ParCorr detects only linear dependence; nonlinear alternatives such as conditional mutual information are discussed in Section 4.10.

4.4 PCMCI Stage 1: PC1 Parent Screening

The PC1 stage estimates, for each target X_t^j, a superset $\hat{\mathcal{P}}(X_t^j)$ of its parents with as few false negatives as possible. It is a variant of the PC algorithm's skeleton phase in which, at each iteration p, every candidate is tested against only the p strongest other candidates rather than against all subsets of size p:

1. Initialise $\hat{\mathcal{P}}^{(0)}$ with all 18,000 candidates $X_{t-\tau}^i$ and set $p = 0$.
2. For every candidate $X_{t-\tau}^i \in \hat{\mathcal{P}}^{(p)}$, let $\mathbf{S}_p$ be the p strongest members of $\hat{\mathcal{P}}^{(p)} \setminus \{X_{t-\tau}^i\}$, ranked by the smallest absolute test statistic recorded for them so far. Test

$$H_0 :\; X_{t-\tau}^i \perp\!\!\!\perp X_t^j \mid \mathbf{S}_p \tag{4.6}$$

   and remove the candidate if $p > \alpha_{\mathrm{PC}}$. Record the minimum absolute partial correlation of each surviving candidate across iterations.
3. Set $p \leftarrow p + 1$ and repeat while $|\hat{\mathcal{P}}^{(p)}| - 1 \geq p$.

Ordering conditions by strength means that strong drivers, including the target's own recent past, are conditioned on first, which removes most autocorrelation-induced and indirect candidates in the first few iterations. Restricting to one conditioning combination per iteration (max combinations = 1, as recommended for high-dimensional climate data [34]) keeps the cost linear in the number of candidates. The significance level $\alpha_{\mathrm{PC}} = 0.05$ is deliberately liberal: false positives at this stage are tolerated because MCI tests every survivor again.

The PC1 stage is also run for each non-target node, but with its candidates restricted to the

node's own past (lags 1–60). The resulting sets $\hat{\mathcal{P}}(X_t^i)$ describe the autocorrelation structure of each source and are used only as conditions in the MCI stage. When the source is itself one of the 50 OLR targets, its full PC1 parent set, which may include other nodes, is used as its condition. This restriction is a modelling choice: it conditions each source on its own memory, which addresses autocorrelation, the most important source of false positives, without estimating a full graph for all 300 nodes. Its consequence for common drivers is discussed in Section 4.10.

4.5 PCMCI Stage 2: Momentary Conditional Independence

The MCI test checks each survivor $X_{t-\tau}^i \in \hat{\mathcal{P}}(X_t^j)$ by conditioning on the parents of the target *and* on the parents of the source, shifted by the lag:

$$\text{MCI}: \quad X_{t-\tau}^i \perp\!\!\!\perp X_t^j \;\Big|\; \hat{\mathcal{P}}(X_t^j) \setminus \{X_{t-\tau}^i\},\, \hat{\mathcal{P}}(X_{t-\tau}^i), \tag{4.7}$$

where $\hat{\mathcal{P}}(X_{t-\tau}^i) = \{X_{t-\tau-s}^k : X_{t-s}^k \in \hat{\mathcal{P}}(X_t^i)\}$. The test statistic is the partial correlation of Eq. (4.5) with this conditioning set, and its value is the link's *MCI strength*:

$$\hat{\rho}_{i\to j}^{\text{MCI}}(\tau) = \hat{\rho}\Big(X_{t-\tau}^i,\, X_t^j \;\Big|\; \hat{\mathcal{P}}(X_t^j) \setminus \{X_{t-\tau}^i\},\, \hat{\mathcal{P}}(X_{t-\tau}^i)\Big). \tag{4.8}$$

Conditioning on the target's parents removes indirect paths and common drivers that are among the target's parents. Conditioning on the source's own parents removes the source's autocorrelation, so that the test measures the *momentary* information that the source contributes at lag τ beyond its own history [34]. This conditioning has two practical effects. First, it calibrates the test: without it, the false-positive rate of a test between two strongly autocorrelated series is far above the nominal level. Second, it sharpens the lag: an autocorrelated source no longer appears at many neighbouring lags. As a result, $\hat{\rho}^{\text{MCI}}$ values are generally smaller than raw correlations and are interpretable as a normalised measure of the direct lagged coupling, given the conditioning sets.

4.5.1 Significance and effect size

A small p-value indicates that a link is unlikely to be zero, not that it is strong. With 7,153 samples, a partial correlation of 0.05 is significant at conventional levels. For this reason, the graph records both the q-value (Section 4.7) and the MCI strength, and later stages rank links by $|\hat{\rho}^{\text{MCI}}|$, not by significance (Section 5.4).

4.6 Lag Range and Candidate Source Pool

The candidate pool is the full set of 300 nodes, so that local and remote sources, and all six variables, are eligible for every target. Restricting the pool by geography would rule out teleconnections; restricting it by variable would encode hypotheses that the analysis should test. The lag range is $\tau_{\min} = 1$ to $\tau_{\max} = 60$ days. The upper bound covers a full BSISO cycle of 30–60 days, so that precursors one cycle ahead, and slow SST-mediated influences, are eligible. Lag 0 is excluded so that every link points forward in time.

The choice of $\tau_{\min}$ has a direct consequence for forecasting. A link at lag τ is usable for a forecast at lead L only if $\tau \geq L$ (Section 5.11). With $\tau_{\min} = 1$ the graph describes all lagged dependencies, most of which, as Chapter 7 shows, act at one or two days. A graph intended specifically for seven-day forecasting would set $\tau_{\min} = L = 7$, so that only links usable at that lead are tested; this variant is part of the future-work programme.

4.7 Statistical Testing and Multiple-Testing Correction

Each target involves 18,000 hypotheses. At $\alpha = 0.05$ without correction, roughly 900 false positives per target would be expected even if no link existed. The Benjamini–Hochberg (BH) procedure [61] controls the false discovery rate, the expected proportion of false links among the selected links:

$$\mathrm{FDR} = \mathbb{E}\left[\frac{V}{\max(S, 1)}\right], \tag{4.9}$$

where S is the number of selected links and V the number of false ones. For the $m = 18{,}000$ p-values of a target, sorted as $p_{(1)} \leq \cdots \leq p_{(m)}$, the adjusted q-values are

$$q_{(h)} = \min_{h' \geq h} \min\left(1,\ \frac{m}{h'}\, p_{(h')}\right), \tag{4.10}$$

and a link is retained if $q \leq \alpha_{\mathrm{FDR}} = 0.05$. Candidates removed by PC1 are counted with $p = 1$, so that the correction is applied over the full set of 18,000 hypotheses per target rather than only over the survivors; this is conservative. BH control is guaranteed under independence and under positive regression dependence of the test statistics [62], which is the typical case for positively correlated climate tests [63]. It is applied per target, so the FDR statement holds for each target's link set.

4.8 Graph Construction and Edge Representation

Each retained link is stored as a record

$$e = \left(r_{\mathrm{src}},\ v_{\mathrm{src}},\ \tau,\ \hat{\rho}^{\mathrm{MCI}},\ p,\ q\right) \tag{4.11}$$

attached to its target region r, where $(r_{\text{src}}, v_{\text{src}})$ identifies the source node. Links whose source region equals the target region ($r_{\text{src}} = r$) are *own-region* links, which include the target's own OLR history; the others are *outside* links. The distinction matters because own-region information is always available to the forecaster, whereas outside links are the information that the graph adds. Subsequent stages attach validation flags, distances and implied speeds to each record (Chapter 5).

4.9 Computational Considerations

The reference implementation of PCMCI in the Tigramite package builds a new data array for every conditional-independence test. With 18,000 candidates per target, several PC1 iterations and 50 targets, this becomes the bottleneck. The implementation developed for this thesis exploits the structure of the problem:

1. **Shared lagged design matrix.** The $n \times 18{,}000$ matrix of all lagged, standardised candidate columns on the season target days is built once and shared by all 50 targets.
2. **Batched screening.** At PC1 iteration p, all candidates that are conditioned on the same set are tested in one batched projection: the conditioning matrix is orthonormalised once by QR decomposition, all candidate columns are residualised by a single matrix product $\mathbf{X}_C - \mathbf{Q}(\mathbf{Q}^\top \mathbf{X}_C)$, and the partial correlations of Eq. (4.5) are computed for all of them at once. Candidates whose strongest-p conditioning set differs because they themselves are among the strongest are handled separately.
3. **Chunking.** Columns are processed in chunks of 4,096 to bound memory.

The cost of one batched test is $O(n\,m\,p)$ for m candidates and p conditions, compared with $O(m)$ separate $O(n\,p^2)$ tests. Correctness is established by testing the implementation link-for-link against Tigramite (Section 6.3). Algorithm 4.1 summarises the procedure.

4.10 Assumptions and Limitations

PCMCI's guarantees hold under assumptions that are only approximately satisfied by regional climate data. Table 4.1 lists them, how they may fail here, and how the failure would show.

Temporal order is not causal direction. Excluding lag 0 guarantees that every link points forward in time, but not that the source causes the target. If an unobserved driver U affects the source at $t-5$ and the target at t, the source at $t-5$ will be dependent on the target at t even after conditioning on all observed parents. The link will be correctly directed in time and still spurious. LPCMCI [51] can represent such latent confounding, at a higher computational cost and with a more complex output.

Algorithm 4.1 Vectorised PCMCI for one OLR target (Stage A)

Require: Node series $\mathbf{X} \in \mathbb{R}^{T \times N}$; season target days $\mathcal{R} \subset \mathcal{T}_{\text{dev}}$ with $\min \mathcal{R} \geq 2\tau_{\max}$; target node j; $\tau_{\min}, \tau_{\max}, \alpha_{\text{PC}}, \alpha_{\text{FDR}}$

Ensure: Links $\{(i, \tau, \hat{\rho}^{\text{MCI}}, p, q)\}$ into X^j

Build lagged design $\mathbf{D} \in \mathbb{R}^{|\mathcal{R}| \times N(\tau_{\max} - \tau_{\min} + 1)}$, column (i, τ) = standardised $X^i_{t-\tau}$, $t \in \mathcal{R}$ ▷ shared by all targets

$y \leftarrow$ standardised X^j_t, $t \in \mathcal{R}$

$C \leftarrow$ all columns; $s_c \leftarrow +\infty$ for all c; $p \leftarrow 0$ ▷ PC1 screening

while $|C| - 1 \geq p$ **do**

  $\mathbf{S} \leftarrow$ the p columns of C with largest s_c

  $\mathbf{Q} \leftarrow \text{QR}(\mathbf{D_S})$; residualise y and $\mathbf{D}_C$ on $\mathbf{Q}$ ▷ one batched projection

  compute $\hat{\rho}_c$ and p_c for all $c \in C$ (Eq. 4.5); retest separately the $c \in \mathbf{S}$ with $\mathbf{S} \setminus \{c\}$ plus the next strongest

  $s_c \leftarrow \min(s_c, |\hat{\rho}_c|)$; $\quad C \leftarrow \{c \in C : p_c \leq \alpha_{\text{PC}}\}$; $\quad p \leftarrow p + 1$

end while

$\hat{\mathcal{P}}(X^j) \leftarrow C$ ordered by s_c

for all $(i, \tau) \in \hat{\mathcal{P}}(X^j)$ **do** ▷ MCI stage

  $\mathbf{Z} \leftarrow (\hat{\mathcal{P}}(X^j) \setminus \{(i, \tau)\}) \cup \{(k, \tau + s) : (k, s) \in \hat{\mathcal{P}}(X^i)\}$ ▷ $\hat{\mathcal{P}}(X^i)$: own-past parents of source

  $(\hat{\rho}^{\text{MCI}}_{i \to j}(\tau), p_{i,\tau}) \leftarrow \text{ParCorr}(X^i_{t-\tau}, X^j_t \mid \mathbf{Z})$

end for

$p \leftarrow 1$ for all candidates removed by PC1; compute q by BH over all $N(\tau_{\max} - \tau_{\min} + 1)$ candidates (Eq. 4.10)

return links with $q \leq \alpha_{\text{FDR}}$

Table 4.1: Assumptions of PCMCI with ParCorr, how they may be violated by regional climate data, and the expected symptom.

Assumption	**Possible violation**	**Symptom**
Causal sufficiency	BSISO phase, ENSO, extratropical forcing and unresolved moisture processes are only partly represented by 300 regional nodes	Links from a node that shares an unobserved driver with the target
Stationarity	Season-to-season and interannual changes in dependence; long-term trends	Links that hold in some years but not others (tested in Stage D)
Linearity	Threshold behaviour of convection with moisture [41]	Missed nonlinear links; diluted strengths
Faithfulness	Cancelling pathways; near-deterministic relations between fields of one region	Missed links; unstable choice between redundant fields
No contemporaneous links	Fast processes acting within one day	Same-day effects appear at lag 1 or are missed
Appropriate scale	Region means mix different processes; boundaries chosen by OLR clustering	Links that depend on the partition (tested by B4)
Gaussian test calibration	Heavy tails and skewness of OLR anomalies	p-values slightly miscalibrated

Restricted source conditioning. Because non-target sources are conditioned only on their own past, a common driver that acts on the source through another node is removed only if it also appears among the target's parents. This is a deliberate trade-off between computational cost and confounding control, and it is one reason why the validation checks of Chapter 5 are needed.

Statistical power. With 18,000 candidates per target, strongly autocorrelated data and FDR control, weak but real links can be missed, and some false links will remain. The FDR guarantee concerns the expected proportion, not each individual link.

Redundant fields within a region. The six fields of a region move together with the same wave. When two fields carry nearly the same information, PCMCI may select one or the other depending on small sampling differences. Claims about which *field* drives a target are therefore weaker than claims about which *region* does, and the forecasting prior is defined at the region level (Section 5.7).

4.11 Reproducibility and Implementation Decisions

The Stage A configuration is fixed and recorded with its outputs (Table 6.2 in Chapter 6): ParCorr test, $\tau_{\min} = 1$, $\tau_{\max} = 60$, $\alpha_{\mathrm{PC}} = 0.05$, one conditioning combination per PC1 iteration, no limit on the dimension of the conditioning sets, BH correction at $\alpha_{\mathrm{FDR}} = 0.05$ over all candidates per target, and own-past-only parents for non-target nodes. Alternative choices that were considered and the reasons for not adopting them are recorded in the decision log (Appendix C.4).

4.12 Chapter Summary

This chapter explained why lagged correlation misidentifies drivers in autocorrelated climate data, and how PCMCI replaces it with a screening stage and a momentary conditional-independence test that conditions on the parents of both target and source. It defined the ParCorr statistic, the Benjamini–Hochberg FDR control applied over all 18,000 candidates per target, the edge representation and the vectorised implementation, and it listed the assumptions under which the resulting graph should be read. The next chapter describes how this graph is validated and used.

Chapter 5

Graph Validation and the Graph-Guided Mixture of Experts

5.1 Why the Discovered Graph Requires Independent Validation

PCMCI with FDR control limits the expected fraction of false links under its assumptions. Those assumptions (Table 4.1) are only approximately met, and the FDR guarantee says nothing about whether a link is physically possible, whether it is stable under a different spatial aggregation, or whether a graph of that size is useful to a forecaster. There is also no ground truth for the real atmosphere against which a discovered graph could be scored. Validation must therefore rely on checks that can *reject* a bad graph even though no check can *prove* a graph correct.

Stage B applies four such checks, in a fixed order, to the Stage A graph (Figure 5.1):

B1 Surrogate decoys: is the pipeline better than chance at rejecting series that have realistic autocorrelation but no real coupling?

B2 Physical lag mask: is each link's implied propagation speed achievable by a process carried by its source variable?

B3 Data-driven top-k: how many outside links per target are worth keeping?

B4 Resolution persistence: do the links reappear when the regions are redrawn at twice the resolution?

The order runs from the cheapest and most general check to the most specific. B1 tests the whole pipeline before any link is trusted; B2 removes physically impossible links before they influence the ranking in B3; B3 sets the size of the prior; and B4, which needs a second full PCMCI run, flags links that depend on the partition. The resulting graph is then frozen and used without further change.

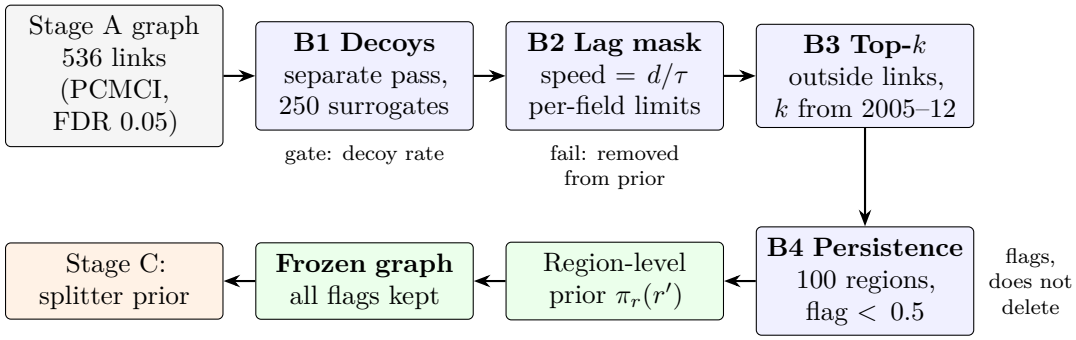


Figure 5.1: The four-stage validation sequence applied to the Stage A graph. Each check can fail. B2 failures are removed from the forecasting prior but kept, with their flags, in the frozen graph file; B4 only flags. The frozen graph is built from 1979–2012 only.

5.2 Check B1: Surrogate Decoys

5.2.1 Rationale and null model

A climate series can be selected as a parent because it is genuinely coupled to the target, or because its autocorrelation and spectral peaks make chance alignment with the target likely. The B1 check asks how often the full pipeline (PC1, MCI and FDR together) selects series that share the spectral properties of real nodes but are, by construction, uncoupled from everything. Such series are *decoys*: they are placed in the candidate pool alongside the real nodes, so they must be rejected by the same tests under the same conditions.

The null model is a stationary linear Gaussian process with the same power spectrum as a real node [65, 66]. A phase-randomised surrogate of a series x_t of length n is generated by Fourier-transforming the demeaned series, replacing the phases with independent uniform random phases, and transforming back:

$$\hat{x}_f = \sum_{t=0}^{n-1} (x_t - \bar{x})\, e^{-2\pi i f t/n}, \qquad \tilde{x}_t = \frac{1}{n} \sum_f |\hat{x}_f|\, e^{i\theta_f}\, e^{2\pi i f t/n}, \qquad \theta_f \sim \mathcal{U}[0, 2\pi), \tag{5.1}$$

with $\theta_0 = 0$ (and $\theta_{n/2} = 0$ for even n) so that $\tilde{x}_t$ is real, and $\theta_{-f} = -\theta_f$ by construction of the real inverse transform. The surrogate is then standardised. Because $|\hat{x}_f|$ is unchanged, the surrogate has the same periodogram and therefore the same sample autocorrelation function as the original, including its 30–60-day intraseasonal peak. Because the phases are random, its cross-spectrum with any other series has random phase, so its expected lagged cross-covariance with every real node is zero.

5.2.2 What phase randomisation preserves and what it does not

The surrogate preserves the power spectrum, the autocorrelation function and the variance of the original series. It does not preserve:

- **the amplitude distribution:** surrogates are close to Gaussian even when the original is skewed, as convective OLR anomalies are;

- **nonstationarity:** the seasonal modulation of variance and of the BSISO's amplitude is spread uniformly over the record;
- **nonlinear structure:** any dependence on phase relations between frequencies;
- **cross-variable structure:** a decoy built from one region's u850 has no relation to that region's other fields, whereas real fields of one region move together.

Amplitude-adjusted and iterated surrogates [66, 67] can restore the amplitude distribution. The other limitations mean that decoys are an easier null than the most dangerous real false positives, which are series coupled to the target only through an unobserved common driver. A low decoy rate therefore shows that the pipeline does not select series merely because of their autocorrelation; it does not eliminate confounding and does not prove that the real links are causal.

5.2.3 Construction and separate pass

$D = 250$ decoys are generated. For each, a real node is drawn at random (with replacement) from the source nodes and phase-randomised with an independent random generator, and the decoy is labelled with its origin. The decoy run is a *separate* PCMCI pass with a pool of $300 + 250 = 550$ nodes, which gives $550 \times 60 = 33{,}000$ candidates per target. Running it separately keeps the main graph free of any effect of the decoys on the FDR correction (the number of hypotheses changes) and on the PC1 condition ordering. In this pass, decoys are allowed only their own past as parents, like other non-target nodes.

5.2.4 Statistics

Let $\mathcal{E}^{\mathrm{dec}}$ be the links selected in the decoy pass and $\mathcal{D}^{\mathrm{sel}}$ the set of decoys that appear as a parent of at least one target. The *decoy rate* is

$$\delta = \frac{|\mathcal{D}^{\mathrm{sel}}|}{D}. \tag{5.2}$$

The *decoy edge share* is the fraction of selected links in the decoy pass whose source is a decoy. The strongest selected decoy defines a strength threshold

$$\kappa_{\mathrm{dec}} = \max_{e \in \mathcal{E}^{\mathrm{dec}}:\, \mathrm{src}(e) \in \mathcal{D}} |\hat{\rho}_e^{\mathrm{MCI}}|, \tag{5.3}$$

and a real link passes B1 if $|\hat{\rho}_e^{\mathrm{MCI}}| > \kappa_{\mathrm{dec}}$. If no decoy is selected, $\kappa_{\mathrm{dec}} = 0$ and the strength comparison is satisfied by every real link; in that case B1's evidence is carried by the decoy rate and by the stability of the real graph between the clean pass and the decoy pass. A more informative comparison, recommended for future runs, is between the distribution of real-link MCI strengths and the distribution of decoy MCI strengths recorded *before* FDR, which is non-degenerate even when no decoy survives.

5.2.5 The acceptance line

The project plan sets a decoy rate of 30% as a warning line. This value is a pragmatic, project-specific threshold and not a statistical standard. Under the decoy null, every decoy is a false positive by construction, so with FDR control at 5% the expected decoy selection rate is low, and a rate approaching 30% would indicate that the tests are badly miscalibrated for series with realistic autocorrelation. The threshold was fixed before the decoy pass was run, and its role is to stop the pipeline if it fails, not to certify the graph if it passes.

5.3 Check B2: Physical Lag Masking

5.3.1 Distance, lag and implied speed

A link from source region r' to target region r at lag τ requires a signal to travel at least the distance between the two regions within τ days. The distance is taken as the great-circle distance between the *closest* pair of cells of the two regions, computed with the haversine formula [111]:

$$d(r', r) = \min_{c' \in \Omega_{r'},\, c \in \Omega_r} 2a \arcsin \sqrt{\sin^2 \tfrac{\varphi_c - \varphi_{c'}}{2} + \cos \varphi_{c'} \cos \varphi_c \sin^2 \tfrac{\lambda_c - \lambda_{c'}}{2}}, \tag{5.4}$$

where φ and λ are latitude and longitude and a is the Earth's radius. For adjacent regions $d = 0$. The implied propagation speed is

$$s_e = \frac{d(r', r)}{\tau\, \Delta t}, \qquad \Delta t = 86{,}400 \text{ s}. \tag{5.5}$$

Because the closest-cell distance is a lower bound on any path between the regions, s_e is a lower bound on the speed that the link requires. A link *fails* B2 if

$$s_e > s^{\max}_{v_{\text{src}}}, \tag{5.6}$$

where $s^{\max}_v$ is the maximum speed at which a process carried by the source variable v can plausibly propagate.

5.3.2 Speed limits by source variable

The limits are deliberately generous. Table 5.1 lists them together with their physical justification. They are guidelines that encode the order of magnitude of the relevant processes, not universal constants.

5.3.3 Distance cutoff versus speed constraint

A geographic-distance cutoff would remove all links between regions more than some fixed distance apart. That would wrongly remove teleconnections: a Kelvin wave can carry an

Table 5.1: Propagation-speed limits used in check B2 and their physical basis. The values are intentionally generous upper bounds.

Source	Limit	Physical basis
SST	$1\,\mathrm{m\,s^{-1}}$	Ocean-mediated influences: oceanic advection and slow coupled processes are of order $1\,\mathrm{m\,s^{-1}}$ or less; the atmospheric response to an SST anomaly is carried by the atmospheric fields, which are separate nodes
OLR, PW	$20\,\mathrm{m\,s^{-1}}$	Convective and moisture signals: the BSISO envelope moves at about $5\,\mathrm{m\,s^{-1}}$; convectively coupled Kelvin waves travel at roughly 15–$20\,\mathrm{m\,s^{-1}}$ [27, 43]
h850, u850, u200	$50\,\mathrm{m\,s^{-1}}$	Dynamical fields: dry or weakly coupled equatorial Kelvin waves and upper-level wave trains can exceed $30\,\mathrm{m\,s^{-1}}$; $50\,\mathrm{m\,s^{-1}}$ bounds the fastest relevant large-scale waves [25, 27]

upper-level wind signal 4,000 km in about a day. A speed constraint instead removes only links that are too *fast*: the same 4,000 km at lag 1 is admissible for u200 ($46\,\mathrm{m\,s^{-1}}$) but not for SST. Remote links at long enough lags, and all links between adjacent regions, pass automatically.

5.3.4 Oscillation aliasing

Because the BSISO is quasi-periodic, a real link at lag τ can reappear at lags $\tau \pm P$ where P is the 30–60-day period, since the source is then in a similar phase. Links at $\tau \geq 25$ days are therefore *flagged* as possible aliases of the oscillation. A flag does not remove the link; it marks it for inspection.

5.3.5 Diagnostic versus exclusion

A B2 failure does not prove that the link is spurious. It may reflect an incorrect lag (the true coupling is at a longer lag that was absorbed by conditioning), an unobserved relay (the signal travels through intermediate regions whose effect is distributed), or an oscillatory alias. Failed links are therefore retained in the frozen graph file with their computed distance, speed and failure flag, so that they can be inspected. For forecasting, however, a link that requires an impossible speed should not drive source selection, so B2 failures are excluded from the region-level prior. This separates the scientific record (all links with all diagnostics) from the operational prior (only physically admissible links).

5.4 Check B3: Data-Driven Top-k Selection

5.4.1 Purpose

A forecasting prior should be small enough to concentrate the splitter on a few sources, but not so small that useful information is lost. B3 ranks the surviving outside links of each target and keeps the k strongest. Own-region links are not counted against the budget: the target region is always a candidate for its own forecaster, so the prior concerns only which *other* regions to add.

5.4.2 Ranking

Links are ranked by $|\hat{\rho}^{\mathrm{MCI}}|$, the conditional effect size, rather than by q-value. A q-value measures how confidently a link is non-zero, which depends heavily on sample size; $|\hat{\rho}^{\mathrm{MCI}}|$ measures how much unique information the source contributes. Other possible measures are the drop in validation error when the link is removed, or bootstrap selection frequency. They are more expensive and could be used in a refined version of the check.

5.4.3 Choosing k from development data

For each candidate $k \in \{0, \ldots, k_{\max}\}$, a ridge regression [112] is fitted for each target on 1979–2004 season days, using as predictors all own-region links plus the top-k outside links, each at its own lag:

$$\hat{\boldsymbol{\beta}}_k = \arg\min_{\boldsymbol{\beta}} \sum_{t \in \mathcal{S} \cap [1979,2004]} \left(X_t^j - \beta_0 - \sum_{e \in \mathcal{E}_j^{\mathrm{own}} \cup \mathcal{E}_j^{\mathrm{out},k}} \beta_e X_{t-\tau_e}^{\mathrm{src}(e)} \right)^2 + \epsilon \|\boldsymbol{\beta}\|^2, \tag{5.7}$$

with a small ridge constant $\epsilon = 10^{-3}$ for numerical stability. Each fit is scored by the coefficient of determination on 2005–2012 season days,

$$R^2_{\mathrm{val}}(k) = \frac{1}{|\mathcal{J}|} \sum_{j \in \mathcal{J}} \left(1 - \frac{\mathrm{Var}\left(X_t^j - \hat{X}_{t,k}^j \right)}{\mathrm{Var}\left(X_t^j \right)} \right), \qquad t \in \mathcal{S} \cap [2005, 2012], \tag{5.8}$$

averaged over targets, and $k^* = \arg\max_k R^2_{\mathrm{val}}(k)$ with $k_{\max} = 10$. The test period plays no part. Because the regression uses each link at its own lag and evaluates the target at time t, this check measures the in-sample explanatory value of the links for the target at their own time scale, not their value at a fixed forecast lead.

5.4.4 Limits of top-k

Keeping the k strongest links does not make them more causal; it only bounds the size of the prior. If PCMCI with FDR already produces a sparse graph, B3 may keep every outside link, in which case it acts as a check that no target has an excessive number of drivers rather than as a filter. A selection criterion matched to the forecast lead, or one that requires the selected links to beat random links of the same size, would make B3 more discriminating and is part of the future-work programme.

5.5 Check B4: Resolution Persistence

5.5.1 Why aggregation matters

Region means are averages of many cells. Changing the partition changes the averaged series: smaller regions have more variance, lower autocorrelation and weaker co-variability with remote

regions. A link that is an artefact of how two particular regions were drawn should be fragile under a different partition, whereas a link that reflects a large-scale process should survive. B4 tests this.

5.5.2 Nested partition and correspondence

Each of the 50 regions is split into two children by k-means with $k = 2$ on its own cells, which gives 100 regions numbered so that the children of region r are $2r$ and $2r + 1$. The full Stage A pipeline is rerun on the 100-region system (596 nodes, $596 \times 60 = 35{,}760$ candidates per target, the same 7,153 season days). The 300-node and 596-node graphs cannot be compared by raw node index; the nesting provides the correspondence rule $\text{parent}(c) = \lfloor c/2 \rfloor$. An outside link $e = (r', v, \tau) \to r$ at 50 regions *persists* if, for some child target $c \in \{2r, 2r + 1\}$, the fine graph contains a link $(c', v', \tau') \to c$ with

$$\lfloor c'/2 \rfloor = r', \qquad v' = v, \qquad |\tau' - \tau| \leq 2. \tag{5.9}$$

The lag tolerance of ± 2 days allows for the slightly different timing of the signal at a sub-region. The persistence score of target r is the fraction of its validated outside links that persist,

$$\Pi_r = \frac{1}{|\mathcal{E}_r^{\text{out}}|} \sum_{e \in \mathcal{E}_r^{\text{out}}} \mathbb{1}[e \text{ persists}], \tag{5.10}$$

undefined for targets with no outside link. A target passes if $\Pi_r \geq 0.5$. Failures are flagged, not deleted.

5.5.3 Interpretation

Persistence supports robustness but does not prove causality: a link driven by a large-scale unobserved common driver will also persist. Conversely, non-persistence can be a genuine scale effect: a link carried by a coherent large-scale signal may be too diluted at sub-region scale to pass FDR with the larger candidate pool (35,760 instead of 18,000 hypotheses per target), or the relevant source signal may sit in only one child region whose link to the target falls just below the threshold. This is why B4 flags rather than deletes.

5.6 Graph Comparison and Robustness Metrics

Several stages compare two link sets A and B, for example the clean and decoy passes in B1, the two resolutions in B4, or consecutive windows in Stage D. The following metrics are used or

recommended:

$$\text{Jaccard similarity:} \quad J(A,B) = \frac{|A \cap B|}{|A \cup B|}, \tag{5.11}$$

$$\text{Persistence (directional):} \quad \Pi(A \to B) = \frac{|\{a \in A : \exists b \in B,\ a \simeq b\}|}{|A|}, \tag{5.12}$$

$$\text{Lag stability:} \quad \Delta\tau = \text{median}_{a \simeq b}\, |\tau_a - \tau_b|, \tag{5.13}$$

$$\text{Strength stability:} \quad \text{corr}\Big(|\hat{\rho}_a^{\text{MCI}}|, |\hat{\rho}_b^{\text{MCI}}|\Big) \text{ over matched pairs } a \simeq b, \tag{5.14}$$

where $a \simeq b$ denotes the correspondence rule of the comparison. The directional persistence of Eq. (5.12) is used in B4 because the fine graph is expected to contain more links than the coarse graph. Jaccard similarity on (source region, field) sets defines the causal-signature distance of Stage D (Section 5.13).

5.7 From Validated Links to a Region-Level Prior

The splitter operates on whole source regions, each carrying all six fields at several lags. The validated links of a target are therefore merged into one prior entry per source region. For target r, let $\mathcal{E}_r^\star$ be its outside links that pass B1 and B2 and are retained by B3. The prior is

$$\pi_r(r') = \max_{e \in \mathcal{E}_r^\star :\, \text{src_region}(e) = r'} |\hat{\rho}_e^{\text{MCI}}|, \qquad r' \neq r, \tag{5.15}$$

defined for the source regions r' that have at least one validated link. Field and lag detail are kept in the graph file. B4 flags are kept but do not remove regions from the prior. The prior size $|\pi_r|$ is the number of source regions for target r; targets with no validated outside link have an empty prior.

5.8 The Graph-Guided Mixture-of-Experts Forecaster

5.8.1 Architecture overview

The forecasting framework assigns one expert to each target region. Each expert predicts its region's OLR from a small set of source regions chosen by a *splitter*. The splitter plays the role of the gate in a Mixture of Experts [72, 74]: instead of routing an input to an expert, it routes source regions to each target's expert, and it decides which sources are relevant through an invariance criterion. Over time, a region can hold several experts trained in different regimes, and a pool of archived experts can be blended (Chapter 8). Figure 5.2 shows the architecture and the three points at which the causal graph can enter.

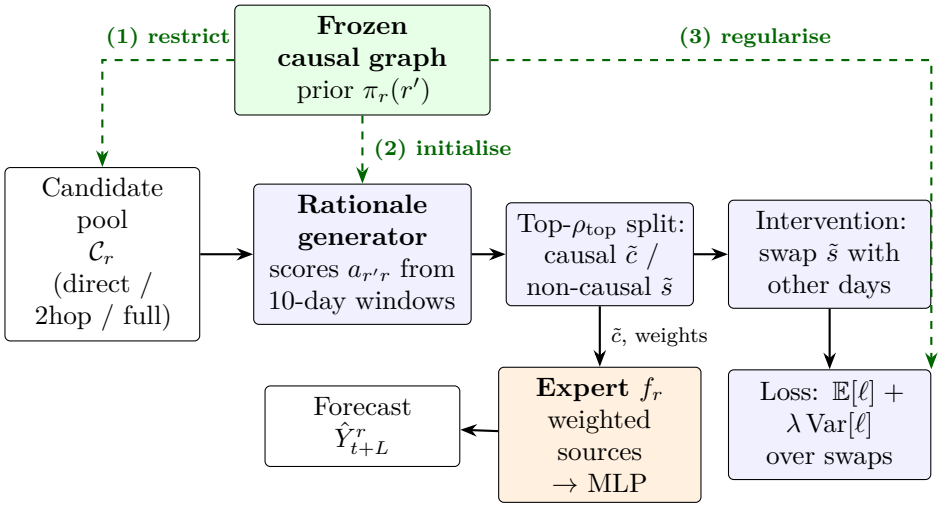


Figure 5.2: The graph-guided splitter and expert for one target region r. The rationale generator scores candidate source regions; the top fraction forms the causal set used by the expert; the remaining edges are swapped with other days to create interventional environments, and the generator is trained to keep the risk low and stable across swaps. The frozen causal graph can (1) restrict the candidate pool, (2) initialise the edge scores, or (3) penalise scores outside the graph.

5.8.2 Candidate pools

Three candidate pools define which source regions the splitter may consider for target r: *direct*, the target and its geographically adjacent regions (typically 6–8 regions); *2hop*, adding neighbours of neighbours (about 15–23 regions); and *full*, all 50 regions. The target region is always a candidate, so its own history can be selected, but it is not supplied to the expert unless selected.

5.8.3 The invariance-trained splitter

The splitter adapts Discovering Invariant Rationales [81] from graph classification to regression on a fixed spatial graph. For a target r on day t:

Scoring. The rationale generator computes a score for each candidate edge $r' \to r$ from a 10-day window of the two endpoints' histories,

$$a_{r'r,t} = \sigma\Big(\mathrm{MLP}_\psi\Big[\mathbf{w}_t^{r'} \,\|\, \mathbf{w}_t^{r}\Big] + b_{r'r}\Big)\,, \tag{5.16}$$

where $\mathbf{w}_t^r$ is the flattened window of region r's state over days $t-9,\ldots,t$, σ is the logistic function, and $b_{r'r}$ is an optional learnable bias (zero unless the initialise mode is used). Using a window rather than a single day lets the generator learn a lagged-association function between the two regions.

Selection. The top fraction $\rho_{\text{top}} = \min(0.5,\, 3/|\mathcal{C}_r|)$ of edges by score forms the causal set $\tilde{c}_t$, and the remainder forms the non-causal set $\tilde{s}_t$. Selection is hard, as in the reference

implementation [81]; gradients flow through the scores used as edge weights.

Intervention. For $M = 6$ interventions, the non-causal edges' messages are replaced by those of the same edges on randomly drawn other days $t'_1, \ldots, t'_M$, while the causal edges keep day t. A shared encoder aggregates the weighted messages and a causal head predicts the target, giving risks $\ell_{t,m}$, $m = 1, \ldots, M$.

Objective. The splitter minimises the mean and the variance of the risk across interventions,

$$\mathcal{L}_{\text{split}} = \frac{1}{M} \sum_{m=1}^{M} \ell_{t,m} \; + \; \lambda(s) \; \text{Var}_m \left[\ell_{t,m} \right] \; + \; \mathcal{L}_{\text{spur}}, \tag{5.17}$$

where $\lambda(s)$ is zero for the first 30% of training steps and then rises linearly to $\lambda_{\max} = 1$, and $\mathcal{L}_{\text{spur}}$ trains a separate spurious head on the non-causal edges with its gradient detached from the generator. The variance term follows the risk-extrapolation principle [80]: if the selected sources truly determine the target, replacing the unselected ones should not change the risk.

Expert. The expert of region r forms a weighted aggregate of the selected sources' feature vectors (Eq. (3.7)) and maps it to the forecast with a small MLP,

$$\hat{Y}^r_{t+L} = \text{MLP}_{\theta_r}\Big(\sum_{r' \in \check{c}_t} a_{r'r,t} \, \mathbf{h}^{r'}_t \Big), \qquad \mathcal{L}_{\text{exp}} = \Big(\hat{Y}^r_{t+L} - Y^r_{t+L} \Big)^2. \tag{5.18}$$

The target's own features enter only if the target region is selected. This design makes the forecast depend on the splitter's choice, so that the choice can be evaluated; if the target's own history were always supplied, the expert could ignore the selected sources entirely, because OLR persistence alone is a strong predictor at short leads.

Discrimination. Whether the splitter actually distinguishes between candidates is measured by the standard deviation σ_a of the candidates' mean scores over the test days. A splitter whose scores cluster around one value has $\sigma_a \approx 0$ and is effectively choosing at random. A splitter is called *informative* if $\sigma_a \geq 0.02$, a threshold fixed in the project's earlier experiments, below which the scores are indistinguishable from noise clustered around a single value.

5.9 Three Modes of Using the Graph

The graph prior π_r of Eq. (5.15) can enter the splitter in three ways, each with a different strength of commitment (Figure 5.2).

(1) Restrict. The candidate pool is replaced by the prior regions and the target itself:

$$\mathcal{C}^{\text{restrict}}_r = \{ r' \, : \, \pi_r(r') \text{ defined} \} \cup \{ r \}. \tag{5.19}$$

This is a hard constraint. It cannot recover a useful source that the graph missed, but it removes all competition from implausible sources.

(2) Initialise. The pool is unchanged, and the learnable edge bias of Eq. (5.16) starts at a value proportional to the prior strength:

$$b_{r'r}^{(0)} = \begin{cases} \beta\, \pi_r(r') / \max_{r''} \pi_r(r'') & r' \in \pi_r, \\ 0 & \text{otherwise,} \end{cases} \qquad \beta = 2. \tag{5.20}$$

Prior regions start with higher scores, but training can override the prior.

(3) Regularise. The pool is unchanged and a penalty discourages scores outside the prior:

$$\mathcal{L} = \mathcal{L}_{\text{split}} + \gamma \frac{1}{|\mathcal{C}_r^{\text{off}}|} \sum_{r' \in \mathcal{C}_r^{\text{off}}} a_{r'r,t}, \qquad \mathcal{C}_r^{\text{off}} = \{r' \in \mathcal{C}_r : r' \notin \pi_r,\, r' \neq r\}, \quad \gamma = 1. \tag{5.21}$$

The target region itself is never biased or penalised in any mode.

These three modes span a range from treating the graph as a constraint, through treating it as a starting point, to treating it as a soft preference. They answer different questions: restrict tests whether the graph contains the right sources; initialise tests whether it is a good starting point for learning; regularise tests whether it is a useful inductive bias that data can still override.

A graph as representation, prior, constraint and predictor. The same object plays four roles that should not be confused. As a scientific *representation*, the graph is a statement about conditional dependence that can be assessed by B1–B4. As a statistical *prior*, it is a belief about which sources are relevant, which data can update. As a computational *constraint*, it reduces the search space of the splitter. Its *predictive utility* is an empirical property of the combination of graph, model and objective. A plausible graph is not automatically a useful prior: an overly restrictive graph removes useful information, a weak graph prior adds noise, and if the splitter's objective rather than its search space is the bottleneck, a random prior of the same size may perform just as well.

5.10 Control Priors

To attribute any effect to the causal content of the graph, each mode is compared with three controls of matched size. For target r and pool $\mathcal{C}_r$, let $K_r = |\pi_r \cap \mathcal{C}_r|$ be the number of prior regions in the pool.

- **No prior:** the splitter uses the full pool with no bias or penalty. This measures what the graph adds over the splitter alone.

- **Random prior:** K_r regions drawn uniformly without replacement from $\mathcal{C}_r \setminus \{r\}$, with the PCMCI strengths randomly permuted onto them, using a seed derived from the run seed and the target. This keeps the prior's size and strength distribution and removes its content. If the random prior matches the causal prior, any gain comes from shrinking the pool, not from choosing the right regions.
- **Correlation prior:** the K_r regions with the largest absolute lagged correlation with the target's OLR, computed on training rows only. This tests whether conditioning (causal discovery) adds anything beyond association.

All priors have the same size per target, the same pool and the same mode, and all runs use identical data, splits, seeds and training budgets.

5.11 Forecast-Time Information Availability

A causal link $X^i_{t-\tau} \to X^j_t$ found by PCMCI states that the source's value τ days before the target carries unique information about it. For a direct forecast issued at time t for the target at $t + L$, the latest available source value is at t, which is L days before the target. Such a link can therefore be exploited directly only if $\tau \geq L$. If $\tau < L$, the source value that carries the information lies in the future at the issue time; the source region may still be useful if its present state predicts its own future state, but the specific link no longer applies. Formally, the information usable at lead L is

$$\mathcal{E}^{(L)}_j = \{(i, j, \tau) \in \mathcal{E} : \tau \geq L\}. \tag{5.22}$$

This condition does not affect the validity of the graph as a description of dependence, but it determines whether the graph is matched to a given forecast lead. A graph whose links are mostly at lags of one or two days is matched to a one-day forecast and not to a seven-day forecast. The remedy is to rebuild the graph with $\tau_{\min} = L$, which restricts PCMCI to links usable at the forecast lead.

5.12 Failure Modes and Ablations

Table 5.2 lists how the build–validate–use chain can fail, how each failure would appear in the results, and the ablation that would diagnose it.

5.13 Temporal Stability of the Causal Structure

Stage D asks whether the causal structure changes over time in a way that tracks known climate regimes. For each target r, PCMCI is rerun on sliding windows of W consecutive BSISO seasons ($W = 1$ or 3), with the candidate set restricted to the target's links that passed B1 and B2 plus its own OLR at lags 1–3, so that each window's test is affordable and focused. The window's

Table 5.2: Failure modes of the graph-guided pipeline and their diagnostics.

Failure mode	Symptom	Diagnostic ablation
Pipeline selects autocorrelated noise	High decoy rate	B1 with alternative surrogates; vary α_{FDR}
Graph lags mismatched to lead	Prior helps at short lead, not at the target lead	Rebuild with $\tau_{\min} = L$; compare leads
Prior too restrictive	Restrict worse than initialise or regularise	Compare the three modes
Gain from pool size, not content	Random prior as good as causal prior	Random-prior control
Gain from association, not conditioning	Correlation prior as good as causal prior	Correlation-prior control
Splitter objective is the bottleneck	No prior, including random, changes skill	Vary objective; synthetic data with known parents
Partition artefact	Low B4 persistence	Different partitions; grid-cell analysis
Data-source artefact	Graph changes with dataset	Second reanalysis or OLR product
Temporal instability	Graph changes between periods	Time-half persistence; Stage D

parent set is the set of significant (source region, field) pairs, $\mathcal{P}_{r,w}$. The causal-signature distance between consecutive windows is

$$\Delta_{r,w} = 1 - J\Big(\mathcal{P}_{r,w}, \mathcal{P}_{r,w-1}\Big), \tag{5.23}$$

and the target-averaged series $\bar{\Delta}_w = \frac{1}{50}\sum_r \Delta_{r,w}$ is tested for regime shifts with STARS [99]. STARS compares each new value with the mean of the current regime and signals a shift when a regime-shift index, the cumulative normalised deviation from that mean, stays positive over a cut-off length. To relate alarms to climate, a Niño-3.4 proxy is computed from the project's SST field (5°S–5°N, 170°W–120°W, season mean, linear trend removed) and each season is labelled El Niño ($> +0.5$ standard deviations), La Niña (< -0.5) or neutral. If the causal structure tracks ENSO, Δ should be larger when the ENSO phase changes between consecutive windows than when it does not, which is tested with a one-sided Mann–Whitney U test.

5.14 Consolidated Algorithm

Algorithm 5.1 gives the complete build–validate–freeze–use procedure. Every step before line 14 uses only the development period.

5.15 Chapter Summary

This chapter defined four validation checks: phase-randomised decoys run as a separate pass, a physical propagation-speed mask with generous per-variable limits, data-driven top-k selection of outside links, and persistence under a nested doubling of the spatial resolution. It then described how the validated links are merged into a region-level prior, how the invariance-trained

Algorithm 5.1 Build, validate, freeze and use the causal graph

Require: 50- and 100-region node series; season mask; development period $\mathcal{T}_{\text{dev}}$; test period $\mathcal{T}_{\text{test}}$

Compute BSISO index and season $\mathcal{S}$ (EOF fitted on $\mathcal{T}_{\text{dev}}$)
for each OLR target j **do** ▷ Stage A
    $\mathcal{E}_j \leftarrow \text{PCMCI}(\mathbf{X}, \mathcal{S} \cap \mathcal{T}_{\text{dev}}, j)$ ▷ Algorithm 4.1
end for
Generate D phase-randomised decoys; run PCMCI on the 550-node pool ▷ B1
$\delta \leftarrow$ decoy rate; **stop** if $\delta \geq 0.30$; $\kappa_{\text{dec}} \leftarrow$ strongest selected decoy
Keep links with $|\hat{\rho}^{\text{MCI}}| > \kappa_{\text{dec}}$
for each link e **do** ▷ B2
    compute d, s_e; flag $\tau \geq 25$; mark fail if $s_e > s_v^{\max}$
end for
Choose k^* by R^2_{val} (fit 1979–2004, score 2005–2012); keep top-k^* outside links per target ▷ B3
Run PCMCI at 100 regions; compute persistence flags and Π_r ▷ B4
Build region prior π_r from B1/B2/B3-passing outside links (Eq. 5.15)
Freeze graph and prior with all flags and configuration
for each pool, prior ∈{none, pcmci, random, corr}, mode ∈{restrict, init, regularise} **do** ▷ Stage C
    Train splitter and experts on $\mathcal{T}_{\text{dev}}$; score on $\mathcal{S} \cap \mathcal{T}_{\text{test}}$
end for
Run sliding-window PCMCI, $\Delta_{r,w}$, STARS and ENSO comparison ▷ Stage D

splitter and per-region experts work, how the prior can restrict, initialise or regularise the splitter, which matched controls are required to attribute an effect to the causal content of the graph, why the graph's lags must match the forecast lead, and how temporal stability is measured. The next chapter describes the implementation and the experimental design.

Chapter 6

Implementation and Experimental Design

6.1 Software and Computational Environment

The system is implemented in Python as a library (`causal_moe`) with one runnable script per experiment and an automated test suite. All experiments run on CPU. Table 6.1 lists the main dependencies with the versions pinned in the project's requirements file.

Table 6.1: Software environment (versions as pinned in `code/requirements.txt`).

Component	Version	Use
NumPy [113]	1.26.4	Arrays, FFT surrogates, vectorised PCMCI
SciPy [114]	1.17.1	Student-t distribution, statistical tests
pandas	2.3.3	Calendar handling, season construction
scikit-learn [115]	1.9.0	k-means partition
PyTorch [116] (CPU)	2.13.0	Splitter and expert models
PyTorch Geometric [117]	2.8.0	Message passing on the region graph
river [118]	0.22.0	ADWIN error-stream drift detector
Matplotlib	3.11.0	Figures
pytest	9.0.3	Automated tests
Tigramite	5.2.10.1	Reference PCMCI for verification only

6.2 Data Pipeline and Preprocessing

The data pipeline consists of four modules. `data/raw.py` loads the six gridded anomaly fields and the ocean mask into a single array of shape $(6, 16071, 25, 144)$. `data/clustering.py` builds the 50-region partition by k-means on OLR, and `data/mesh.py` derives the adjacency of regions, with longitude wrap-around. `causal_graph/nodes.py` forms the 300 regional-mean node series (SST over ocean cells only), re-standardises each node over 1979–2012, constructs the nested 100-region split, and computes closest-cell haversine distances between all pairs of

regions. `causal_graph/season.py` computes the BSISO index and the per-year season mask. `data/windows.py` builds the 49-dimensional feature vectors and lead-L targets and caches them to disk, so that all forecasting experiments read identical inputs.

6.3 PCMCI Implementation and Verification

`causal_graph/pcmci.py` implements Algorithm 4.1. The lagged design for all 300 nodes at 60 lags on the 7,153 season target days is built once (7,153 × 18,000 single-precision values, about 0.5 GB) and shared by all targets. Partial correlations are computed by orthonormalising the conditioning set with a QR decomposition and residualising all candidate columns in one matrix product, processed in chunks of 4,096 columns.

Correctness was verified by dedicated tests (`tests/test_pcmci.py`) on synthetic vector-autoregressive data with known parents, including a common driver and a pure-noise node. The tests check that the implementation recovers the true parents and rejects the common-driver and noise nodes, and they compare it with Tigramite's `PCMCI.run_pcmci` using ParCorr and equivalent link assumptions (any lagged parent for the targets; own past only for the other nodes), for one target and for two targets that are also sources of each other. The PC1 parent sets must be identical, the MCI strengths must agree to within 2×10^{-3} and the p-values to within 5% relative error. All tests pass. The full 50-target Stage A run then takes about 23 minutes on a multi-core CPU. Table 6.2 gives the configuration.

Table 6.2: Stage A (PCMCI) configuration as recorded in the frozen graph file.

Parameter	**50 regions**	**100 regions (B4)**
Nodes	300	596
Candidates per target	18,000	35,760
Targets	50 OLR nodes	100 OLR nodes
Target days	7,153 season days, 1979–2012	
Lags $\tau_{\min}$–$\tau_{\max}$	1–60 days (lag 0 excluded)	
CI test	ParCorr (partial correlation, Student-t)	
α_{PC}	0.05	
PC1 combinations per iteration	1	
Max. conditioning dimension	unlimited	
Non-target node parents	own past only	
Multiple-testing correction	Benjamini–Hochberg over all candidates per target	
α_{FDR}	0.05	

6.4 Surrogate Generation

`causal_graph/validate.py` generates decoys with NumPy's real FFT: the demeaned series is transformed, the zero-frequency phase (and the Nyquist phase for an even length) is fixed at zero, the remaining phases are drawn from $\mathcal{U}[0, 2\pi)$, the inverse transform is taken, and the result is standardised (Eq. (5.1)). The 250 decoys are built from real source nodes drawn with

replacement, using a fixed random seed, and the origin of every decoy is recorded. The decoy pass reuses the PCMCI implementation with the 550-node pool; decoy links are marked with a source region of -1.

6.5 Physical-Mask Implementation

Closest-cell distances are precomputed for all region pairs from cell-centre coordinates with the haversine formula. For each link, the implied speed of Eq. (5.5), the pass/fail result against the limits in Table 5.1, and the oscillation flag for $\tau \geq 25$ days are added to the link record.

6.6 Top-k Selection and Resolution Comparison

The top-k search evaluates $k = 0, \ldots, 10$ by ridge regression with constant 10^{-3}, fitted on 1979–2004 and scored on 2005–2012 season days (Eq. (5.8)). The resolution comparison reruns Stage A on the 100-region nodes and evaluates the persistence rule of Eq. (5.9) with a lag tolerance of ± 2 days. The frozen output, `results/phase1/causal_graph_50.json`, holds the configuration of every stage, the summary statistics, and for each target its validated links (source region, field, lag, strength, p, q, distance, speed and all flags) together with the region-level prior. Table 6.3 gives the validation configuration.

Table 6.3: Stage B configuration.

Check	Setting
B1 decoys	250 phase-randomised surrogates; separate PCMCI pass (550 nodes, 33,000 candidates per target); warning line: decoy rate $\geq 30\%$; strength bar: strongest selected decoy
B2 lag mask	Closest-cell haversine distance; limits SST $1\,\mathrm{m\,s^{-1}}$, OLR and PW $20\,\mathrm{m\,s^{-1}}$, h850, u850 and u200 $50\,\mathrm{m\,s^{-1}}$; flag $\tau \geq 25$ days; failures removed from the prior, kept in the graph file
B3 top-k	Outside links ranked by $\lvert\hat{\rho}^{\mathrm{MCI}}\rvert$; $k \in \{0, \ldots, 10\}$; ridge fit 1979–2004, R^2 on 2005–2012 season days
B4 persistence	100 nested regions (k-means, 2 per region); same field, lag within ± 2 days, any child pair; pass if $\Pi_r \geq 0.5$; flag only
Prior	One entry per source region, strength $= \max \lvert\hat{\rho}^{\mathrm{MCI}}\rvert$

6.7 MoE Integration

The splitter (`splitter/dirgnn.py`) and the expert (`experts/expert.py`) are trained jointly with one Adam optimiser. `causal_graph/prior.py` builds the priors (`pcmci`, `random`, `corr`) and applies the modes (`restrict`, `init`, `regularize`). All prior options are off by default, so that the earlier, prior-free behaviour is reproduced exactly when no prior is requested. Table 6.4 lists the training configuration, which is identical for every condition compared in this thesis.

Table 6.4: Splitter and expert training configuration, identical across all compared conditions.

Setting	Value
Epochs	5
Training samples	first 4,000 contiguous season samples of the training period
Optimiser, learning rate	Adam, 10^{-3}
Batch size	16
Hidden width (splitter encoder, expert MLP)	16
Rationale generator window	10 days
Interventions per sample M	6
Variance weight λ	0 for the first 30% of steps, then linear ramp to 1
Selected fraction ρ_{top}	$\min(0.5,\ 3/\lvert\mathcal{C}_r\rvert)$
Target's own features supplied unconditionally	no (only if selected)
Initialise-mode scale β	2
Regularise-mode weight γ	1
Informative-splitter threshold on σ_a	0.02
Seed	0
Lead times	7 days (primary), 1 day (secondary)
Train / validation / test	1979–2004 / 2005–2012 / 2013–2022
Scored test days	1,989 season target days

6.8 Experimental Conditions and Ablations

The Stage C design crosses three priors (`pcmci`, `random`, `corr`) with three modes (restrict, initialise, regularise) and three pools (direct, 2hop, full), giving 27 prior configurations, plus a no-prior baseline per pool. Each configuration trains one splitter and expert per region, 50 per configuration, or 1,350 trainings for the prior grid. Table 6.5 shows the grid and the configurations reported in this thesis.

Table 6.5: Stage C experimental grid. **measured** = results reported in Chapter 7; *planned* = defined, implemented and unit-tested, results to follow. All cells use identical data, splits, seed and training budget.

Prior	Mode	direct	2hop	full
none	–	**measured** (7 d)	**measured** (7 d)	**measured** (7 d, 1 d)
pcmci	restrict	*planned*	*planned*	**measured** (7 d, 1 d)
pcmci	initialise	*planned*	*planned*	*planned*
pcmci	regularise	*planned*	*planned*	*planned*
random	restrict / initialise / regularise	*planned*	*planned*	*planned*
corr	restrict / initialise / regularise	*planned*	*planned*	*planned*

The full pool is the most informative setting for the first comparison, because it is the one in which the splitter alone fails to discriminate (Section 7.6) and in which a prior has the most room to help. The restrict mode is the strongest form of the prior. The reported comparisons are therefore the most direct test of whether the validated graph can rescue source selection in a large pool. The random-prior control at the same pool and mode is the next comparison in the

programme, because it separates the effect of the graph's content from the effect of a smaller pool. Further ablations defined for the graph itself are listed in Appendix B.4: the graph rebuilt with $\tau_{\min} = 7$, top-k selected by forecast-horizon skill, a full-year graph, persistence across the two halves of the development period, and a second data source.

6.9 Train, Validation and Holdout Discipline

Every decision that could be tuned (the season definition, the PCMCI parameters, the validation thresholds, the choice of k and the training configuration) was fixed using 1979–2012 data only, and the graph was frozen before any forecast on 2013–2022 was scored. Test-period scores were not used to select among configurations. Each reported configuration is a pre-defined cell of the grid in Table 6.5.

6.10 Metrics

6.10.1 Forecast accuracy

For region r with test days $t \in \mathcal{T}_r$ and forecasts $\hat{Y}_t$ of targets Y_t (lead suppressed for brevity):

$$\mathrm{MSE} = \frac{1}{|\mathcal{T}_r|}\sum_t (\hat{Y}_t - Y_t)^2, \qquad \mathrm{RMSE} = \sqrt{\mathrm{MSE}}, \qquad \mathrm{MAE} = \frac{1}{|\mathcal{T}_r|}\sum_t |\hat{Y}_t - Y_t|. \tag{6.1}$$

Because the target is a standardised anomaly, these errors are in units of the target's development-period standard deviation.

6.10.2 Skill scores

Skill is measured relative to two reference forecasts [119, 120]. *Persistence* predicts that the anomaly at the issue time persists, $\hat{Y}^{\mathrm{pers}}_{t+L} = Y_t$. *Climatology* predicts the mean of the training targets, $\hat{Y}^{\mathrm{clim}} = \bar{Y}_{\mathrm{train}}$, which is close to zero for anomalies. The MSE skill scores are

$$S_{\mathrm{pers}} = 1 - \frac{\mathrm{MSE}}{\mathrm{MSE}_{\mathrm{pers}}}, \qquad R^2_{\mathrm{clim}} = 1 - \frac{\mathrm{MSE}}{\mathrm{MSE}_{\mathrm{clim}}}. \tag{6.2}$$

A positive score means the model beats the reference. The two references are complementary. At short leads persistence is very hard to beat and climatology easy; at seven days OLR keeps almost no memory, so persistence becomes a weak reference, while climatology, which is the optimal constant forecast under squared error, becomes the demanding one. Both scores are reported together. The anomaly correlation coefficient

$$\mathrm{ACC} = \frac{\sum_t (\hat{Y}_t - \bar{\hat{Y}})(Y_t - \bar{Y})}{\sqrt{\sum_t (\hat{Y}_t - \bar{\hat{Y}})^2 \sum_t (Y_t - \bar{Y})^2}} \tag{6.3}$$

measures phase agreement independently of amplitude and is part of the evaluation set for future runs.

6.10.3 Source-selection quality

The splitter's discrimination is measured by σ_a, the standard deviation of a target's mean candidate scores over the test days, and summarised as the number of informative splitters ($\sigma_a \geq 0.02$) out of 50. The selected sources and their overlap with the prior are recorded for every region.

6.10.4 Graph metrics

Stage A and B are summarised by link counts by field, region and lag, the decoy rate, the number of B2 failures, the validation R^2 curve, the persistence scores, the prior size per target and the Jaccard distance between windows (Section 5.6).

6.11 Statistical Analysis

Comparisons between two conditions are *paired*: both are evaluated on the same 50 regions and the same test days. For each region the difference of a score between the two conditions is computed, and the distribution of the 50 differences is summarised by the number of regions that improve, the median and the mean. Two tests of a zero median difference are reported: the sign test, which uses only the direction of each difference, and the Wilcoxon signed-rank test, which also uses their ranks. These tests treat regions as independent, which they are not: neighbouring regions share weather and their errors are correlated. Their p-values are therefore indicative and likely too small.

Temporal dependence within a region matters for any test performed on daily errors. Consecutive forecast errors are autocorrelated, so a test that treats the 1,989 test days as independent would overstate significance [44]. The appropriate tools are the Diebold–Mariano test with a heteroscedasticity- and autocorrelation-consistent variance [121] for paired error series, and the moving-block bootstrap [68] for confidence intervals of skill scores. Variability due to training randomness is measured by repeating each configuration over several seeds. These analyses are part of the evaluation protocol for the full grid; the comparisons reported in this thesis use a single seed and are interpreted accordingly. Results that are neutral or negative are reported with the same detail as positive ones.

6.12 Runtime, Memory and Scalability

Table 6.6 gives approximate wall-clock costs on the project's CPU workstation. The dominant cost of Stage C is the number of trainings, $3 \times 3 \times 3 \times 50 = 1{,}350$ for the full grid, which runs in parallel across regions.

Table 6.6: Approximate wall-clock costs (CPU only).

Step	Cost
Stage A, 50 regions (18,000 candidates × 50 targets)	about 23 min
Stages A + B in full (season, A50, decoy pass, A100, validation)	about 2.3–2.5 h on 20 threads
One Stage C configuration (50 regions)	about 20 min on 12 worker processes
One region, one configuration	roughly 0.5–1.5 min
Full Stage C grid (27 configurations)	about 6–8 h on 12 worker processes
Stage D, one window length	about 1 h

6.13 Reproducibility and Artifact Organisation

All results are produced by scripts and saved as JSON files with their configuration; figures are regenerated from these files without retraining. The Phase 1 commands are:

```
python scripts/phase1_build_graph.py # season, A50, B1, A100, validation
python scripts/phase1_run_stage_c.py --pools full --priors none,pcmci --modes restrict
python scripts/phase1_run_stage_c.py --pools full --priors none,pcmci --modes restrict \
    --cache-path cache/windowed_clustered50_lead1_v2.npz --out-dir results/phase1/
        stage_c_lead1
python scripts/phase1_stage_d_drift.py --window-seasons 1
python scripts/phase1_stage_d_drift.py --window-seasons 3
python scripts/generate_phase1_figures.py
```

Appendix C lists the repository structure, the result files behind every number in Chapter 7, and step-by-step instructions.

6.14 Chapter Summary

This chapter described the software, the data pipeline, the verified vectorised PCMCI, the decoy, mask, top-k and persistence implementations, and the integration of the prior into the splitter. It defined the 27-configuration Stage C grid and identified the configurations reported here, the metrics for forecast skill and source selection, the paired statistical analysis and its limits, and the computational costs. The next chapter presents the results.

Chapter 7

Results and Discussion

7.1 Evaluation Overview

This chapter reports the Phase 1 results in the order of the research questions. Every number is taken from a saved result file of the project (Appendix C maps each number to its file), and every figure in this chapter, except where stated, was generated from those files. Table 7.1 summarises the evidence available for each stage.

Table 7.1: Evidence reported in this chapter, by stage.

Stage	Evidence	RQ
Season	BSISO index and season from own data	–
A: discovery	PCMCI graph at 50 regions; walk-through for region 22	RQ1
B: validation	B1 decoys, B2 lag mask, B3 top-k, B4 persistence at 100 regions	RQ2
Pool-size baseline	No-prior splitter in three pools, year-round and season	RQ3 context
C: graph as prior	PCMCI restrict prior vs no prior, full pool, 7-day and 1-day leads	RQ3, RQ4 (partial)
Ablations	Alternative splitter objectives; training budget; season vs year-round	–
D: drift	One- and three-season windows vs ENSO	RQ5

7.2 BSISO Index and Season Diagnostics

The BSISO index reproduced from the project's OLR and u850 anomalies (Section 3.6) has leading modes that explain 11.2% and 7.7% of the combined variance on May–October days of 1979–2012. Lee *et al.* [11] report approximately 10% and 7% for their BSISO1 modes. The small excess is consistent with the smaller domain used here (northern edge at 30°N instead of 40°N), which concentrates the analysis on the region of strongest intraseasonal convection. Table 7.2 gives the comparison. The resulting amplitude has the expected seasonal cycle: low in boreal winter and spring, rising above 1 in May and remaining high until about November (Figure 7.1, top).

Table 7.2: Variance explained by the leading combined EOF modes of the BSISO index.

Mode	This work	Lee et al. (2013) [11]
EOF 1	11.2%	≈10%
EOF 2	7.7%	≈7%

The activity-based season lasts on average 208 days per year, with a range of 82 to 286 days, and begins between April and July (Figure 7.1, bottom). The variability is substantial: a fixed May–October window would include weeks of inactive BSISO in some years and miss active weeks in others. The development years provide 7,153 season target days for discovery, and the test years provide 1,989 scored season target days for the forecasting comparison.

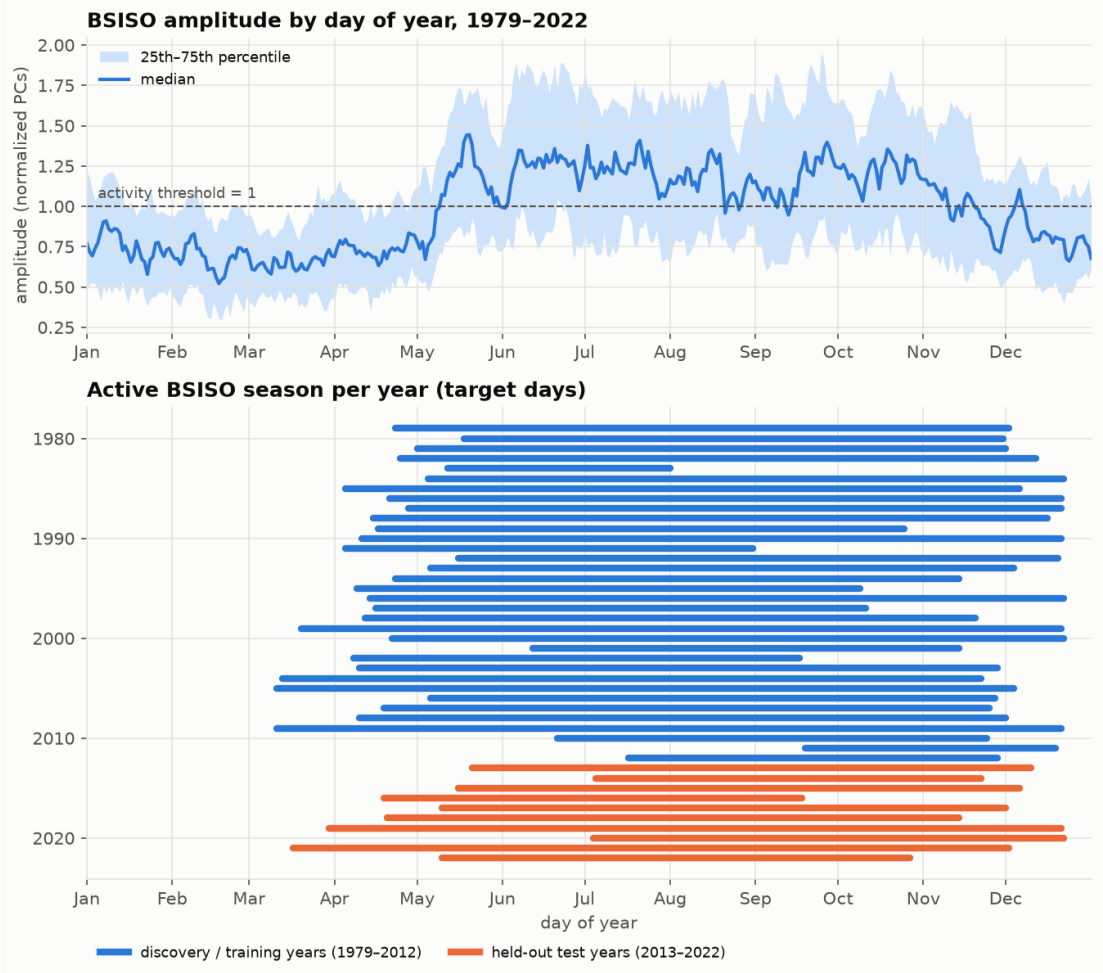


Figure 7.1: BSISO index and activity-based season. Top: median and interquartile range of the BSISO amplitude by day of year over 1979–2022, with the activity threshold of 1. Bottom: the season (target days) of each year; blue for discovery and training years (1979–2012), orange for held-out test years (2013–2022).

7.3 Stage A: PCMCI Discovery (RQ1)

7.3.1 Size and composition of the graph

PCMCI at an FDR of 0.05 retains **536 links** into the 50 OLR targets, 10.7 per target on average, out of 900,000 candidates. The graph is therefore very sparse: about 0.06% of candidates survive. Table 7.3 breaks the links down.

Table 7.3: Composition of the Stage A graph (50 regions, FDR 0.05).

Quantity	**Value**
Significant links	536
Links per target (mean)	10.7
Own-region links	376
Outside links	160
from OLR	72
from u850	37
from u200	22
from h850	18
from PW	11
from SST	0
Median lag of outside links	1 day
Outside links with lag $\leq$ 2 days	90%
Outside links with lag $\geq$ 25 days	0

Three features stand out.

1. **Most links are within the target's own region**, and most of these are the target's own OLR at lags of 1–7 days. This is the expected autoregressive structure of convection: the best single predictor of a region's OLR is its own recent past.

2. **Outside links are short-lag.** The median lag is one day and 90% are at two days or less. No outside link has a lag of 25 days or more. Slow precursors, such as SST one BSISO cycle ahead, do not survive. This is consistent with how PCMCI works: once the target's short-lag parents are conditioned on, the information that a slow precursor carries about the target has largely been transmitted through those faster intermediaries, so the precursor has little *direct* link left. This is a property of conditional rather than marginal dependence, not evidence that slow precursors are irrelevant.

3. **There are no outside SST links.** The atmospheric fields respond to SST over days to weeks, and these responses are captured by the atmospheric nodes themselves. SST's direct contribution, after conditioning on them, is too small to pass FDR.

Among the outside drivers, OLR is the most frequent field, followed by the low-level wind u850, the upper-level wind u200 and the low-level height h850. This ordering matches the physical picture of propagating convection: convection in a neighbouring region, together with

the low-level flow that carries moisture and the upper-level outflow and wave response, precede convection in the target.

7.3.2 Spatial structure

Figure 7.2 maps the validated outside drivers. Most links connect neighbouring regions, along the equatorial Indian Ocean and Maritime Continent, across the western Pacific and across the southern subtropical Pacific. A handful of longer links, such as 22→32, 46→28 and 8→38, are carried by wind and height fields and require 1–2 days at equatorial-wave speeds. The map also shows which links reappear at 100 regions (Section 7.4.4).

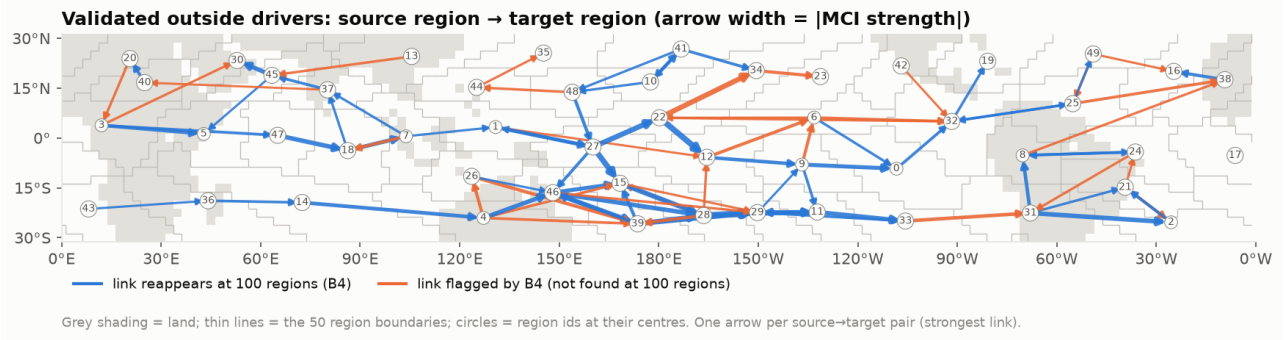


Figure 7.2: Validated outside drivers of the 50 OLR targets. Each arrow joins a source region to a target region (one arrow per pair, strongest link), with width proportional to $|\hat{\rho}^{\mathrm{MCI}}|$. Blue: the link reappears at 100 regions (B4); orange: flagged by B4. Grey shading is land; thin lines are region boundaries; circles mark region identifiers at region centres.

7.3.3 A worked example: region 22

Region 22 lies over the equatorial western Pacific near the Maritime Continent, where several circulation regimes meet. Table 7.4 follows its links through the pipeline. PCMCI finds 13 links: ten within the region (its own OLR at lags 1–3 and 5–7, and its u850 and h850 at lags 1–2) and three outside links (h850 from region 27 at lags 1 and 3, and PW from region 6 at lag 2). Both source regions are about 280 km away at their closest cells, so the implied speeds of 1–3 $\mathrm{m\,s^{-1}}$ are physically easy. The resulting prior is region 27 (strength 0.21) and region 6 (strength 0.05). Both region-27 links persist at 100 regions; the region-6 PW link does not and is flagged, giving a persistence score of 0.67.

Independently of the graph, when the splitter is trained on region 22 with no prior in the direct pool, it selects regions 6, 12 and 27. Both prior regions are among its choices. The two methods use different data representations, objectives and algorithms, so their agreement on the same two drivers is a useful, if anecdotal, piece of corroborating evidence.

7.3.4 Answer to RQ1

PCMCI identifies a sparse set of conditional lagged dependencies whose fields, lags and geography are physically interpretable. The answer to RQ1 is positive, with the qualification that the

Table 7.4: The links of target region 22 through Stages A and B.

Step	Links left	What happened
PCMCI	13	10 own-region links (OLR lags 1–3, 5–7; u850 and h850 lags 1–2); h850 from region 27 at lags 1 and 3; PW from region 6 at lag 2
B1 decoys	13	No decoy selected anywhere; no strength threshold
B2 lag mask	13	Outside sources ≈280 km away; implied speeds 1–3 $\mathrm{m\,s^{-1}}$
B3 top-k	3 outside	All three outside links retained; own-region links do not count against k
B4 persistence	score 0.67	Both region-27 links persist; region-6 PW link flagged
Prior	2 regions	Region 27 (0.21), region 6 (0.05)

identified dependencies are dominated by short-lag, largely regional coupling rather than by long-lead teleconnections.

7.4 Stage B: Validation (RQ2)

Figure 7.3 shows how many links survive each check, and Table 7.5 summarises the outcome.

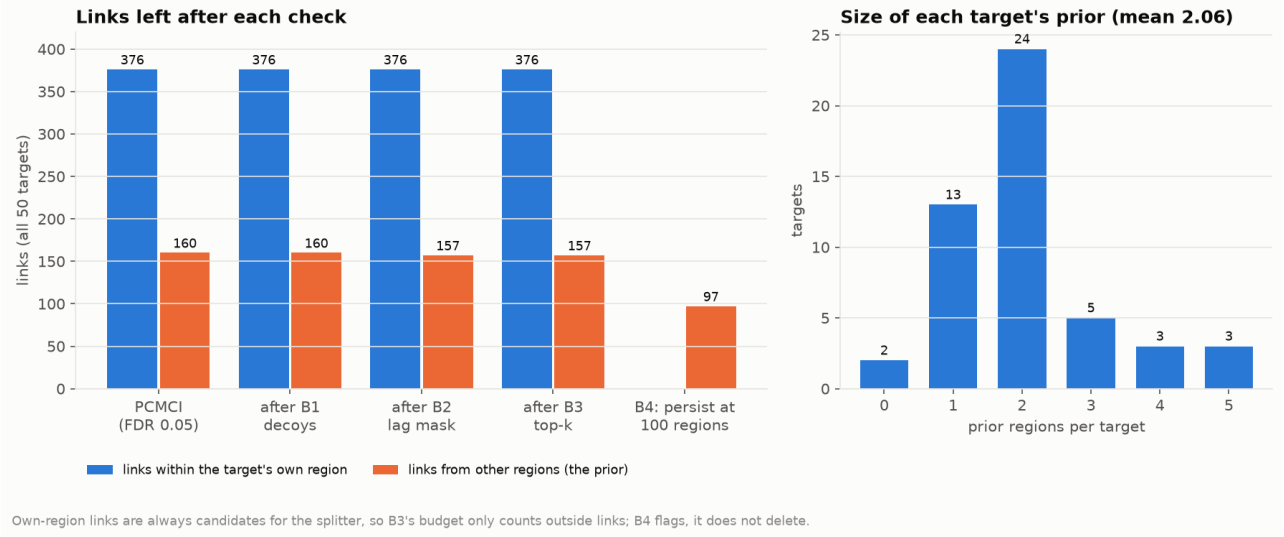


Figure 7.3: Left: links remaining after each validation check, split into own-region links (blue) and outside links (orange). B4 does not delete links; its bar shows how many outside links persist at 100 regions. Right: distribution of the number of source regions in each target's prior (mean 2.06).

7.4.1 B1: decoys

In the separate pass with 250 phase-randomised decoys in the pool (550 nodes, 33,000 candidates per target), **no decoy is selected as a parent of any target**. The decoy rate is 0%, far below the 30% warning line. At the same time, the real part of the graph is almost unchanged: 517 real links (10.3 per target) compared with 536 (10.7 per target) in the clean pass. The small reduction is expected, because the larger pool increases the number of hypotheses in the BH correction and slightly raises the effective threshold. Figure 7.4 shows that the strength

Table 7.5: Outcome of the four validation checks.

Check	Result	Outcome
B1 decoys	0 of 250 decoys selected; real graph 10.3 vs 10.7 links per target with and without decoys	Pass (rate 0% < 30%)
B2 lag mask	3 of 536 links exceed the speed limit (64, 33 and 26 $\mathrm{m\,s^{-1}}$, all at lag 1); no lag $\geq$ 25 days	Pass
B3 top-k	Validation R^2 rises from 0.9619 ($k = 0$) to 0.9633 ($k = 10$); $k^* = 10$, which retains all 157 surviving outside links	Prior size set at 2.06 regions per target
B4 persistence	Mean persistence 0.62; 33 of 48 scored targets ≥ 0.5	Pass on average; 15 targets flagged

distributions of real links in the two passes are nearly identical, including the cluster of strong own-OLR links near $|\hat{\rho}^{\mathrm{MCI}}| \approx 0.85$.

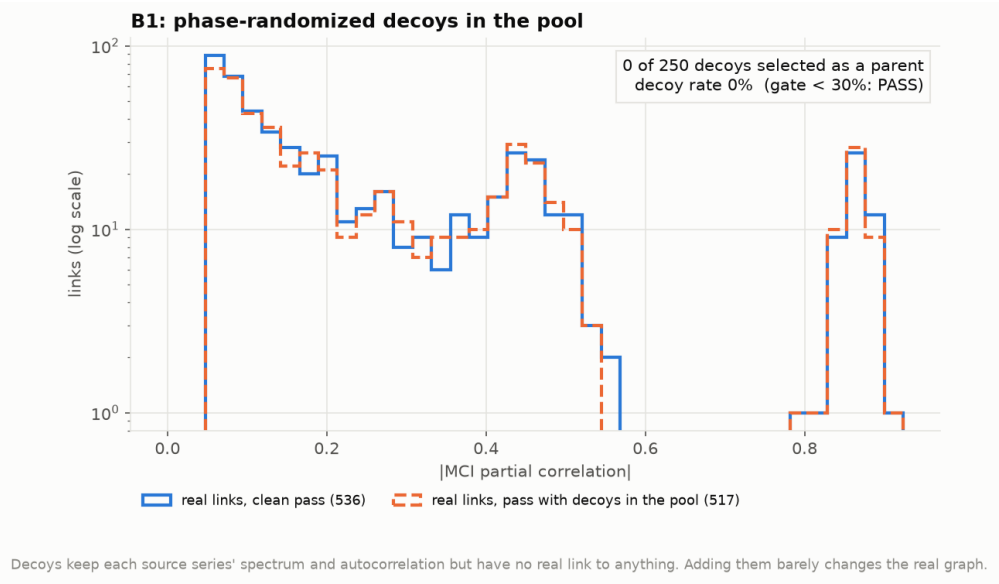


Figure 7.4: B1: distribution of $|\hat{\rho}^{\mathrm{MCI}}|$ for real links in the clean pass (536 links) and in the pass with 250 decoys in the pool (517 links). No decoy was selected, so the decoy rate is 0%.

Because no decoy survives FDR, the strength bar of Eq. (5.3) is zero and removes no real link. The evidence from B1 is therefore carried by the decoy rate and by the stability of the real graph. The result shows that the pipeline does not select series merely because they share the autocorrelation and the 30–60-day spectral peak of real climate nodes. Within the limits of the surrogate null (Section 5.2), this addresses the most common source of spurious links in climate data.

7.4.2 B2: physical lag mask

Only **3 of 536** links need a speed above the limit of their source field: h850 from region 22 to region 32 (5,531 km at lag 1, 64 $\mathrm{m\,s^{-1}}$), OLR from region 28 to region 33 (2,889 km at lag 1, 33 $\mathrm{m\,s^{-1}}$) and OLR from region 3 to region 47 (2,215 km at lag 1, 26 $\mathrm{m\,s^{-1}}$). All three are at lag 1, which is where the speed requirement is highest. They are removed from the prior and kept,

flagged, in the graph file. Of the outside links that pass, 28 span 500–5,500 km, and the fastest of them requires 46 m s^{-1}; all such fast links come from the wind and height fields, consistent with equatorial-wave propagation. Figure 7.5 shows that the implied speeds of OLR and PW links lie mostly below 5 m s^{-1}, the speed of the BSISO envelope, while dynamical fields reach faster speeds at larger distances. No link reaches the 25-day oscillation flag.

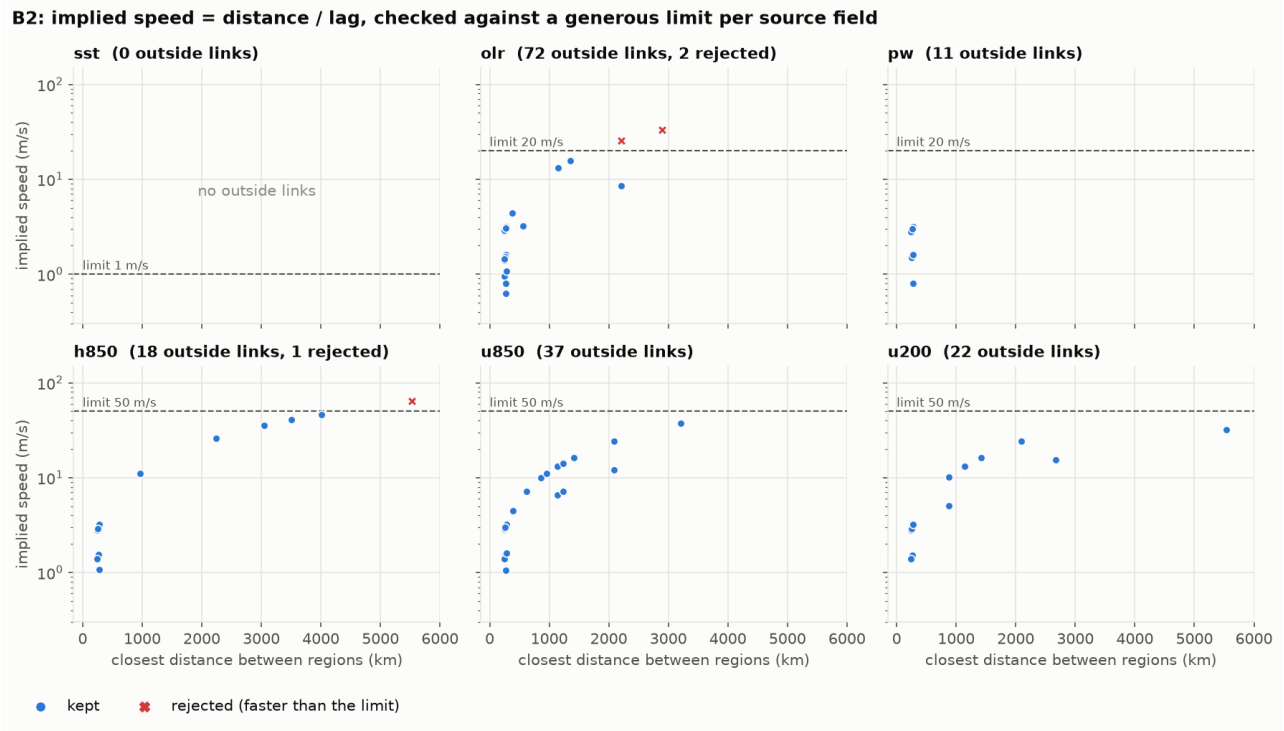


Figure 7.5: B2: implied speed (closest-cell distance divided by lag) against distance for every outside link, by source field, with the speed limit of each field (dashed). Crosses are links that exceed the limit.

7.4.3 B3: top-k selection

The validation R^2 of Eq. (5.8), averaged over targets, increases monotonically with k, from 0.9619 with own-region links only to 0.9633 with ten outside links (Figure 7.6). The gain from each additional outside link decreases from about 3.6×10^{-4} for the first to below 10^{-5} for the ninth and tenth. The maximum is at $k^* = 10$, which equals the largest number of outside links of any target, so B3 retains all 157 outside links that passed B2.

Two observations explain this outcome. First, PCMCI with FDR is already sparse: with about three outside links per target on average, there is little to prune. Second, the high baseline R^2 shows that own-region links, evaluated at their own lags, explain most of the target's variance, and every outside link adds a small amount of independent information on development data. B3 therefore confirms that no target has an excessive number of drivers and fixes the prior at its natural size of **2.06 source regions per target**, distributed as shown in Figure 7.3 (right): 2 targets have no outside driver (regions 13 and 42), 13 have one, 24 have two, and 11 have three to five.

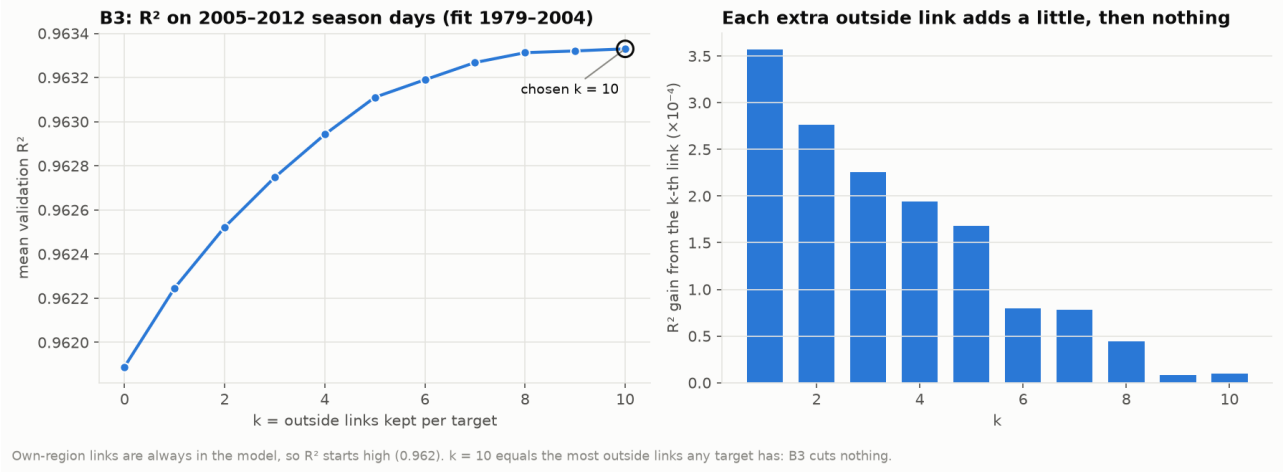


Figure 7.6: B3: mean validation R^2 on 2005–2012 season days (fit 1979–2004) as a function of the number k of outside links per target (left), and the gain from the k-th link (right). Own-region links are always included, so R^2 starts high.

7.4.4 B4: resolution persistence

At 100 regions, the mean persistence score across the 48 targets that have outside drivers is **0.62**, and **33 of 48** targets reach the 0.5 threshold (Figure 7.7). Of the 157 outside links, 97 reappear at the finer resolution. Persistence differs markedly by source field: OLR links persist best (56 of 70), followed by u200 (14 of 22), u850 (17 of 37), h850 (6 of 17) and PW (4 of 11).

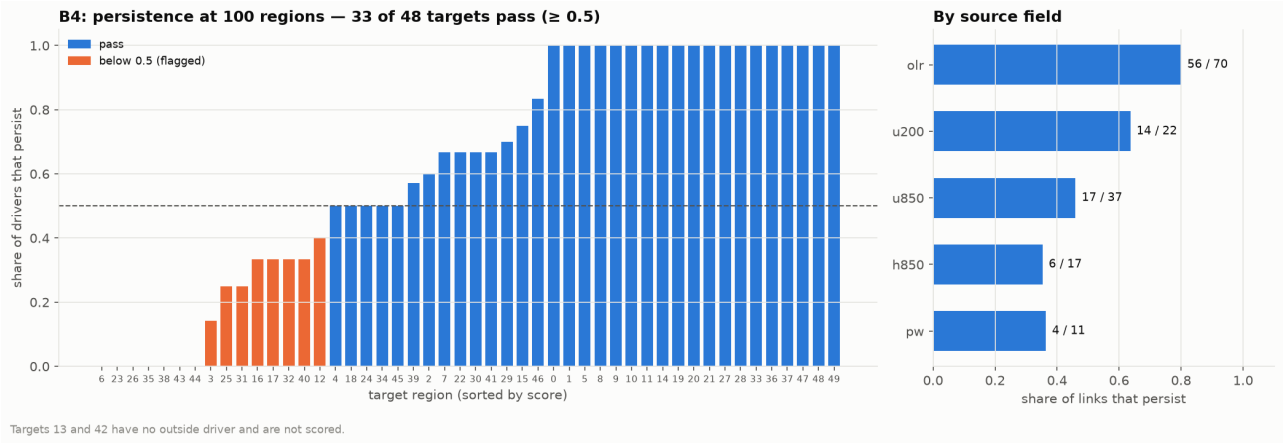


Figure 7.7: B4: persistence score of each target at 100 regions, sorted (left; orange: below 0.5, flagged), and the share of links that persist by source field (right). Targets 13 and 42 have no outside driver and are not scored.

The field dependence is physically sensible. Links between convective regions (OLR to OLR) reflect the large-scale propagation of convective envelopes and survive a change of partition. Upper-level wind anomalies are large in scale and also persist well. Moisture (PW) and low-level height links are more local and noisier at sub-region scale, so they are more sensitive to how regions are drawn. Seven targets have a persistence score of zero; these are mostly targets with one or two weak outside links.

7.4.5 Answer to RQ2

The graph passes the decoy test with no decoy selected, the physical-speed test with 3 failures in 536 links, and the persistence test on average with 62% of outside links reappearing at doubled resolution. B3 confirms that the graph is already compact. Hypotheses 1 and 2 are supported. The answer to RQ2 is that the discovered relationships are robust under every check that can reject them, with the qualification that persistence is weaker for moisture and low-level height links, and that these checks bound the rate of certain kinds of error rather than prove the links causal.

7.5 Graph Properties: Sparse, Short-Lag and Regional

Taken together, Stages A and B describe the dependence structure of BSISO-season OLR at the scale of 50 regions as follows:

- **Sparse:** about ten links per target, of which about three come from other regions, merging into about two source regions.
- **Short-lag:** outside links act at one or two days; own-region memory extends to about a week.
- **Regional:** most outside drivers are neighbouring regions; the few remote links are carried by dynamical fields at wave speeds.
- **Convectively coherent:** OLR-to-OLR links dominate and are the most persistent.

This structure has a direct implication for forecasting. By the availability condition of Eq. (5.22), almost none of the outside links can be exploited directly by a seven-day forecast, whereas nearly all of them can be exploited by a one-day forecast. The graph identifies *which* regions are coupled to a target; the coupling acts on time scales of a day or two.

7.6 Source Selection Without a Prior: Pool Size and Discrimination

Before using the graph, it is necessary to establish what the splitter achieves on its own. Table 7.6 reports the no-prior splitter in the three candidate pools at the 7-day lead, both year-round (from the project's earlier experiments on the same data) and on BSISO-season days (the Phase 1 baseline).

The pattern is the same in both settings (Figure 7.8). Forecast skill relative to persistence is essentially independent of the pool, at about +0.40, and every region beats persistence. The splitter's ability to discriminate between candidates, however, collapses as the pool grows: from 27–30 informative regions in the direct pool to 8–9 in the 2hop pool and none in the full pool. The splitter must score every candidate in one pass with a shared network, and with 50 similar-looking candidates its scores converge to a common value. Accuracy is maintained

Table 7.6: No-prior splitter by candidate pool, 7-day lead, 50 regions, test period 2013–2022. Informative: splitters with $\sigma_a \geq 0.02$.

Days	Pool	Informative	Beat persistence	Mean S_{pers}	Beat climatology	Mean R^2_{clim}
Year-round	direct	27/50	50/50	+0.401	–	−0.014
	2hop	8/50	50/50	+0.395	–	−0.023
	full	0/50	50/50	+0.396	–	−0.020
Season	direct	30/50	50/50	+0.412	15/50	−0.020
	2hop	9/50	50/50	+0.413	18/50	−0.018
	full	0/50	50/50	+0.415	14/50	−0.015

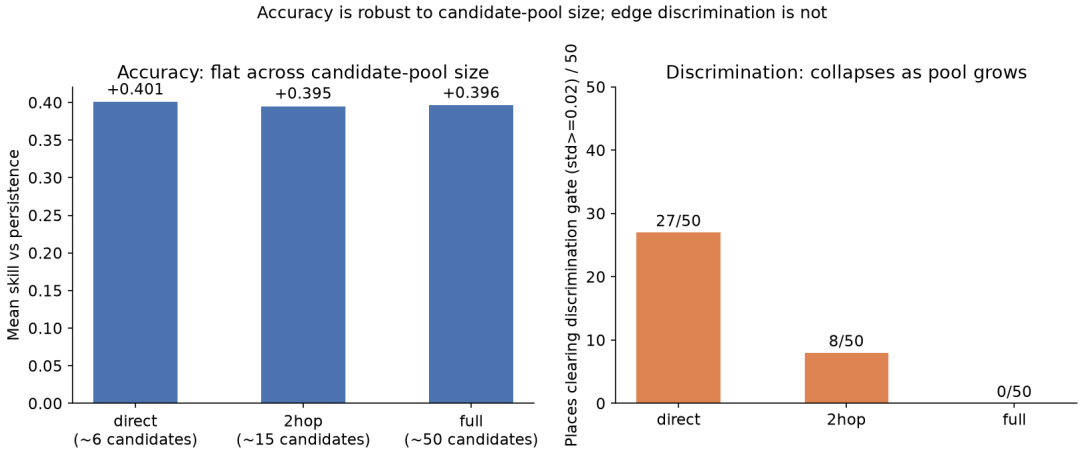


Figure 7.8: No-prior splitter, year-round, 7-day lead: forecast skill relative to persistence is flat across the three candidate pools (left), whereas the number of regions with informative splitter scores collapses as the pool grows (right).

because the expert can still extract information from whichever sources are passed to it, but the selection is no longer meaningful.

This result defines the problem that the causal prior is meant to address: in a large pool, where remote drivers can be considered, the splitter alone cannot tell which candidates matter. In the project's earlier experiments on region 22, shrinking the full pool to its eight most correlated candidates did not restore discrimination, which already suggested that *how* the pool is reduced matters.

The table also shows that the model loses to climatology on average at seven days ($R^2_{\text{clim}} \approx -0.015$ to -0.023), although 14–18 of 50 regions beat it on season days. Beating persistence is a low bar at seven days, because OLR keeps almost no memory over a week; climatology is the demanding reference at this lead.

7.7 Stage C: The Causal Graph as a Splitter Prior (RQ3, RQ4)

7.7.1 Seven-day lead

Table 7.7 and Figure 7.9 compare the PCMCI prior in restrict mode with the no-prior splitter in the full pool.

Table 7.7: PCMCI prior (restrict) versus no prior, full pool, 2013–2022 season days (1,989 days, 50 regions, single seed).

Lead	Prior	Informative	Beat pers.	Mean S_{pers}	Beat clim.	Mean R^2_{clim}
7 days	none	0/50	50/50	+0.415	14/50	−0.0153
	PCMCI	**24/50**	50/50	+0.416	19/50	−0.0151
1 day	none	0/50	0/50	−1.527	50/50	+0.720
	PCMCI	**36/50**	3/50	**−1.327**	50/50	**+0.745**

At seven days the prior has **no measurable effect on skill**. Per region, 23 improve and 27 worsen in R^2_{clim}; the median change is -0.010, the mean change -0.0002 in R^2_{clim} and $+0.001$ in S_{pers}, and the Wilcoxon signed-rank test gives $p = 0.56$. The number of regions that beat climatology rises from 14 to 19, but with mean R^2_{clim} unchanged this reflects small movements of regions near zero rather than a systematic gain. Hypothesis 3 is not supported.

The prior does have a large effect on *source selection*: the number of informative splitters rises from 0 to 24 of 50. With the prior, the restricted pool contains the target and its prior regions, about three candidates instead of fifty, which is the regime in which the splitter can discriminate (Section 7.6). This shows that the graph turns an uninformative full-pool splitter into one that makes distinct choices, without loss of accuracy. It is, however, expected from the pool size alone, and it does not by itself show that the graph's choice of regions is better than another choice of the same size.

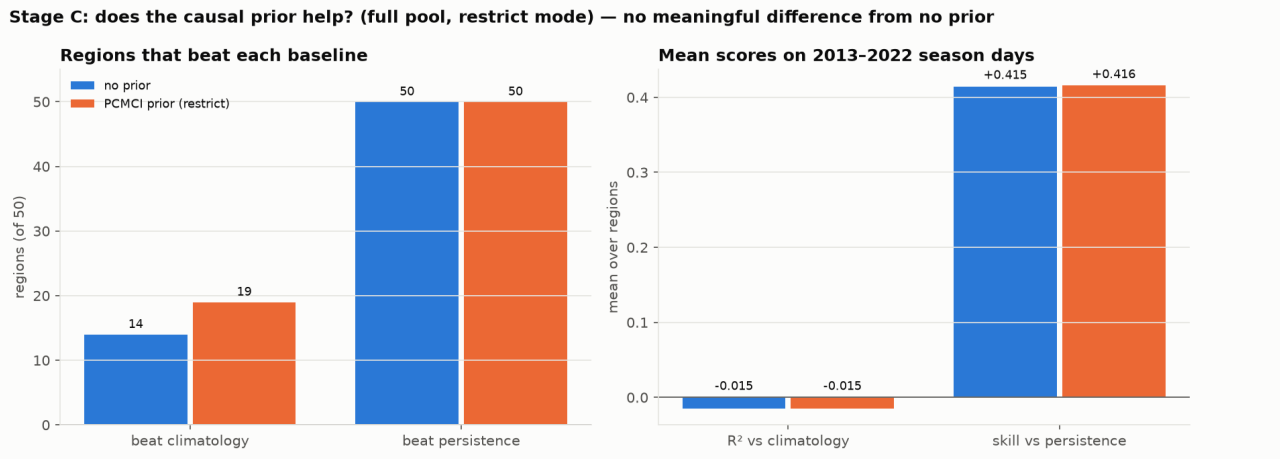


Figure 7.9: Stage C at the 7-day lead, full pool: number of regions that beat climatology and persistence (left) and mean skill scores (right), with and without the PCMCI restrict prior.

7.7.2 One-day lead

At one day, the picture changes (Table 7.7, Figure 7.10). The prior improves skill relative to persistence in **32 of 50 regions**, with a median improvement of +0.40 in S_{pers} and +0.048 in R^2_{clim}; the mean S_{pers} improves from −1.527 to −1.327 and the mean R^2_{clim} from 0.720 to 0.745. Across regions, the Wilcoxon signed-rank test gives $p = 0.049$ for S_{pers} and the sign test $p = 0.065$. The median ratio of the expert's MSE with and without the prior is 0.83. Region 22, the worked example, improves from $S_{\text{pers}} = -0.60$ to -0.05. The number of informative splitters rises from 0 to 36.

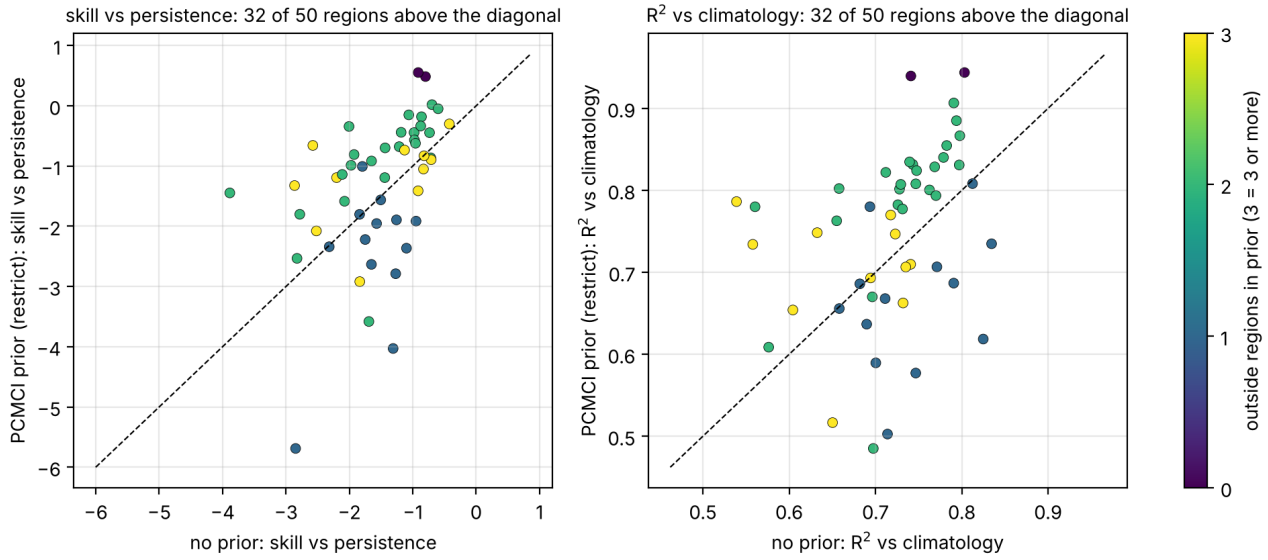


Figure 7.10: Stage C at the 1-day lead, full pool: per-region skill with the PCMCI restrict prior against skill without a prior, for skill relative to persistence (left) and R^2 relative to climatology (right). Points above the diagonal improve with the prior; colour shows the number of outside regions in the prior. Generated for this thesis from the saved Stage C result files.

Negative skill against persistence at one day is expected: OLR tomorrow is very close to

OLR today, and neither model fully exploits this because the target's own history reaches the expert only when the splitter selects it. Both models are nonetheless clearly skilful relative to climatology ($R^2_{\text{clim}} > 0.7$ on average, 50 of 50 regions).

7.7.3 Is the one-day gain due to the graph or to the smaller pool?

Table 7.8 groups the regions by the number of outside regions in their prior. The two targets with an empty prior, which are restricted to their own region only, improve most: their mean S_{pers} changes from -0.857 to $+0.519$, and they account for two of the three regions that beat persistence with the prior. Targets with a single outside driver become worse on average, from -1.628 to -2.477, whereas targets with two or more improve.

Table 7.8: Mean 1-day skill relative to persistence by number of outside regions in the prior, full pool.

Outside regions in prior	Targets	No prior	PCMCI restrict
0	2	-0.857	$+0.519$
1	13	-1.628	-2.477
2	24	-1.528	-0.908
≥ 3	11	-1.530	-1.218

This pattern shows that the size of the restricted pool, and in particular whether the target's own region is effectively forced into the forecast, strongly influences the outcome. A target restricted to itself is a near-persistence model, which is very hard to beat at one day. A target restricted to itself and one other region has a selected fraction of $\rho_{\text{top}} = 0.5$, so the splitter selects exactly one of the two, and when it selects the outside region the target's own strong persistence is lost. These effects are properties of the restricted pool, not of which regions the graph chose. Consequently, the one-day improvement over the no-prior model is established, but its attribution to the causal content of the graph requires the matched random-prior control, in which each target receives the same number of randomly chosen regions in the same mode and pool. That comparison answers RQ4 and is the first item of the continuing programme (Section 9.8).

7.7.4 Why the graph helps at one day and not at seven

The contrast between leads follows from the graph's structure. Its outside links act at one or two days. At a one-day lead, a source region's state at the issue time is exactly the information that the link describes, so restricting the forecaster to the graph's sources concentrates it on relevant, usable information. At a seven-day lead, the same sources are coupled to the target on a time scale much shorter than the lead, and their state at the issue time says little more about the target a week later than the other candidates do. The graph identifies the right *neighbours* but not the right *precursors* for a week ahead. A graph built with $\tau_{\min} = 7$, which

tests only links usable at that lead, is the direct test of whether week-ahead precursors exist in this representation.

7.7.5 Answers to RQ3 and RQ4

For RQ3, at the seven-day lead for which the graph was intended, the validated graph does not change forecast skill, while it does make the splitter's source selection discriminative. At one day, it improves skill over the no-prior model in a majority of regions. For RQ4, the comparison with the no-prior model is established; the comparisons with random and correlation priors of equal size, which determine whether the causal content of the graph is responsible, remain to be measured. The core Phase 1 hypothesis, that a validated causal graph improves source selection *and* forecast skill on unseen years, is therefore not yet confirmed: the graph is reliable, and its forecasting benefit is so far demonstrated only against the no-prior model and only at short lead.

7.8 Regional and Temporal Breakdown

At seven days the per-region changes are small and symmetric around zero, with no consistent geographic pattern. At one day, improvements are spread across the domain and concentrate among targets with two or more prior regions; deteriorations concentrate among targets with a single prior region. Region-by-region differences come from a single training seed and include seed-to-seed variability that has not yet been quantified. A breakdown by test year and by BSISO phase would require enough scored days per category to give stable estimates; with 1,989 test days over ten seasons, a year-by-year breakdown is possible but noisy, and it is included in the protocol for the multi-seed runs.

7.9 Ablation Analysis

7.9.1 Alternative splitter objectives

Two alternative selection mechanisms adapted from the invariant-learning literature, one sharpening score discrimination directly (CIA) and one using iterative sparsification (GSINA), were compared with the plain invariance-trained splitter on region 22 under identical settings (Table 7.9). The plain splitter has the lowest forecast error and is the only one that clears the discrimination threshold, so it is the splitter used throughout.

7.9.2 Training budget

With the direct pool at seven days, training for 2 epochs instead of 5 reduces the number of informative splitters from 27 to 5 of 50, while forecast skill barely changes. Discrimination therefore needs a sufficient training budget, and evaluating a selection mechanism after too

Table 7.9: Alternative splitter mechanisms, region 22, direct pool, identical settings.

Splitter	**Forecast MSE**	σ_a **(informative** $\geq$ **0.02)**
Plain invariance-trained splitter	**0.3207**	**0.0203**
CIA (weight 1.0)	0.3324	0.0026
GSINA (20 iterations)	0.3364	0.0087

little training can make it appear to fail. All Phase 1 comparisons use the same budget of 5 epochs.

7.9.3 Season versus year-round training

Restricting target days to the BSISO season changes the no-prior results only slightly (Table 7.6): skill relative to persistence rises from about +0.40 to +0.41, and the number of informative splitters in the direct pool from 27 to 30. The seasonal restriction is therefore not the source of any large change in the forecasting results; its main role is to make the graph specific to the BSISO regime.

7.9.4 Graph-level ablations

The validation pipeline itself provides ablations: removing the decoys changes the graph by 19 links (536 vs 517); removing B2 would add 3 links to the prior; B3 does not remove links at $k^* = 10$; and B4 identifies 60 of 157 outside links as resolution-sensitive. The graph-level ablations listed in Section 6.8 extend this analysis.

7.10 Stage D: Temporal Stability of the Causal Structure (RQ5)

The Niño-3.4 proxy computed from the project's own SST matches the known ENSO phase in 22 of 24 labelled years; 1989 and 2008 are misclassified. Table 7.10 and Figure 7.11 report the causal-signature distance between consecutive windows.

Table 7.10: Stage D: causal-signature distance between consecutive windows and its relation to ENSO phase changes.

Window	**Parents per window**	$\bar{\Delta}$**: ENSO phase changed vs not**	**Mann–Whitney** p	**STARS alarms**
1 season	1.1–1.3	0.164 vs 0.151	0.20	Mean series: 2004, 2011, 2016. Per target: 55 alarms, 76% in phase-change years (67% of years have one); 13 in stable years
3 seasons	1.6–2.0	0.200 vs 0.207	0.76	Mean series: none. Per target: 21 alarms, 43% in phase-change years; 12 in stable years

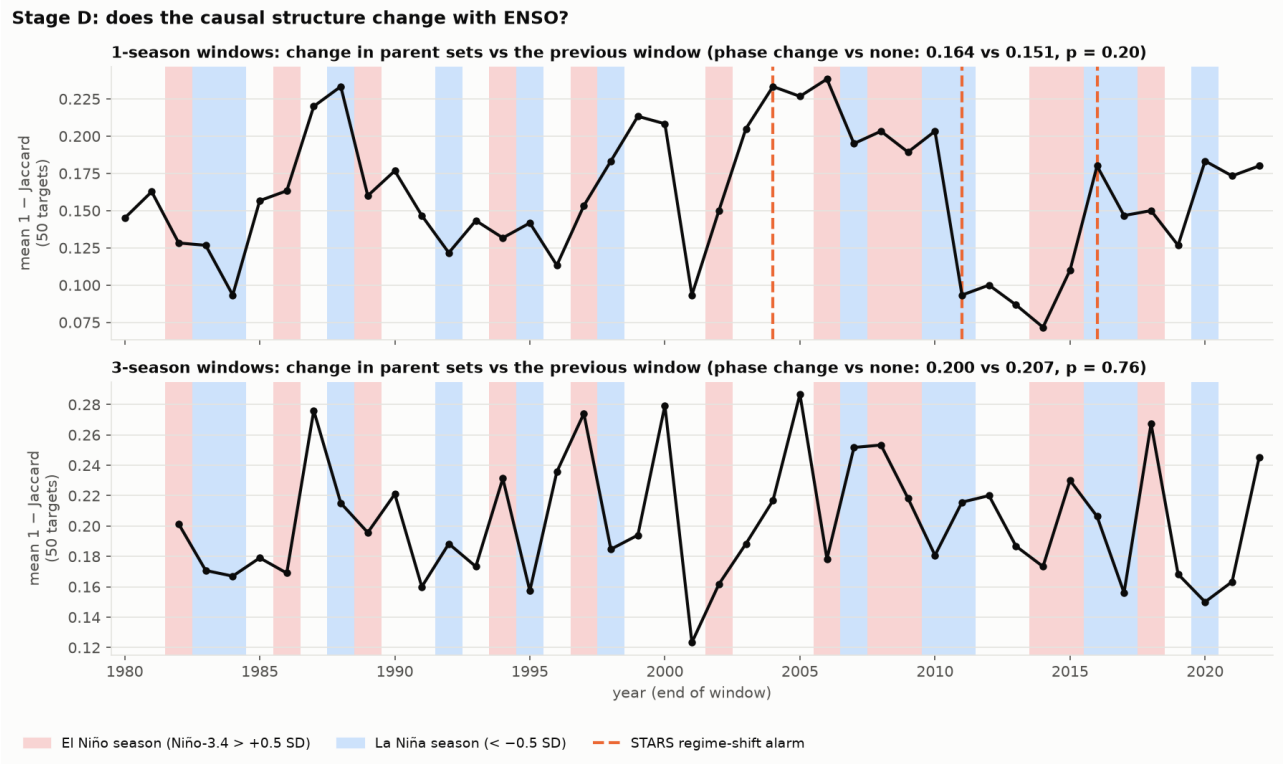


Figure 7.11: Stage D: mean causal-signature distance (1−Jaccard of parent sets, averaged over 50 targets) between consecutive one-season (top) and three-season (bottom) windows, with El Niño (red) and La Niña (blue) seasons shaded and STARS regime-shift alarms marked.

The distance is not significantly larger when the ENSO phase changes than when it does not, for either window length. With one-season windows, 76% of per-target alarms fall in years with a phase change, but 67% of all years have one, so the alarms are only marginally more frequent than chance. With three-season windows, the relation disappears. The 2016 alarm of the mean series coincides with the end of the 2015–16 El Niño, but a single coincidence is not evidence.

The reason is visible in the parent counts: each window has only one or two parents per target, mostly the target's own OLR plus whichever outside link happens to pass FDR in that window. With so few parents, the set changes from window to window mainly through sampling noise, and Jaccard distance between such small sets is dominated by single additions or removals. Hypothesis 5 is not supported. As constructed, the causal-signature distance is not reliable enough to trigger expert adaptation, which shapes the Phase 2 design (Chapter 8). Longer windows with a fixed or rolling reference, and a distance based on link strengths rather than set membership, are the natural refinements.

7.11 Computational Cost and Scalability

The vectorised PCMCI makes the full-pool analysis practical: 900,000 candidates for 50 targets in about 23 minutes, and the complete graph pipeline, including a 550-node decoy pass and a 596-node second resolution, in about 2.5 hours on one workstation. The cost grows roughly linearly with the number of candidates per target and with the number of targets, which makes a

100-region or multi-dataset analysis feasible on the same hardware. The forecasting experiments cost about 20 minutes per 50-region configuration with 12 workers; the full 27-configuration grid is a few hours of computation, and multi-seed repetition multiplies this by the number of seeds.

7.12 Error Analysis and Failure Cases

Seven-day forecasts and climatology. At seven days, every model loses to climatology on average by about 0.015–0.02 in R^2. A seven-day OLR anomaly in a single region of about 275 km scale is difficult to predict, and a small model with a 4,000-sample training budget cannot exploit the slow, weak signals that remain. The relevant comparison in this thesis is between source-selection strategies under identical conditions, but the absolute level indicates the room that a better forecaster or a lead-matched graph has to work with.

Targets without outside drivers. Regions 13 and 42 have no validated outside driver. Their restricted forecasters see only their own region, which at one day is the strongest information available and at seven days is little better than any other.

Targets with a single driver. At one day, the 13 targets with one outside driver deteriorate under the restrict prior. With two candidates and $\rho_{\text{top}} = 0.5$, the splitter must choose between the target's own history and its single driver, and choosing the driver discards the strong persistence signal. This is a property of the interaction between the selection rule and a very small pool, and it suggests that the initialise and regularise modes, or a rule that always allows the target's own history, may be better suited to very small priors.

Resolution-sensitive links. Fifteen targets fall below the persistence threshold, and PW and h850 links persist least. These links are kept in the prior with a flag. A stricter prior that excludes B4-flagged links is a further variant that can be tested with the same infrastructure.

7.13 Interpretation of Positive, Neutral and Negative Outcomes

Table 7.11 summarises the outcome of each hypothesis.

The neutral and negative outcomes are informative. The neutral seven-day result, combined with the short lags of the graph, points to a mismatch between the time scale of the discovered dependencies and the forecast lead, rather than to a failure of the graph or of the splitter. The negative drift result shows that set-based causal signatures from short windows are dominated by sampling noise. Both findings have concrete consequences for the method, which are carried into Phase 2.

Table 7.11: Outcome of the Phase 1 hypotheses.

Hypothesis	Evidence	Outcome
H1 Sparsity and credibility	Decoy rate 0%; 3 of 536 links (0.6%) exceed speed limits	Supported
H2 Resolution persistence	33 of 48 targets with $\Pi_r \geq 0.5$; mean 0.62	Supported
H3 Predictive usefulness (7 days)	No change in skill (Wilcoxon $p = 0.56$); discrimination $0 \rightarrow 24$	Not supported
at 1 day (secondary)	Skill improves in 32 of 50 regions; discrimination $0 \rightarrow 36$	Supported vs no prior
H4 Causal specificity	Random- and correlation-prior comparisons pending	Open
H5 Regime sensitivity	$p = 0.20$ and 0.76	Not supported

7.14 Threats to Validity and Limitations

No ground truth. The validation checks can reject a bad graph but cannot prove a graph correct. Every causal statement in this thesis is a statement about conditional dependence under the stated assumptions.

Statistical power. With 18,000 candidates per target on strongly autocorrelated daily data, some real links will be missed and some false links retained, even with FDR control.

Hidden common drivers. PCMCI assumes that all common drivers are among the nodes. BSISO phase, ENSO, extratropical influences and sub-regional moisture processes are only partly captured by 300 regional means.

Field choice within a region. The fields of a region move together with the same wave, so which field PCMCI selects can change with small changes in the data. Claims are therefore made at the level of source regions.

Partly circular regions. Regions were clustered on OLR co-variability and then used to find OLR links. The effect is to make between-region OLR links harder to find, but the partition is not independent of the target.

Climatology overlap. The upstream anomaly climatology was computed over 1979–2014, which overlaps the test period by two years. All compared models share it, so the comparisons between them are unaffected, but absolute skill values may be very slightly optimistic.

Single seed. All forecasting comparisons use one training seed. Per-region differences of the size seen at seven days are within the range that seed variability can produce, and the one-day differences, although consistent across most regions, should be confirmed with several seeds.

Weak seven-day skill. At seven days the models lose to climatology on average, so the comparison between priors takes place in a regime of low absolute skill.

Region-scale statistical claims. A statement such as "h850 in region 27 drives OLR in region 22 at one to three days" is a statistical statement about region means, not a description of a physical mechanism.

External validity. The graph is estimated from one data archive, one partition and one season definition. Its stability across data sources, time periods and the full year is part of the continuing programme.

7.15 Implications for Phase 2

The Phase 1 results lead to three groups of conclusions for the India-focused extension:

- **Keep:** the validation pipeline (decoys, physical lag mask and resolution persistence), which runs at scale and gives interpretable diagnostics; the activity-based season restriction for target days; and a region-level prior, which matches the granularity at which source claims are reliable.
- **Change:** build the graph with lags matched to the forecast horizon ($\tau_{\min} \geq L$); select the prior size by forecast-horizon skill or by comparison with random sources; and keep the target's own history available when the prior is very small.
- **Do not yet rely on:** set-based causal-signature distance from short windows as a trigger for expert adaptation; error-based drift signals, or strength-based structural distances over longer windows, are better candidates.

7.16 Chapter Summary

The Phase 1 graph is sparse, short-lag and largely regional, and it passes every validation check that can reject it: no decoy is selected, 3 of 536 links violate physical speed limits, and 62% of outside links persist at doubled resolution. Used as a restrict prior in the full pool, it makes the splitter's source scores discriminative at both leads. It leaves seven-day skill unchanged and improves one-day skill over the no-prior model in 32 of 50 regions. Whether the one-day gain reflects the graph's choice of regions, rather than the reduced pool, is determined by the matched random-prior control. The causal structure estimated from short windows does not track ENSO. These results answer RQ1 and RQ2 positively, answer RQ3 with a lead-dependent result, leave RQ4 open, and inform RQ5, which the next chapter develops.

Chapter 8

Phase 2: India-Focused Forecasting and Adaptive Experts

This chapter describes Phase 2, which carries the Phase 1 lessons into an India-focused regional forecasting system with causal gating and adaptive experts. It combines the regional-forecasting and adaptation components of the research plan into one extension. Section 8.12 separates what has been implemented from what is designed and what is future work; unless stated otherwise, the content of this chapter is a design specification.

8.1 Motivation: From Regional OLR to Forecasting over India

Phase 1 studied a deliberately simplified problem: a single convective proxy, daily resolution, 50 large regions and a small forecaster. This made it possible to test the causal graph with full control. Its findings, however, matter most if they help a forecast that people use. India is a natural setting for the extension:

- The BSISO studied in Phase 1 drives the active and break spells of the Indian summer monsoon [19, 20], so the large-scale drivers identified there are physically relevant.
- High-resolution regional reanalysis (IMDAA [104]) and dense station records [105] make kilometre-scale, hourly evaluation possible.
- Near-surface heat, humidity and wind over India have strong societal impact, and their predictability changes with monsoon regime and ENSO state [30], which makes adaptation to shift a real requirement.

The question for Phase 2 is: *can the causal architecture contribute to a competitive hourly forecasting system over India up to 72 hours ahead, and can it adapt when the regional climate relationships change?*

8.2 Forecasting Targets, Resolution and Horizon

Table 8.1 lists the planned targets. Gridded targets are near-surface fields on a regional grid of about 0.12° (about 12 km), matching the IMDAA reanalysis. Point targets are station observations, used to evaluate the gridded forecasts where they matter most and, optionally, as direct forecast targets.

Table 8.1: Planned Phase 2 forecast targets.

Target	Type	Notes
2 m temperature	Gridded, station	Diurnal cycle and heat-wave relevance
2 m relative humidity	Gridded	Heat stress; monsoon moisture
10 m zonal and meridional wind	Gridded	Monsoon flow; wind energy
Mean sea-level pressure	Gridded	Synoptic systems, monsoon trough and depressions
2 m dewpoint	Station	Moisture at observation sites
Temporal resolution: hourly. Horizon: 1–72 hours. Domain: Indian region at about 0.12° (≈12 km).		

Two features distinguish this task from Phase 1. The horizon is short (up to three days), so the short-lag dependencies that Phase 1 found are exactly the ones that matter. And the targets are dominated by the diurnal cycle and by local surface conditions, so a strong shared model of local dynamics is required before any remote driver can add value.

8.3 Proposed Datasets and Their Roles

Table 8.2 lists the candidate datasets. Their versions, periods, licences and acquisition status must be confirmed before they are treated as final choices; the decision log (Appendix C.4) records them as open.

Table 8.2: Candidate Phase 2 datasets. All entries are planned; versions, periods and licences are to be confirmed.

Dataset	Characteristics	Role
IMDAA [104]	Regional reanalysis for the Indian monsoon region; about 12 km; hourly; satellite era	Primary gridded inputs and targets over India
ERA5 [93]	Global reanalysis; about 31 km (0.25°); hourly; 1940 onward	Boundary context outside the regional domain; source of candidate large-scale drivers (for example BSISO, ENSO and Indian Ocean indices, remote circulation)
NOAA ISD [105]	Hourly and synoptic station observations worldwide	Independent point verification; station targets (2 m temperature, dewpoint)

Station data require explicit quality control (duplicate removal, outlier and gap handling, station relocation checks) and a policy for matching stations to grid cells. Reanalysis data require consistent regridding, unit harmonisation and alignment of hourly time stamps. As in Phase 1, all statistics used to normalise the data must be estimated on training years only.

8.4 Shared Forecasting Backbone

Phase 1 used one small expert per region and no shared model. At kilometre scale and hourly resolution this would discard the common physics shared by all locations. Phase 2 therefore introduces a shared backbone B_ϕ that maps the recent regional state to a latent representation for each grid node,

$$\mathbf{z}_{n,t} = B_\phi\Big(\mathbf{x}_{\mathcal{N}(n),\, t-H:t}\Big), \tag{8.1}$$

where $\mathcal{N}(n)$ is a local neighbourhood of node n on the regional mesh and H the input history. A graph-based limited-area model [92] or a regional adaptation of a message-passing global model [12, 13] is a natural choice. The backbone carries the bulk of forecasting skill; the experts and the causal gate add regime-specific and driver-specific corrections.

8.5 Candidate Climate Drivers and Lagged Causal Discovery

A *causal detector* identifies which large-scale drivers, and at what lags, carry information about each regional target after conditioning on local history. The candidate drivers come from ERA5 and Phase 1-style regional means: intraseasonal indices (BSISO1/2 amplitude and phase), ENSO and Indian Ocean SST indices, upstream moisture transport, and the state of the large-scale regions identified as drivers in Phase 1. The detector applies the Phase 1 pipeline with three changes motivated by the Phase 1 results:

1. **Lead-matched lags.** For a forecast at lead L, drivers are tested only at lags $\tau \geq L$, so that every retained link is usable at the issue time (Eq. (5.22)).
2. **Driver-level priors.** Claims are made at the level of drivers and source regions rather than individual fields, as in the region-level prior of Phase 1.
3. **Validation retained.** Decoys, the physical speed mask and resolution persistence are applied as in Phase 1; persistence is tested between the regional grid and a coarser aggregation.

Because hourly data are much more strongly autocorrelated than daily data, the effective sample size per hour of data is low. Discovery can be carried out on daily or six-hourly aggregates of the drivers, with the target aggregated to the same resolution, and the resulting graph applied to the hourly model.

8.6 Graph-Guided Gating and Expert Architecture

Figure 8.1 shows the planned architecture. Experts $\{E_k\}_{k=1}^{K}$ are small networks that refine the backbone's representation for particular regimes or sub-regions. A gate g assigns weights to the experts for each node and time,

$$\hat{\mathbf{y}}_{n,t+L} = D_\omega(\mathbf{z}_{n,t}) + \sum_{k=1}^{K} g_k(\mathbf{z}_{n,t}, \mathbf{c}_{n,t})\, E_k(\mathbf{z}_{n,t}, \mathbf{c}_{n,t}), \qquad \sum_k g_k = 1, \tag{8.2}$$

where D_ω is the backbone's decoder and $\mathbf{c}_{n,t}$ is a *causal context* vector: the values, at the issue time, of the drivers that the causal detector retained for node n, weighted by their link strengths. The gate is sparse (top-1 or top-2 routing [74, 76]) with a load-balancing penalty.

This differs from the Phase 1 use of the graph in an important way. In Phase 1 the graph selected *predictors*: it decided which source regions entered the forecast. In Phase 2 the graph conditions the *gate*: it decides which expert's correction to apply, given the state of the drivers. A driver that is too slow or too remote to add predictive information directly can still be useful for identifying the regime, for example whether the BSISO is in an active phase over India, in which a particular expert is reliable.

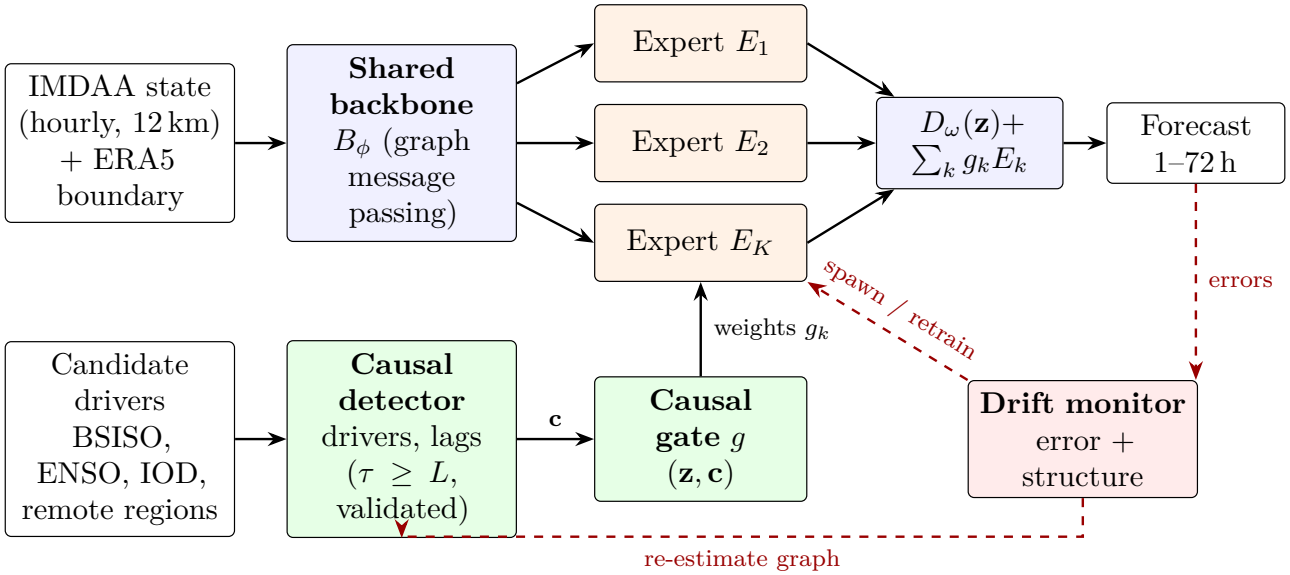


Figure 8.1: Planned Phase 2 architecture (schematic). A shared backbone encodes the regional state; a causal detector provides validated, lead-matched drivers that condition a sparse gate over experts; a drift monitor watches forecast errors and causal structure, and triggers re-estimation of the graph and the creation, retraining or reactivation of experts.

Open design choices. The design documents admit several expert structures, and none has been finalised: (i) *regional experts*, one per sub-region of India (for example by homogeneous rainfall zones); (ii) *per-node experts* with shared parameters and node embeddings; (iii) *regime experts* shared across space and selected by the causal context; or (iv) a hybrid of a shared expert and regime experts. Option (iii) uses the causal context most directly; option (i) is

closest to Phase 1. The choice will be made on validation data, and the alternatives are recorded in the decision log.

8.7 Drift Monitoring and Causal-Signature Changes

A deployed regional model faces three kinds of change: *seasonal* changes that recur every year, *interannual regime* changes such as ENSO phases, and *long-term* changes due to climate trends or changes in the observing system. The model should not react to the first (it can be learned), should adapt to the second, and should be retrained for the third.

Two monitoring channels are planned:

1. **Error channel.** The rolling forecast error of each expert is monitored with ADWIN [98], which keeps a variable-length window and signals a change when two sub-windows have significantly different means, or with CUSUM [97]:

$$C_t = \max\Big(0,\ C_{t-1} + (e_t - \mu_0 - \kappa)\Big), \qquad \text{alarm if } C_t > h, \tag{8.3}$$

   where e_t is the error, μ_0 its reference mean, κ a slack and h a threshold.

2. **Structure channel.** The causal detector is rerun on rolling windows and the change in the driver set or driver strengths is monitored with STARS [99]. Phase 1 showed that a set-based Jaccard distance on short windows is dominated by sampling noise, so Phase 2 uses a strength-based distance over longer windows against a fixed reference,

$$\Delta_w^{\mathrm{str}} = \frac{\|\mathbf{m}_w - \mathbf{m}_{\mathrm{ref}}\|_1}{\|\mathbf{m}_{\mathrm{ref}}\|_1}, \tag{8.4}$$

   where $\mathbf{m}_w$ is the vector of MCI strengths of the candidate links in window w.

STARS computes, for each new observation, a regime-shift index

$$\mathrm{RSI}_t = \sum_{s=t}^{t+m} \frac{x_s - \bar{x}_{\mathrm{new}}}{\ell\, \hat{\sigma}_\ell}, \tag{8.5}$$

where $\bar{x}_{\mathrm{new}}$ is the critical level that a new regime's mean must exceed (set from a t-test at a chosen significance level), ℓ is the cut-off length and $\hat{\sigma}_\ell$ the average standard deviation of ℓ-length intervals; a shift is accepted if the index keeps its sign over the cut-off length [99]. The two channels can be combined with OR, which reacts quickly at the cost of more false alarms, or with AND, which is more conservative.

8.8 Regime-Shift Detection and Selective Expert Adaptation

When a drift alarm is raised for a region or regime, the lifecycle of the affected expert is decided in tiers:

- **Tier 0, no action:** the alarm is not confirmed by the second channel, or the error is within its normal range.
- **Tier 1, retrain:** the expert is fine-tuned on recent data, with an elastic-weight-consolidation penalty [101] to limit forgetting of earlier regimes.
- **Tier 2, spawn or reactivate:** the current expert is archived together with its causal signature, and either a new expert is spawned or an archived expert is reactivated.

Archived experts are retrieved by the similarity of their causal signature to the current one. Two retrieval rules are compared: *single best match*, which reactivates the most similar archived expert, and *pool blend*, which blends the k most similar experts with weights proportional to their similarity. Because climate regimes such as El Niño recur, reactivation reuses knowledge rather than relearning it, and adding experts instead of overwriting them avoids catastrophic forgetting [102, 103].

8.9 Groundwork Implemented on the Phase 1 Dataset

The drift-monitoring and expert-lifecycle machinery has been implemented and tested on the Phase 1 daily OLR data, using the splitter's edge-score pattern as the structural signal (rather than the PCMCI parent sets of Stage D), at three regions chosen to span low, medium and high splitter discrimination (regions 22, 12 and 29). These experiments are the groundwork for the Phase 2 lifecycle.

Drift detection. Two test shifts were used: the abrupt 1997–98 El Niño and a gradual multi-year shift. Table 8.3 shows the results for region 22. The error channel alone missed the abrupt shift and detected the gradual one slowly; the structure channel detected both quickly; their OR combination detected both quickly; their AND combination missed the abrupt shift. At regions 12 and 29, both shifts were detected under all signal and combination settings, with region-specific delays; at region 29 the structure channel detected the gradual shift after 117 days, compared with 1,158 days for the error channel.

Expert retrieval. Table 8.4 compares the two retrieval rules and the choice between warm-starting a reactivated expert and spawning a fresh one. Blending the three most similar archived experts beat reactivating the single best match on forecast error in 61–69% of drift events and on forgetting at all three regions, which is the most consistently repeated finding of these experiments. Whether warm-starting helps is region-dependent.

Table 8.3: Drift detection on two test shifts, region 22 (Phase 1 data, splitter-score structure signal).

Detector	Abrupt shift (1997–98)	Gradual shift
Error channel only	missed	detected, slowly
Structure channel only	detected, quickly	detected
OR combination	detected, quickly	detected
AND combination	missed	detected, slowly

Table 8.4: Expert lifecycle comparisons on Phase 1 data at three regions. Fractions are drift events in which the first option has lower forecast error.

Comparison	Region 22	Region 12	Region 29
Pool blend beats single best (forecast error)	30/49 (61%)	42/61 (69%)	41/61 (67%)
Pool blend better on forgetting	yes	yes	yes
Warm start beats fresh spawn (forecast error)	28/34 (82%)	mixed	24/36 (67%)
Warm start better on forgetting	no	no	yes

Reconciling the two structural signals. The splitter-score signal detected the two test shifts, whereas the PCMCI parent-set distance of Stage D did not track ENSO across all years. The two results are not contradictory: they use different signals (continuous splitter scores versus discrete parent sets), different tests (two named events at three regions versus all phase changes at fifty regions) and different window lengths. Together they suggest that continuous, strength-based structural signals are more promising than set-based ones, which is the choice made for Phase 2 (Section 8.7).

8.10 Evaluation under Distribution Shift

The Phase 2 evaluation protocol follows the Phase 1 discipline of chronological splits, frozen choices and paired comparisons, and adds evaluation that is specific to shift:

- **Temporal holdout:** the most recent years are held out entirely; normalisation, graph discovery and model selection use training years only.
- **Regime-stratified scores:** errors reported separately for El Niño, La Niña and neutral seasons, active and break monsoon phases, monsoon onset and withdrawal, and pre-monsoon heat.
- **Leave-regime-out tests:** training without one regime type and testing on it, to measure generalisation to an unseen regime.
- **Detector metrics:** detection delay after a known shift, false-alarm rate in periods without a shift (with known recurring events counted as true shifts, not false alarms), and the number of lifecycle actions.
- **Adaptation metrics:** error after adaptation, forgetting (the increase of error on earlier

regimes after adaptation), and compute cost of retraining.

- **Accuracy metrics:** RMSE, MAE and bias per variable, lead and region; station-based verification against ISD; anomaly correlation for the diurnal-cycle-removed signal.

8.11 Planned Baselines and Benchmarks

Table 8.5 lists the planned baselines. The ablations isolate each component: the backbone alone, the backbone with experts but an uninformed gate, the gate with a random or correlation context in place of the causal context, and the full system with and without adaptation.

Table 8.5: Planned Phase 2 baselines and ablations.

Baseline / ablation	**Purpose**
Persistence; diurnal persistence (same hour yesterday)	Lower bounds for hourly forecasts
Hourly climatology by day of year	Reference for skill scores
Backbone only	Value of experts and gate
Backbone + experts, learned gate without causal context	Value of the causal context
Gate with random / correlation context of equal size	Specificity of causal information (as in Phase 1)
Full system without adaptation	Value of drift-triggered adaptation
Periodic retraining on a fixed schedule	Adaptation without drift detection
Published global ML forecasts, e.g. GraphCast [13], GenCast [15], interpolated to the regional grid	External benchmark, where forecasts are available for the test period

8.12 Implementation Status and Unresolved Design Choices

Table 8.6 records the status of each component.

8.13 Relationship to the Phase 1 Contribution

Phase 2 depends on Phase 1 in three ways. First, it inherits the validated components: the decoy test, the physical speed mask and the persistence test give a way to trust a driver set before using it, and they scale to large candidate pools. Second, it inherits two design corrections: lags must match the forecast lead, and set-based structural signals from short windows are too noisy to drive adaptation. Third, it inherits the evaluation discipline: graphs frozen before testing,

Table 8.6: Phase 2 component status.

Component	Status	Notes
Causal discovery and validation pipeline	Implemented (Phase 1)	To be extended with lead-matched lags and driver indices
Error and structure drift channels; STARS; ADWIN	Implemented (Phase 1 data)	Strength-based structure distance to be added
Expert archive, reactivation, pool blending	Implemented (Phase 1 data)	Tested at three regions
India datasets (IMDAA, ERA5, ISD)	Planned	Versions, periods and licences to be confirmed
Shared backbone	Designed	Architecture choice open
Causal gate and expert structure	Designed	Regional, per-node, regime or hybrid experts: open
Shift-aware evaluation	Designed	Protocol in Section 8.10
External benchmarks	Planned	Depends on forecast availability for the test period

matched random and correlation controls, and reporting of neutral and negative results. The question left open by Phase 1, whether the causal content of a graph improves on a random graph of the same size, is equally central to Phase 2, where it becomes the comparison between a causal and a random gate context.

8.14 Future Scale-Up Considerations

If Phase 2 shows regional value, the approach could be extended to larger domains and eventually to a global mesh. Three issues would then dominate: the cost of causal discovery, which grows with the number of targets times the number of candidates and would require hierarchical or dimension-reduced candidate sets; the number of experts, which must stay bounded through archiving and merging; and evaluation against global data-driven systems on standard benchmarks such as WeatherBench 2 [87]. Global forecasting is not part of this thesis and is discussed only as a long-term direction (Section 9.8).

8.15 Chapter Summary

Phase 2 extends the causal-graph approach to hourly, kilometre-scale forecasting of near-surface variables over India up to 72 hours ahead. It combines a shared backbone, a causal detector with lead-matched and validated drivers, a gate conditioned on causal context, and drift-triggered expert adaptation with archiving and reactivation. The drift and lifecycle machinery has been implemented and tested on Phase 1 data, where blending archived experts outperformed single-expert reactivation in a majority of drift events and on forgetting at all three regions tested; the remaining components are designed, with their open choices and evaluation protocol recorded.

Chapter 9

Conclusion and Scope for Further Research

9.1 Summary of the Scientific Problem

Forecasting models choose their predictors by association, and in the tropical atmosphere association is inflated by autocorrelation, shared seasonal cycles and common drivers. This thesis asked whether a graph of lagged dependencies, found by causal discovery, validated by independent checks and frozen before the test years, can tell a forecasting model which source regions to use. The setting was daily OLR, a proxy for deep convection, forecast for 50 tropical regions during the BSISO season, with 1979–2012 for development and 2013–2022 held out.

9.2 Summary of the Methods

The work built a 300-node representation of six variables over 50 OLR-coherent regions and an activity-based BSISO season from a reproduced BSISO index. It applied PCMCI with partial-correlation tests and Benjamini–Hochberg FDR control to 18,000 lagged candidates per target, using a vectorised implementation verified against Tigramite. It validated the graph with four checks: phase-randomised decoys in a separate pass, a physical propagation-speed mask, data-driven top-k selection, and persistence at doubled spatial resolution. It merged the validated links into a region-level prior for an invariance-trained splitter and per-region experts, defined three modes of use and three matched control priors, evaluated the prior on held-out years at 7-day and 1-day leads, and measured the temporal stability of the causal structure against ENSO. Finally, it designed an India-focused hourly extension with causal gating and drift-triggered expert adaptation.

9.3 Objective-by-Objective Assessment

Table 9.1 summarises the outcome of each objective. Six of the seven objectives are achieved. Objective O5 is partly achieved: the most informative first comparison has been made, and the

Table 9.1: Assessment of the research objectives.

Objective	Outcome	Status
O1 Data representation	300 nodes, 1979–2022, development/test split; BSISO index with 11.2%/7.7% leading variances; season of 208 days per year on average	Achieved
O2 Graph discovery	536 links over 900,000 candidates at FDR 0.05; vectorised PCMCI verified against Tigramite; 23 minutes for 50 targets	Achieved
O3 Graph validation	0/250 decoys; 3/536 speed violations; $k^* = 10$; mean persistence 0.62 (33/48 targets ≥ 0.5)	Achieved
O4 Graph use	Region-level prior (2.06 regions per target); restrict, initialise and regularise modes and random and correlation controls implemented and unit-tested	Achieved
O5 Forecast evaluation	Restrict prior vs no prior in the full pool at 7 and 1 days reported; remaining cells of the 27-configuration grid to follow	Partly achieved
O6 Temporal stability	Causal-signature distance computed; no alignment with ENSO	Achieved (negative result)
O7 Extension design	Phase 2 architecture, data, baselines and shift-aware evaluation specified; lifecycle groundwork tested	Achieved (design)

remaining comparisons use the same, already tested infrastructure.

9.4 Contributions and Evidence

1. **A trustworthy graph.** The Phase 1 graph passes every validation check that binds: no decoy selected among 250, 3 of 536 links physically impossible, 62% mean persistence at doubled resolution, and independent agreement with the splitter for region 22.
2. **A reproduced BSISO index and season** from the project's own data that matches the published variance split.
3. **A verified, fast PCMCI implementation** that matches Tigramite and makes an 18,000-candidate-per-target analysis take 23 minutes.
4. **The graph makes source selection discriminative.** With the restrict prior, the number of informative full-pool splitters rises from 0 to 24 of 50 at 7 days and from 0 to 36 at 1 day.
5. **A short-lead forecast improvement over the no-prior model.** At 1 day, the restrict prior improves skill in 32 of 50 regions (mean skill relative to persistence $-1.53 \rightarrow -1.33$; mean R^2 relative to climatology $0.720 \rightarrow 0.745$).
6. **A characterisation of the graph's time scale.** Outside links act at one or two days, which explains why the graph helps at 1 day and not at 7.
7. **A negative result on structural drift.** Set-based causal signatures from one- and three-season windows do not track ENSO.

8. **A Phase 2 design** that transfers the validated components and the lessons on lags and drift signals.

9.5 What the Experiments Do and Do Not Establish

The experiments establish that PCMCI at full scale produces a sparse, interpretable and robust graph for BSISO-season OLR; that using this graph as a restrict prior makes the splitter's choices discriminative; that at the 7-day lead the prior does not change forecast skill; and that at the 1-day lead it improves skill relative to the no-prior model in most regions.

They do not establish that the graph's links are causal in a mechanistic sense, which no observational check can do. They do not yet establish that the one-day gain comes from the graph's choice of regions rather than from the reduced size of the candidate pool; this is the question answered by the random-prior control. And they do not establish that the causal structure changes with climate regimes in a detectable way. The central Phase 1 hypothesis, that a validated causal graph improves both source selection and forecast skill on unseen years, is therefore *not yet confirmed*: the graph is reliable, and its forecasting benefit is so far shown only against the no-prior model and only at short lead.

9.6 Scientific and Engineering Limitations

The main limitations are those listed in Section 7.14: the absence of ground truth; finite statistical power with 18,000 candidates per target; hidden common drivers; the instability of field choice within a region; the partly circular construction of the regions; the two-year overlap of the upstream climatology with the test period; a single training seed; low absolute skill at seven days, where the models lose to climatology on average; the restriction to linear, lagged dependence; and a single data source. On the engineering side, the forecaster is deliberately small, and the full grid of priors, modes and pools, with multiple seeds, requires several hours of computation per seed.

9.7 Lessons for the Phase 2 Extension

- Validate before use: the decoy, speed and persistence checks are cheap relative to the value of trusting a driver set, and they scale.
- Match the graph to the forecast: test only lags that are usable at the forecast lead.
- Claim at the right granularity: region-level or driver-level priors are more reliable than field-level ones.
- Always include matched random controls: a reduction of the candidate pool can improve a model on its own.

- Prefer continuous, strength-based structural signals over discrete parent sets for drift detection.
- Use the graph for routing as well as for selection: drivers that do not add predictive information directly can still identify the regime in which an expert is reliable.

9.8 Future Directions

The future work is ordered by its importance for the central question.

1. **Matched controls.** Run the random- and correlation-prior controls in restrict mode, full pool, at 1 and 7 days. This determines whether the one-day gain is due to the graph's content.
2. **Lead-matched graph.** Rebuild the graph with $\tau_{\min} = 7$ and repeat the comparison at 7 days, to test whether week-ahead precursors exist in this representation and whether they improve the forecast.
3. **Forecast-aware top-k.** Choose the prior size by forecast-horizon skill, or by requiring the selected links to outperform random links of the same size.
4. **Decoy strengths before FDR.** Record the full distribution of decoy MCI strengths to give the B1 strength comparison a non-degenerate threshold.
5. **The full grid with several seeds.** Run the initialise and regularise modes and the direct and 2hop pools, with three to five seeds, block-bootstrap confidence intervals and Diebold–Mariano tests.
6. **Sensitivity analyses.** A full-year graph, persistence between the two halves of the development period, a second reanalysis or OLR product, and a climatology refitted on 1979–2012.
7. **Nonlinear and latent-variable discovery.** Conditional-mutual-information tests and LPCMCI [51] to address nonlinear coupling and unobserved drivers.
8. **Strength-based structural drift.** Longer windows with a fixed or rolling reference and strength-based distances, evaluated against known ENSO and monsoon regime changes.
9. **Phase 2 implementation.** Acquire and preprocess IMDAA, ERA5 and ISD; implement the backbone, causal gate and lifecycle; and evaluate with the shift-aware protocol.

9.9 Potential Wider-Scale Applications

If the regional system shows value, the same protocol could be applied over larger domains, other monsoon regions, or a global mesh. At that scale, causal discovery would need hierarchical candidate sets, the expert pool would need active management, and evaluation would follow

standard global benchmarks such as WeatherBench 2 [87] against global data-driven models [13, 15]. Beyond weather forecasting, the build–validate–freeze–use protocol applies to any setting in which a model must choose among many lagged, autocorrelated candidate predictors, such as hydrology, air quality or energy demand.

9.10 Concluding Remarks

This thesis set out to test whether a causal graph can guide a forecasting model. It found that a graph of lagged climate dependencies can be built at full scale and that it survives every independent check that could have rejected it. It also found that a credible graph is not automatically a useful prior: its benefit depends on whether its time scale matches the forecast, and an apparent benefit must be compared with random alternatives of the same size before it is credited to causal structure. Separating the credibility of a graph from its usefulness, and reporting both honestly, is the central methodological lesson of this work. It is also the principle on which the India-focused extension is designed.

Appendix A

Mathematical Notation and Supplementary Derivations

A.1 Notation Conventions

Scalars are italic (x), vectors bold lower case ($\mathbf{x}$), matrices bold upper case ($\mathbf{X}$) and sets calligraphic ($\mathcal{E}$). The time index t counts days in Phase 1 and hours in Phase 2. A node is a pair $i = (r, v)$ of region and variable; j denotes a target node. Lags τ and leads L are positive integers in units of the time step. Estimated quantities carry a hat. The full symbol list is given in the front matter (List of Symbols).

A.2 Partial Correlation as Correlation of Regression Residuals

Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ be centred samples and $\mathbf{Z} \in \mathbb{R}^{n \times c}$ a conditioning matrix with orthonormal basis $\mathbf{Q}$ (so $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$ and $\mathbf{P} = \mathbf{Q}\mathbf{Q}^\top$ is the orthogonal projector onto the column space of $\mathbf{Z}$). The least-squares residuals of $\mathbf{x}$ and $\mathbf{y}$ on $\mathbf{Z}$ are $\mathbf{r}_x = (\mathbf{I} - \mathbf{P})\mathbf{x}$ and $\mathbf{r}_y = (\mathbf{I} - \mathbf{P})\mathbf{y}$. The sample partial correlation is the correlation of these residuals,

$$\hat{\rho}(x, y \mid \mathbf{Z}) = \frac{\mathbf{r}_x^\top \mathbf{r}_y}{\|\mathbf{r}_x\| \|\mathbf{r}_y\|} = \frac{\mathbf{x}^\top (\mathbf{I} - \mathbf{P})\mathbf{y}}{\sqrt{\mathbf{x}^\top (\mathbf{I} - \mathbf{P})\mathbf{x}} \sqrt{\mathbf{y}^\top (\mathbf{I} - \mathbf{P})\mathbf{y}}}, \tag{A.1}$$

using the idempotence and symmetry of $\mathbf{I} - \mathbf{P}$. For m candidate columns $\mathbf{X}_C$ sharing the same conditioning set, the numerators for all candidates are the vector $\mathbf{X}_C^\top (\mathbf{I} - \mathbf{P})\mathbf{y} = (\mathbf{X}_C - \mathbf{Q}\mathbf{Q}^\top \mathbf{X}_C)^\top \mathbf{r}_y$, which is the batched computation used in Algorithm 4.1. Under joint Gaussianity and the null hypothesis $\rho = 0$, the statistic $t = \hat{\rho} \sqrt{(n - 2 - c)/(1 - \hat{\rho}^2)}$ follows a Student-t distribution with $n - 2 - c$ degrees of freedom, because $\hat{\rho}$ is then the ordinary correlation of two independent Gaussian vectors in the $(n - c)$-dimensional residual space [28].

A.3 Phase Randomisation Preserves the Autocorrelation Function

Let $\hat{x}_f$ be the discrete Fourier transform of a demeaned series of length n. By the Wiener–Khinchin theorem, the circular sample autocovariance is the inverse transform of the periodogram,

$$\hat{\gamma}(h) = \frac{1}{n}\sum_{t=0}^{n-1} x_t\, x_{(t+h) \bmod n} = \frac{1}{n^2}\sum_{f} |\hat{x}_f|^2\, e^{2\pi i f h/n}. \tag{A.2}$$

A phase-randomised surrogate has Fourier coefficients $\hat{\hat{x}}_f = |\hat{x}_f| e^{i\theta_f}$, so $|\hat{\hat{x}}_f|^2 = |\hat{x}_f|^2$ and its circular autocovariance is identical to that of the original. The cross-periodogram between a surrogate and any other fixed series y is $\hat{\hat{x}}_f \overline{\hat{y}_f} = |\hat{x}_f||\hat{y}_f| e^{i(\theta_f - \arg \hat{y}_f)}$; with θ_f uniform and independent, its expectation is zero at every non-zero frequency, so the expected circular cross-covariance at every lag is zero. The linear (non-circular) autocovariance differs from the circular one by end effects of order h/n, which are negligible for $h \le 60$ and $n \approx 16{,}000$.

A.4 Benjamini–Hochberg as a Step-Up Procedure

Given m p-values sorted as $p_{(1)} \le \cdots \le p_{(m)}$, the BH procedure rejects the hypotheses $H_{(1)}, \ldots, H_{(h^*)}$ with $h^* = \max\{h : p_{(h)} \le h\alpha/m\}$. The adjusted values of Eq. (4.10), $q_{(h)} = \min_{h' \ge h} \min(1, m\, p_{(h')}/h')$, satisfy $q_{(h)} \le \alpha \iff h \le h^*$, so thresholding q at α reproduces the procedure. Under independence of the null p-values, $\mathbb{E}[V/\max(S,1)] = \alpha m_0/m \le \alpha$, where m_0 is the number of true nulls [61]; the bound continues to hold under positive regression dependence on the subset of nulls [62]. Counting candidates removed by PC1 with $p = 1$ increases m without adding rejections, which makes the procedure conservative relative to correcting over the survivors only.

A.5 Implied Speed as a Lower Bound

Any signal travelling from region r' to region r must cover at least the shortest great-circle distance between a point of r' and a point of r. The closest-cell distance $d(r', r)$ of Eq. (5.4) approximates this minimum from cell centres, with an error of at most about half a cell diagonal (about 200 km at 2.5°). A signal that takes at most τ days therefore needs a speed of at least $d(r', r)/(\tau\, \Delta t)$, so s_e is a lower bound on the required speed, and the B2 test rejects a link only if even this most favourable speed is too high. For adjacent regions, $d = 0$ and every lag passes.

A.6 Relation Between the Two Skill Scores

For a standardised anomaly target with training mean close to zero, $\mathrm{MSE}_{\mathrm{clim}} \approx \mathrm{Var}(Y)$. For persistence at lead L, $\mathrm{MSE}_{\mathrm{pers}} = \mathbb{E}[(Y_{t+L} - Y_t)^2] = 2\,\mathrm{Var}(Y)\big(1 - \rho_Y(L)\big)$ for a stationary series

with lag-L autocorrelation $\rho_Y(L)$. Hence

$$1 - R^2_{\text{clim}} = \frac{\text{MSE}}{\text{Var}(Y)} = 2\Big(1 - \rho_Y(L)\Big)\Big(1 - S_{\text{pers}}\Big). \tag{A.3}$$

At a short lead, $\rho_Y(L)$ is close to one and persistence is very accurate, so a model can have a large R^2_{clim} and still a strongly negative S_{pers}, as at the 1-day lead in Table 7.7. At a long lead, $\rho_Y(L) \approx 0$, so $\text{MSE}_{\text{pers}} \approx 2\,\text{Var}(Y)$; a model with $S_{\text{pers}} \approx 0.4$ then has $1 - R^2_{\text{clim}} \approx 1.2$, consistent with the slightly negative R^2_{clim} at the 7-day lead. Both outcomes follow directly from the persistence properties of OLR rather than from any defect of the models.

A.7 Counting Candidate Combinations

With N nodes, lags $1, \ldots, \tau_{\max}$ and J target nodes, the number of lagged candidates per target is $N\tau_{\max}$ (each node, including the target itself, at each lag), and the total for all targets is $JN\tau_{\max}$. For $N = 300$, $\tau_{\max} = 60$ and $J = 50$ this is 18,000 per target and 900,000 in total. If all N nodes were targets and self-links were excluded, the count would be $N(N-1)\tau_{\max} = 5{,}382{,}000$. For the decoy pass, $N = 550$ gives 33,000 candidates per target; for the 100-region pass, $N = 596$ gives 35,760.

Appendix B

Configuration, Variable Dictionary, and Experiment Registry

B.1 Variable Dictionary

Table B.1: Variable and derived-quantity dictionary.

Name (code)	Definition	Notes
`sst`	Sea surface temperature anomaly	Region mean over ocean cells only
`h850`	850 hPa geopotential height anomaly	Region mean over all cells
`u200`	200 hPa zonal wind anomaly	Region mean
`pw`	Precipitable water anomaly	Region mean
`u850`	850 hPa zonal wind anomaly	Region mean; BSISO index field
`olr`	Outgoing longwave radiation anomaly	Region mean; forecast target; BSISO index field
`is_ocean`	Ocean fraction of a region	Static feature appended to each feature vector
Node (r, v)	Standardised region mean of variable v	Standardised over 1979–2012
Feature vector	6 fields $\times$ lags $\{0, 1, 2, 3, 5, 7, 10\}$ + ocean fraction	49 values per region per day

Name (code)	Definition	Notes
BSISO amplitude	$\sqrt{\mathrm{PC1}^2 + \mathrm{PC2}^2}$	Combined OLR/u850 EOF, May–Oct 1979–2012
Season mask	Activity-based target-day set	31-day mean amplitude > 1 for ≥ 20 days, $\pm$21-day pad
Niño-3.4 proxy	Season-mean SST, 5°S–5°N, 170°W–120°W, detrended	El Niño $> +0.5$ SD; La Niña < -0.5 SD
MCI strength	$\hat{\rho}^{\mathrm{MCI}}_{i \to j}(\tau)$	Signed partial correlation
Implied speed	$d(r', r)/(\tau \cdot 86{,}400\,\mathrm{s})$	Closest-cell haversine distance
Persistence score	Π_r	Fraction of outside links reappearing at 100 regions
Prior strength	$\pi_r(r') = \max \lvert\hat{\rho}^{\mathrm{MCI}}\rvert$	Over validated links from region r'

B.2 Hyperparameter Summary

Table B.2 collects every fixed setting of Phase 1 in one place; the stage-specific tables are Table 6.2 (Stage A), Table 6.3 (Stage B) and Table 6.4 (Stage C).

Table B.2: All fixed Phase 1 settings.

Stage	Setting	Value
Data	Regions / nodes	50 / 300 (100 / 596 for B4)
	Development / test	1979–2012 / 2013–2022
Season	Index domain	10°S–30°N, 40°–160°E
	Search window, threshold, run, pad	1 Apr–30 Nov; 1; 20 days; 21 days
A	Lags; test; α_{PC}; α_{FDR}	1–60; ParCorr; 0.05; 0.05 (BH)
B1	Decoys; gate	250; decoy rate $< 30\%$
B2	Speed limits	SST 1, OLR/PW 20, h850/u850/u200 $50\,\mathrm{m\,s^{-1}}$
	Oscillation flag	$\tau \geq 25$ days
B3	k range; ridge; fit/score	0–10; 10^{-3}; 1979–2004 / 2005–2012
B4	Lag tolerance; pass	$\pm$2 days; $\Pi_r \geq 0.5$
C	Epochs; samples; lr; batch	5; 4,000; 10^{-3}; 16

Stage	Setting	Value
	Hidden; window; swaps; $\lambda_{\max}$	16; 10 days; 6; 1 (30% warm-up)
	ρ_{top}; β; γ; seed	$\min(0.5, 3/\lvert\mathcal{C}_r\rvert)$; 2; 1; 0
D	Windows; candidate set	1 and 3 seasons; B1+B2 links + own OLR lags 1–3
	Distance; detector	1−Jaccard of (region, field) sets; STARS

B.3 Physical-Mask Assumptions and References

The B2 limits (Table 5.1) rest on three bodies of evidence. The eastward speed of the intraseasonal convective envelope is about $5\,\mathrm{m\,s^{-1}}$ [8, 11]. Convectively coupled Kelvin waves propagate at roughly 15–$20\,\mathrm{m\,s^{-1}}$, and dry or weakly coupled Kelvin waves faster [27, 43]; equatorial wave theory [25] and the Gill response [26] explain how localised heating projects onto these waves. SST anomalies evolve through slow oceanic and air–sea processes, so a direct SST-to-target link faster than about $1\,\mathrm{m\,s^{-1}}$ is implausible; fast atmospheric responses to SST are carried by atmospheric nodes. The limits are upper bounds chosen to be generous: a link that fails them would need a speed above any of these processes even along the shortest path.

B.4 Experiment Registry and Ablation Matrix

Table B.3: Experiment registry. **measured**: results in Chapter 7 or 8; *planned*: defined, to be run.

ID	Experiment	Question	Status
E1	BSISO index and season	Reproduce published index	**measured**
E2	Stage A, 50 regions	RQ1	**measured**
E3	B1 decoy pass	RQ2	**measured**
E4	B2 lag mask	RQ2	**measured**
E5	B3 top-k	RQ2	**measured**
E6	Stage A at 100 regions; B4	RQ2	**measured**
E7	No prior, 3 pools, 7 d, season	Baseline	**measured**
E8	PCMCI restrict, full, 7 d	RQ3	**measured**
E9	No prior and PCMCI restrict, full, 1 d	RQ3	**measured**
E10	Stage D, 1- and 3-season windows	RQ5	**measured**

ID	Experiment	Question	Status
E11	Alternative splitters (CIA, GSINA), region 22	Splitter choice	**measured**
E12	Drift channels and lifecycle, regions 12, 22, 29	Phase 2 groundwork	**measured**
E13	Random prior, restrict, full, 1 and 7 d	RQ4	*planned*
E14	Correlation prior, restrict, full, 1 and 7 d	RQ4	*planned*
E15	Graph with $\tau_{\min} = 7$; rerun E8, E13, E14	RQ3 at matched lead	*planned*
E16	Top-k by forecast-horizon skill or vs random	B3 sharpness	*planned*
E17	Decoy strengths before FDR	B1 strength bar	*planned*
E18	Initialise and regularise modes, all pools	RQ3, RQ4	*planned*
E19	3–5 seeds for all Stage C cells	Uncertainty	*planned*
E20	Full-year graph	Season specificity	*planned*
E21	Persistence across development-period halves	Temporal robustness	*planned*
E22	Second reanalysis / OLR product	Data-source robustness	*planned*
E23	Climatology refitted on 1979–2012	Leakage sensitivity	*planned*
E24	Strength-based drift, longer windows	RQ5	*planned*
E25	Prior excluding B4-flagged links	B4 as a filter	*planned*

B.5 Data Sources and Licences

Table B.4: Data sources. Licence terms must be confirmed with each provider before redistribution of derived data.

Source	Phase	Licence / access
BSISO monitoring archive (six fields)	1	[confirm with provider]
IMDAA regional reanalysis [104]	2	Distributed by the Indian national meteorological agencies; registration required [confirm terms]
ERA5 [93]	2	Copernicus Climate Change Service licence [confirm version]
NOAA ISD [105]	2	NOAA open data; WMO data-exchange restrictions may apply to some stations [confirm]

Appendix C

Reproducibility, Artifacts, and Decision Log

C.1 Repository Structure

All code lives under `code/` in the project repository. The Phase 1 components are:

```
code/
  causal_moe/
    data/ raw.py, clustering.py, mesh.py, windows.py, splits.py,
                candidate_edges.py (direct / 2hop / full pools)
    causal_graph/ season.py BSISO index and season mask
                nodes.py 300 / 596 node series, distances
                pcmci.py Stage A (PC1 + MCI, ParCorr, BH-FDR)
                validate.py Stage B (decoys, lag mask, top-k, persistence)
                prior.py Stage C (pcmci / random / corr; restrict / init / regularize)
    splitter/ dirgnn.py (invariance-trained splitter), cia.py, gsina.py
    experts/ expert.py (per-region expert), pool.py (expert blending)
    drift/ rodionov.py (STARS), channels.py, archive.py, forgetting.py
    baselines/ simple.py
  scripts/ phase1_build_graph.py, phase1_run_stage_c.py,
                phase1_stage_d_drift.py, generate_phase1_figures.py, ...
  tests/ automated tests (including test_pcmci.py)
  results/phase1/ frozen graph, check outputs, Stage C and D results, figures
thesis/ this LaTeX project
```

C.2 Result Files Behind the Reported Numbers

Table C.1: Mapping from reported results to result files (paths relative to `code/results/`).

Result	File
BSISO variance, season statistics (Table 7.2, Fig. 7.1)	`phase1/bsiso_season.json`, `.npz`
Stage A graph (Table 7.3)	`phase1/stage_a_50.json`
B1 decoys (Fig. 7.4)	`phase1/b1_decoy_pass.json`
B2, B3, prior, flags (Figs. 7.3–7.6)	`phase1/causal_graph_50.json`
B4 (Fig. 7.7)	`phase1/stage_a_100.json`, `causal_graph_50.json`
No-prior pools, season (Table 7.6)	`phase1/stage_c/{direct,2hop,full\}\_none.json`
No-prior pools, year-round (Table 7.6)	`step7_all_places_direct_lead7.json`, `step7_all_places_2hop.json`, `step7_all_places_full.json`
Stage C, 7 days (Table 7.7, Fig. 7.9)	`phase1/stage_c/full_{none,pcmci_restrict\}.json`
Stage C, 1 day (Tables 7.7, 7.8, Fig. 7.10)	`phase1/stage_c_lead1/full_{none,pcmci_restrict\}.json`
Stage D (Table 7.10, Fig. 7.11)	`phase1/stage_d_drift_w{1,3\}.json`
Alternative splitters (Table 7.9)	`step4_place22.json`, `step_cia_place22_direct_w1.0.json`, `step_gsina_place22_direct_seed0.json`
Drift detection (Table 8.3)	`step5_drift_place{22,12,29\}\_direct.json`
Expert lifecycle (Table 8.4)	`step6_lifecycle_place{22,12,29\}\_direct.json`

The figures of Chapter 7 were produced by `scripts/generate_phase1_figures.py`, except Figure 7.8, produced by the project's earlier figure script, and Figure 7.10, produced for this thesis by `thesis/scripts/make_stage_c_figures.py` from the Stage C 1-day result files. The paired tests of Section 7.7 were computed from the per-region values in the same files.

C.3 Step-by-Step Reproduction Guide

The following steps reproduce every Phase 1 result, starting from the repository root. Install the pinned dependencies, and Tigramite for the PCMCI verification tests; place the gridded archive under `data/raw/rt_preprocessed_archive/` (six `{field}_rt_1979_2022.npz` files and `parameters/sst_ocean_mask.npz`); then run, from `code/`:

```
pip install -r requirements.txt tigramite
python -m pytest tests # automated tests
python scripts/build_mesh_cache.py # 7-day feature cache
python scripts/build_mesh_cache.py --lead-time-days 1 \
    --out cache/windowed_clustered50_lead1_v2.npz # 1-day feature cache
python scripts/phase1_build_graph.py # Stages A and B (~2.5 h)
python scripts/phase1_run_stage_c.py --pools full --priors none,pcmci --modes restrict
python scripts/phase1_run_stage_c.py --pools full --priors none,pcmci --modes restrict \
    --cache-path cache/windowed_clustered50_lead1_v2.npz \
    --out-dir results/phase1/stage_c_lead1 # 1-day lead
python scripts/phase1_run_stage_c.py --workers 12 # full 27-configuration grid
python scripts/phase1_stage_d_drift.py --window-seasons 1
python scripts/phase1_stage_d_drift.py --window-seasons 3
python scripts/generate_phase1_figures.py # figures from saved results
```

The thesis itself is built from `thesis/` with `latexmk -pdf mythesis.tex` (see `thesis/README.md`).

C.4 Decision Log

Table C.2: Methodological decisions: the choice made, the alternatives, and open items.

ID	Decision	Choice and rationale	Alternatives / status
D1	Prior granularity	Region level; the splitter selects whole regions and field choice within a region is unstable	Field- or lag-level prior; kept in graph file
D2	CI test	ParCorr; fast, exact for linear-Gaussian, verified	CMI-kNN, GPDC for nonlinear dependence (future)
D3	Source conditioning	Own past for non-target nodes; full parents for target nodes	Full graph for all 300 nodes (higher cost)
D4	FDR scope	Per target over all 18,000 candidates	Over survivors only (less conservative); over all targets jointly
D5	Lag range	1–60 days, lag 0 excluded	$\tau_{\min} = L$ for lead-matched graph (E15)
D6	B2 failures	Removed from prior, kept in graph	Diagnostic only; or hard delete
D7	B3 ranking	$\lvert\hat{\rho}^{\mathrm{MCI}}\rvert$	Ablation-based importance; bootstrap frequency

ID	Decision	Choice and rationale	Alternatives / status
D8	B4 failures	Flag, keep in prior	Exclude flagged links (E25)
D9	Region means	Unweighted cell means	Area-weighted ($\cos\varphi$); difference <13% in weights
D10	Season	Activity-based, target days only	Fixed May–Oct; full-year (E20)
D11	Anomaly climatology	Upstream, 1979–2014	Refit on 1979–2012 (E23)
D12	Forecast type	Direct, separate model per lead	Autoregressive roll-out
D13	First Stage C comparison	Restrict, full pool, 7 and 1 days	Remaining grid cells (E13–E19)
D14	Data provenance	Archive as received	Confirm originating products and licence
D15	Phase 2 expert structure	Open	Regional, per-node, regime or hybrid experts
D16	Phase 2 datasets	IMDAA, ERA5, ISD (planned)	Versions, periods and licences to confirm

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List of Publications

(Based on the Research Work)

Papers Published

[None at the time of submission. Replace this line with published papers, if any.]

Papers Communicated

[None at the time of submission. Replace this line with communicated papers, if any.]